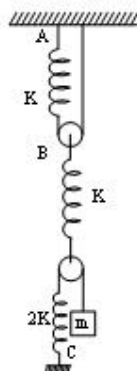


11. The potential energy of a particle of mass 0.1 kg, moving along x-axis under the action of a single force, is given by $U = 5x(x-4)$ J where x is in metres. It can be concluded that
- A) The particle is acted upon by a constant force
 - B) The speed of the particle is maximum at $x = 2$ m
 - C) The particle executes simple harmonic motion
 - D) The period of oscillation of the particle is $\pi/5$ s
12. In the given figure, the block is attached with a system of three ideal springs A,B,C. The block is displaced by a small distance x from its equilibrium position vertically downwards and released. T represents the time period of small vertical oscillations of the block. Then (Pulleys are ideal)



A) $T = 2\pi\sqrt{\frac{11m}{2K}}$

B) The deformation of the spring A due to displacement of block is $(2/11)$ times the displacement of the Block

C) The deformation of the spring C due to displacement of block is $(1/11)$ times the displacement of the block

D) The deformation of the spring B due to displacement of block is $(4/11)$ times the displacement of the block

13. An ideal spring has natural length 40 cm and spring constant 500 N/m. A block of mass 1 kg is attached at one end of the spring and other end of the spring is attached to ceiling. The block released from the position, where the spring has length 45 cm.

A) the block will perform SHM of amplitude 5 cm.

B) the block will have maximum velocity $30\sqrt{5} \text{ cm/sec}$

C) the block will have maximum acceleration 15 m/s^2 .

D) the minimum potential energy of the spring will be zero.

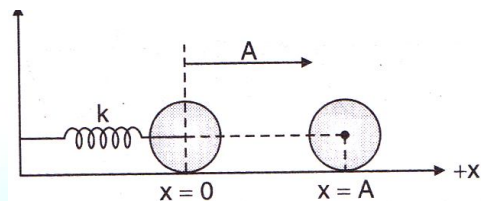
14. Two particles execute SHM with amplitudes A and $2A$ respectively and angular frequencies ω and 2ω respectively. At $t = 0$ they start with some initial phase difference. At, $t = \frac{2\pi}{3\omega}$. They are in same phase. Their initial phase difference is:
- A) $\frac{\pi}{3}$ B) $\frac{2\pi}{3}$ C) $\frac{4\pi}{3}$ D) π

SECTION – III
(PARAGRAPH TYPE)

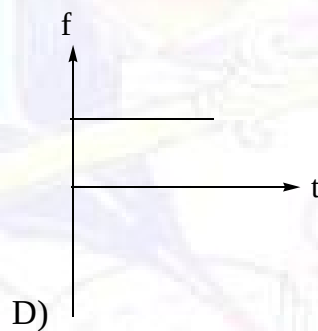
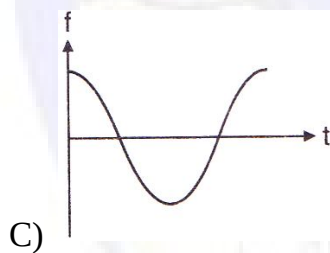
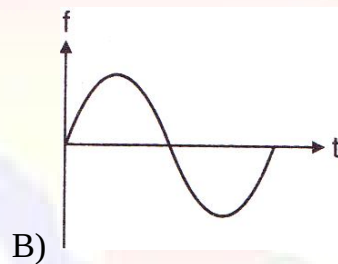
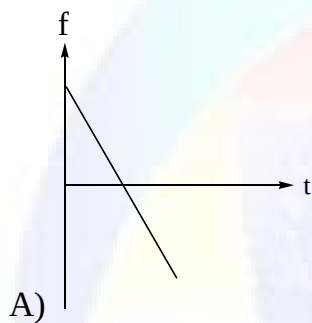
This section contains **2 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct. **Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.**

Paragraph For Questions 15 and 16

A disc of mass m and radius R is attached with a spring of force constant k at its centre as shown in figure. At $x = 0$, the spring is unstretched. The disc is moved to $x = A$ and then released. There is no slipping between disc and ground. Let f be the force of friction on the disc from the ground. Based on the information given, answer the following questions.



15. The f versus t (time) graph will be as



16. If $k = 10 \text{ Nm}^{-1}$, $m = 2 \text{ kg}$, $R = 1 \text{ m}$ and $A = 2 \text{ m}$, then the linear speed of the disc at mean position is

A) $\sqrt{\frac{40}{3}} \text{ ms}^{-1}$ B) $\sqrt{20} \text{ ms}^{-1}$ C) $\sqrt{\frac{10}{3}} \text{ ms}^{-1}$ D) $\sqrt{\frac{50}{3}} \text{ ms}^{-1}$

Paragraph For Questions 17 and 18

A forced action on a block is $F = (-4x + 8)$, where F is in Newton and x the position of block on x -axis in meters. The energy of oscillation is 18J. Based on the information given, answer the following questions.

17. The motion of the block is
- A) Periodic but not simple harmonic
B) Not periodic
C) Simple harmonic about origin, $x = 0$
D) Simple harmonic about $x = 2 \text{ m}$
18. The block will oscillate between the points
- A) $x = 0$ and $x = 4 \text{ m}$ B) $x = -1 \text{ m}$ and $x = 5 \text{ m}$
C) $x = -2 \text{ m}$ and $x = 6 \text{ m}$ D) $x = 1 \text{ m}$ and $x = 3 \text{ m}$

CHEMISTRY

Max Marks: 62

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.

19. Which is incorrect product?

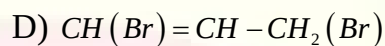
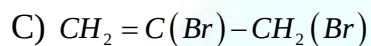
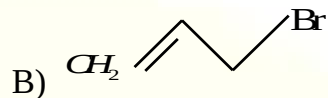
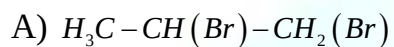
- A) $\text{CH}_3 - \text{C} \equiv \text{C} - \text{CH}_3 \xrightarrow{\text{Na / Liq. NH}_3} \begin{array}{c} \text{H}_3\text{C} \quad \text{CH}_3 \\ \diagdown \quad \diagup \\ \text{C} = \text{C} \\ \diagup \quad \diagdown \\ \text{H} \quad (\text{cis}) \quad \text{H} \end{array}$
- B) $\text{CH}_3 - \text{C} \equiv \text{C} - \text{CH}_3 \xrightarrow[\text{Pd / BaSO}_4]{\text{H}_2} \begin{array}{c} \text{H}_3\text{C} \quad \text{CH}_3 \\ \diagdown \quad \diagup \\ \text{C} = \text{C} \\ \diagup \quad \diagdown \\ \text{H} \quad (\text{cis}) \quad \text{H} \end{array}$
- C) $\begin{array}{c} \text{H} \quad \text{CH}_3 \\ \diagdown \quad \diagup \\ \text{C} = \text{C} \\ \diagup \quad \diagdown \\ \text{H} \quad \text{CH}_3 \end{array} \xrightarrow{\text{Br}_2} \begin{array}{c} \text{CH}_3 \\ | \\ \text{H} - \text{C} - \text{Br} \\ | \\ \text{Br} - \text{C} - \text{H} \\ | \\ \text{CH}_3 \end{array} + \begin{array}{c} \text{CH}_3 \\ | \\ \text{Br} - \text{C} - \text{H} \\ | \\ \text{H} - \text{C} - \text{Br} \\ | \\ \text{CH}_3 \end{array}$
cis-But-2-ene (±) -2, 3-Dibromobutane
- D) $\begin{array}{c} \text{H} \quad \text{CH}_3 \\ \diagdown \quad \diagup \\ \text{C} = \text{C} \\ \diagup \quad \diagdown \\ \text{H}_3\text{C} \quad \text{H} \end{array} \xrightarrow{\text{Br}_2} \begin{array}{c} \text{CH}_3 \\ | \\ \text{H} - \text{C} - \text{Br} \\ | \\ \text{H} - \text{C} - \text{Br} \\ | \\ \text{CH}_3 \end{array}$
trans-But-2-ene Meso -2, 3-dibromobutane

Sec: Jr. Super60-I

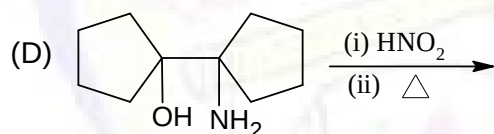
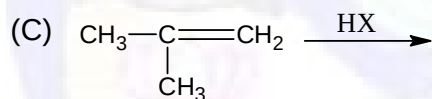
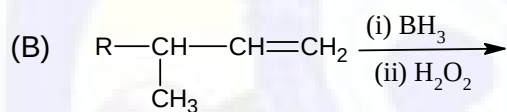
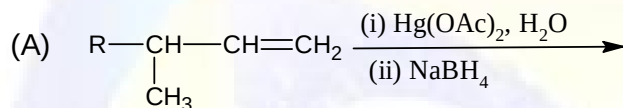
space for rough work

Page 13

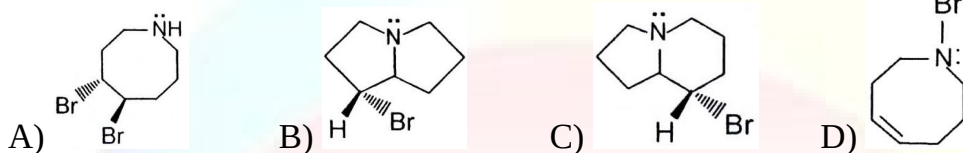
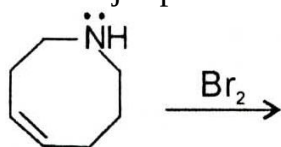
20. $\text{CH}_2 = \text{CH} - \text{CH}_3 \xrightarrow[\text{CCl}_4]{\text{NBS, } 500^\circ\text{C}}$ A. The chief product A is



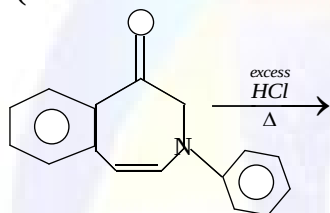
21. Which of the following reaction takes place by formation and rearrangement of carbocation



22. The major product of the following reaction would be

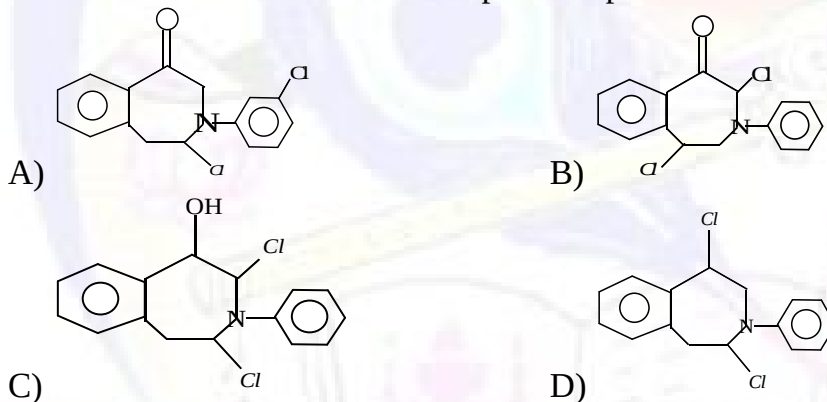


(Assume solvent to be nonpolar)

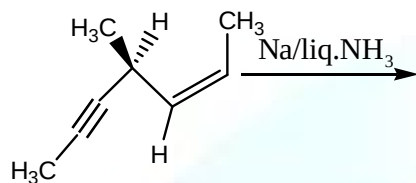


23.

The most possible product of the above reaction is



24. Hydrogenation of the adjoining compound in the presence of Na in liq. NH_3 Gives.



x is number of organic products obtained(including stereo isomers)

A) 2

B) 4

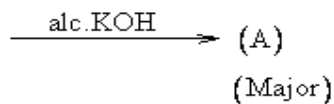
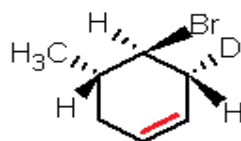
C) 5

D) 1

SECTION – II
(MULTIPLE CORRECT ANSWER TYPE)

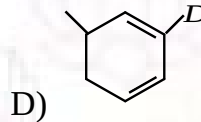
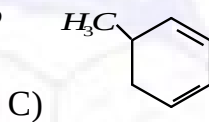
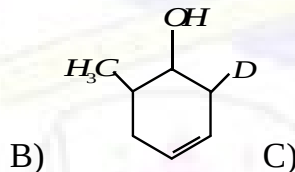
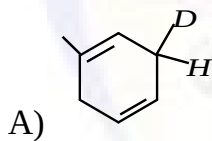
This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -2 in all other cases

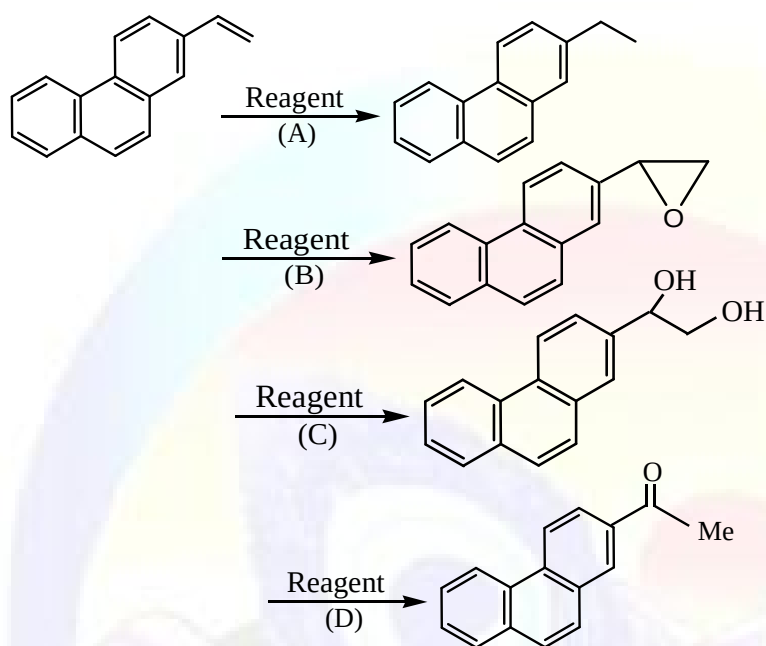


25.

Product (A) is not a



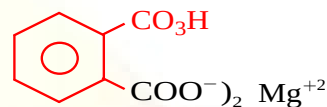
26.



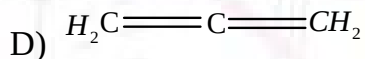
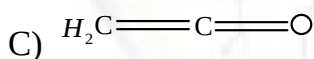
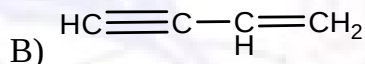
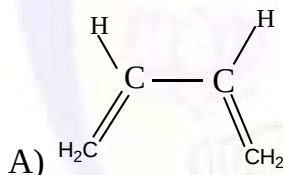
Which statements are correct for reagents A, B, C and D?

	(A)	(B)	(C)	(D)
a.	H_2/Pt	MMPP	Cold alk. KMnO_4	$\text{O}_2 + \text{PdCl}_2$ + CuCl_2 + H_2O
b.	$\text{H}_2/\text{Pd} + \text{BaSO}_4$ + quinoline	HCO_3H	Hot alk. KMnO_4	Dil. H_2SO_4 + Hg^{2+}
c.	SiH_2BH + CH_3COOH	PhCO_3H	$\text{OsO}_4/$ NaHSO_4	$\text{O}_2 + \text{PdCl}_2$ + CuCl_2 + H_2O
d.	$\text{BH}_3 + \text{THF}$ + CH_3COOH	MCPBA	OsO_4 H_2O_2	$\text{O}_2 + \text{PdCl}_2$ + CuCl_2 + H_2O

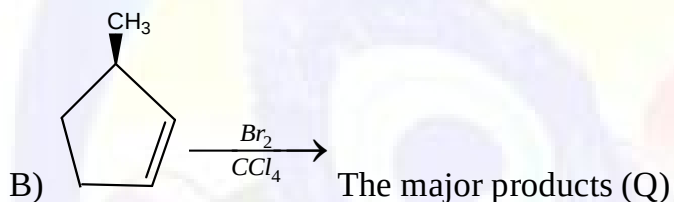
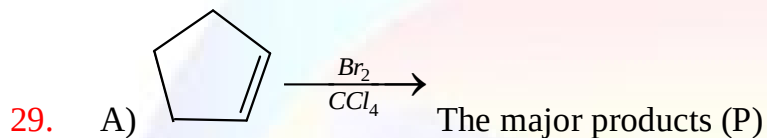
MMPP = Magnesium mono peroxy phthalate



27. Amongst the given options, the compounds in which all the atoms are in one plane in all the possible conformations are

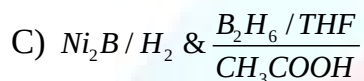
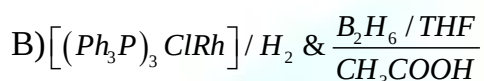
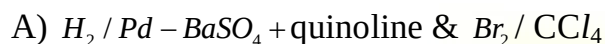


28. When cis-but-2-ene reacts with Baeyer's reagent
- A) an optically inactive compound is formed.
 - B) a brown inorganic precipitate is formed along with the organic product.
 - C) it undergoes oxidation.
 - D) an optically inactive mixture of two chiral compounds is formed.



- A) 'A' gives meso compound 'B' gives diastereoisomers
- B) Both 'A' & 'B' gives enantiomers
- C) 'A' gives enantiomers where as 'B' gives diastereoisomers
- D) 'P' & 'Q' are optically inactive and active respectively

30. Which of the following pair of reagents gives syn-addition with alkenes individually



31. Sleet the correct statements.

a) Addition of Br_2 on trans -2-butene gives erythro product

b) Addition of Cl_2 on cis-2-butene gives threo product

c) Addition of Br_2 on cis-2-butene gives racemic mixture

d) Addition of D_2 on cis-2-butene gives meso product

32. Which of the following will give R-X?

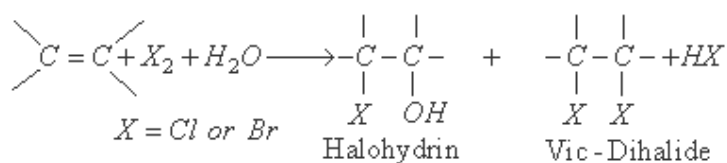


SECTION – III (PARAGRAPH TYPE)

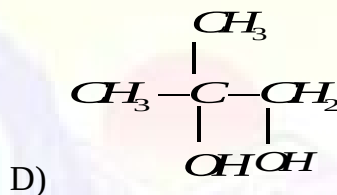
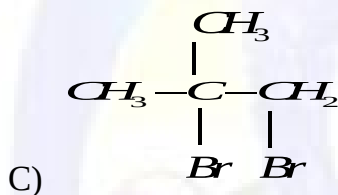
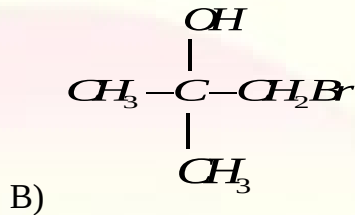
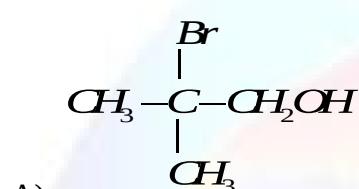
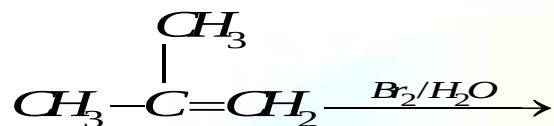
This section contains **2 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph For Questions 33 and 34

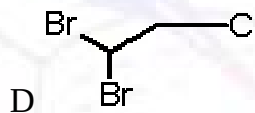
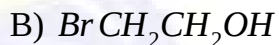
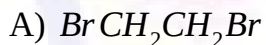
When the halogenation of an alkene is carried out in aqueous solution, rather than in a non-nucleophilic solvent, the major product is a halohydrin (also called a halo alcohol) instead of a vic-dihalide. In this case, molecules of water are in high concentration (since it is the solvent) and therefore react with the halonium ion intermediate as the predominant nucleophile. The result is formation of a halohydrin as the major product. If the halogen is bromine, it is called a bromohydrin and if chlorine, a chlorohydrin.



33. What is the major product in the following reaction?

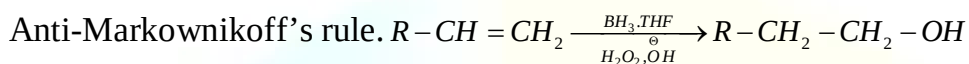


34. When ethene gas is passed into an aqueous solution containing bromine and sodium chloride, which of the following product is not formed in the reaction ?

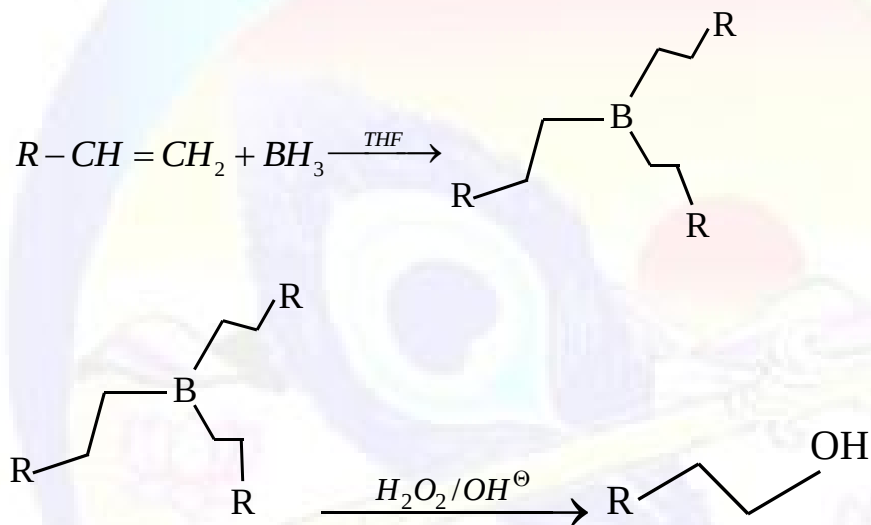


Paragraph For Questions 35 and 36

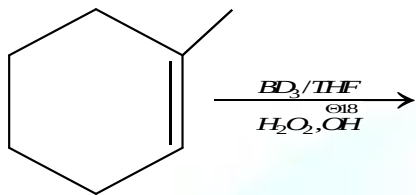
Hydroboration oxidation reaction is a process of addition of H_2O according to



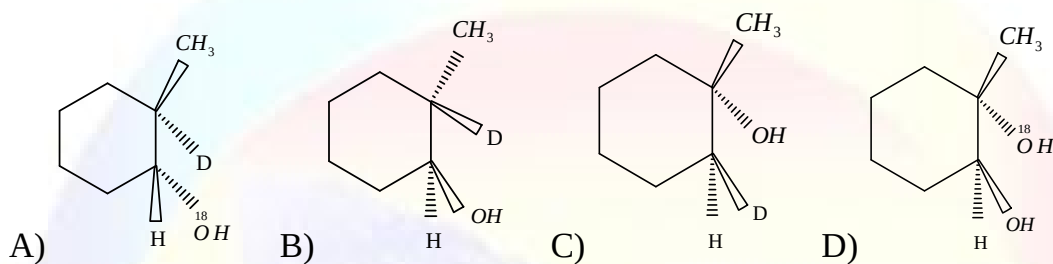
Reaction is regioselective. Regioselectivity of reaction is increased by using hindered boranes. THF (Tetrahydrofuran) is used to control reactivity of borane.



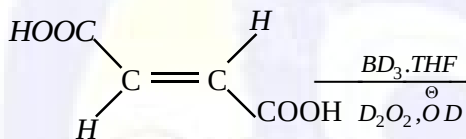
35.



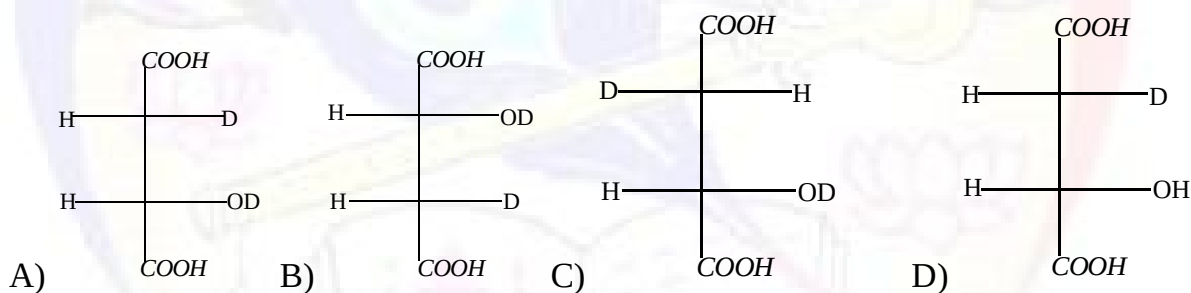
Major product:



36.



Product, possible fischer projection of product is :



MATHEMATICS**Max Marks: 62****SECTION – I
(SINGLE CORRECT ANSWER TYPE)**

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.

37. If the sides of a triangle are $\sin \alpha, \cos \alpha, \sqrt{1 + \sin \alpha \cdot \cos \alpha}$, $0 < \alpha < \frac{\pi}{2}$, the largest angle is
- A) 60° B) 120° C) 90° D) 150°
38. In a triangle ABC, if $\tan \frac{A}{2} \tan \frac{C}{2} = \frac{1}{3}$ and $ac = 4$, then the least value of b is :
- A) 1 B) 2 C) 4 D) 6
39. In triangle ABC, if $2b = a + c$ and $A - C = 90^\circ$, then $\sin B$ equals:
- A) $\frac{\sqrt{7}}{5}$ B) $\frac{\sqrt{5}}{8}$ C) $\frac{\sqrt{7}}{4}$ D) $\frac{\sqrt{5}}{3}$
40. Triangle ABC is right angled at A. The points P and Q are on the hypotenuse BC such that $BP = PQ = QC$. If $AP = 3$ and $AQ = 4$ then the length BC is equal to:
- A) $\sqrt{27}$ B) $\sqrt{36}$ C) $\sqrt{45}$ D) $\sqrt{54}$

41. In triangle ABC if $A:B:C=1:2:4$, then $(a^2-b^2)(b^2-c^2)(c^2-a^2)=\lambda a^2b^2c^2$, where $\lambda =$ (where notations have their usual meaning)
 A) 1 B) 2 C) 4 D) 9
42. Let AD be a median of the $\triangle ABC$. If AE and AF are median of the triangle ABD and ADC respectively and $AD=m_1$, $AE=m_2$, $AF=m_3$ then $\frac{a^2}{8} =$
 A) $m_2^2 + m_3^2 - 2m_1^2$ B) $m_1^2 + m_2^2 - 2m_3^2$
 C) $m_1^2 + m_3^2 - 2m_2^2$ D) $m_1^2 + m_2^2 - m_3^2$

SECTION – II**(MULTIPLE CORRECT ANSWER TYPE)**

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -2 in all other cases

43. If Δ represents the area of acute angled triangle ABC, then
 $\sqrt{a^2b^2-4\Delta^2} + \sqrt{b^2c^2-4\Delta^2} + \sqrt{c^2a^2-4\Delta^2} =$
 A) $a^2 + b^2 + c^2$ B) $\frac{a^2 + b^2 + c^2}{2}$
 C) $ab \cos C + bc \cos A + ca \cos B$ D) $ab \sin C + bc \sin A + ca \sin B$

44. If in a $\triangle ABC$; $\angle C = \frac{\pi}{8}$; $a = \sqrt{2}$; $b = \sqrt{2 + \sqrt{2}}$ then the measure of $\angle A$ can be:

- A) 45° B) 135° C) 30° D) 150°

45. Sides a, b, c of a triangle ABC are in AP and $\cos \theta_1 = \frac{a}{b+c}$, $\cos \theta_2 = \frac{b}{a+c}$, $\cos \theta_3 = \frac{c}{a+b}$

then $\tan^2 \frac{\theta_1}{2} + \tan^2 \frac{\theta_3}{2} =$

- A) $\frac{1}{2}$ B) $\frac{2}{3}$ C) $\frac{40}{60}$ D) $\frac{2}{6}$

46. If in a triangle ABC , $BC = 5$ $AC = 4$, $AB = 3$, D, E are points on BC such that $BD = DE = EC$, $\angle CAE = \theta$ then

- A) $AE^2 = \frac{73}{3}$ B) $AE^2 = \frac{73}{9}$ C) $\tan \theta = \frac{3}{8}$ D) $AD = \frac{2\sqrt{13}}{3}$

47. If the area of cyclic quadrilateral of $ABCD$ is $\frac{3\sqrt{3}}{4}$. The radius of its circumcircle is 1. If

$AB = 1$, $BD = \sqrt{3}$ then $BC \cdot DC =$

- A) 1 B) 2 C) 3 D) 4

48. In $\triangle ABC$ if $AB = x, BC = x+1, c = \frac{\pi}{3}$ then the possible integer value of 'x' is
- A) 6 B) 7 C) 8 D) 9
49. In a triangle ABC with given base BC, the vertex A moves such that $\cos B + \cos C = 4 \sin^2 \frac{A}{2}$. If a, b and c denote the lengths of the sides of triangle opposite of the angles A, B and C respectively then
- A) $c + a = 2b$
B) $b + c = 2a$
C) $\operatorname{cosec}\left(\frac{A}{2}\right) = 2 \cdot \sec\left(\frac{B-C}{2}\right)$
D) Locus of point A is a pair of straight lines
50. In a $\triangle ABC$, a, b and $\angle A$ are given, c_1, c_2 are possible values of 3rd side then
- A) $c_1 + c_2 = 2b \cos A$
B) $c_1 c_2 = b^2 - a^2$
C) $c_1^2 + c_2^2 - 2c_1 c_2 \cos 2A = 4a^2 \cos^2 A$
D) $c_1 + c_2 = 2a \cos B$

SECTION – III
(PARAGRAPH TYPE)

This section contains **2 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct. **Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.**

Paragraph For Questions 51 and 52

a, b, c are the length of sides BC, CA, AB respectively of $\triangle ABC$ satisfying

$$\log\left(1 + \frac{c}{a}\right) + \log a - \log b = \log 2. \text{ Also the quadratic equation } a(1 - x^2) + 2bx + c(1 + x^2) = 0$$

has two equal roots.

51. Measure of angle C is:

- A) 30° B) 45° C) 60° D) 90°

52. The value of $(\sin A + \sin B + \sin C)$ is equal to:

- A) $\frac{5}{2}$ B) $\frac{12}{5}$ C) $\frac{8}{3}$ D) 2

Paragraph For Questions 53 and 54

Let $a = 6, b = 3$ and $\cos(A - B) = \frac{4}{5}$

53. Angle C is equal to

- A) $\frac{3\pi}{4}$ B) $\frac{\pi}{4}$ C) $\frac{\pi}{2}$ D) $\frac{\pi}{6}$

54. Area (in Sq units) of the triangle ABC is equal to

- A) 9 B) 12 C) 11 D) 10



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A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr. Super60-I

Time: 07:30AM to 10:30AM

WAT-28

2016_P2

Date: 09-12-17

Max.Marks: 186

KEY SHEET

PHYSICS

1	D	2	C	3	B	4	B	5	B
6	A	7	BD	8	ABCD	9	AC	10	BC
11	BCD	12	ABCD	13	BCD	14	BC	15	C
16	A	17	D	18	B				

CHEMISTRY

19	A	20	B	21	D	22	B	23	C
24	D	25	ABD	26	ACD	27	ABCD	28	ABC
29	CD	30	BC	31	CD	32	ABC	33	B
34	D	35	B	36	C				

MATHS

37	B	38	B	39	C	40	C	41	A
42	A	43	BC	44	A	45	BC	46	BCD
47	AB	48	BCD	49	BC	50	ABC	51	D
52	B	53	C	54	A				

SOLUTIONS

PHYSICS

1. $E = \frac{1}{2}m(A\omega)^2$

$$E \propto (A\omega)^2$$

$$(A_1\omega_1)^2 = (A_2\omega_2)^2$$

$$4 \times 10 = 5\omega$$

$$\Rightarrow \omega = 8 \text{ units}$$

2. $y = A \sin \omega t$

$$\omega t_1 = \sin^{-1} \frac{4}{5}$$

$$\omega t_2 = \sin^{-1} \frac{3}{5}$$

$$\Rightarrow t_2 - t_1 = \frac{8}{45} = 0.18 \text{ s}$$

3. $T = 2\pi \sqrt{\frac{m}{k}}$

$$f = \frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{20}} = 2$$

$$1 = \frac{1}{2\pi} \sqrt{\frac{k}{20+m}}$$

$$\sqrt{\frac{k}{20}} = 2(2\pi)$$

$$\sqrt{k} = 2\sqrt{20} \cdot 2\pi$$

$$k = 4(20) \cdot 4\pi^2 = 320\pi^2$$

$$\therefore 1 = \frac{1}{2\pi} \sqrt{\frac{320\pi^2}{20+m}}$$

$$4\pi^2(20+m) = 320\pi^2$$

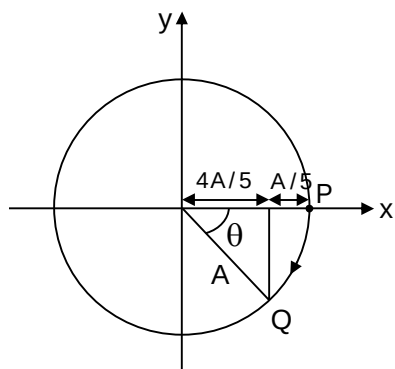
$$20 + m = 80$$

$$M = 60 \text{ kg}$$

4. **Method: I**

Particle is starting from rest, i.e. from one of its extreme position.

As particle moves a distance $\frac{A}{5}$, we can represent it on a circle as shown.



$$\cos \theta = \frac{4A/5}{A} = \frac{4}{5} \quad \theta = \cos^{-1}\left(\frac{4}{5}\right)$$

$$\omega t = \cos^{-1}\left(\frac{4}{5}\right) \quad t = \frac{1}{\omega} \cos^{-1}\left(\frac{4}{5}\right)$$

$$= \frac{T}{2\pi} \cos^{-1}\left(\frac{4}{5}\right)$$

Method: II

As starts from rest i.e. from extreme position $x = A \sin (\omega t + \phi)$

$$\text{At } t = 0; x = A \Rightarrow \phi = \frac{\pi}{2} \quad \therefore A - \frac{A}{5} = A \cos \omega t$$

$$\frac{4}{5} = \cos \omega t \Rightarrow \omega t = \cos^{-1} \frac{4}{5} \quad t = \frac{T}{2\pi} \cos^{-1}\left(\frac{4}{5}\right)$$

5. : Here, $V(x) = A(1 - \cos px)$

$$\text{Force, } F = \frac{dv}{dx} = -\frac{d}{dx}(A - A \cos px)$$

$$\text{For small } x, F = -Ap^2x$$

$$\text{Acceleration, } a = \frac{F}{m} = \frac{Ap^2x}{m} \quad \dots\dots\dots (i)$$

$$\text{The standard equation of SHM is, } a = -\omega^2x$$

$$\text{Comparing equations (i) and (ii), we get } \dots\dots\dots (ii)$$

$$\omega^2 = \frac{Ap^2}{m} \text{ or } \omega = \sqrt{\frac{Ap^2}{m}}$$

$$\text{Period of oscillation, } T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{Ap^2}}$$

6. $x(x-4) = 1 - 5\cos\omega t$

or $x^2 - 4x + 4 = 5 - 5\cos\omega t$

or $(x-2)^2 = 5(1 - \cos\omega t) = 10\sin^2\frac{\omega t}{2}$

or $x-2 = \sqrt{10}\sin\frac{\omega t}{2}$

or $y = x-2 = \sqrt{12}\sin\frac{\omega t}{2}$

$$\frac{dy}{dt} = \sqrt{10}\cos\frac{\omega t}{2} \times \left(\frac{\omega}{2}\right)$$

$$\frac{d^2y}{dt^2} = -\sqrt{10}\left(\frac{\omega^2}{4}\right)\sin\left(\frac{\omega t}{2}\right)$$

$$\frac{d^2(x-2)}{dt^2} = -\frac{\omega^2}{4}(x-2)$$

i.e., motion is simple harmonic.

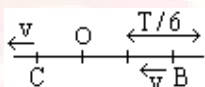
7. $x = 3\sin 100t + 8\cos^2 50t$

$$= 3\sin 100t + \frac{8[1 + \cos 100t]}{2}$$

$$x = 4 + 3\sin 100t + 4\cos 100t$$

$$(x-4) = 5\sin(100t + \phi) \left\{ \tan \phi = \frac{4}{3} \right\}$$

Amplitude = 5 units Maximum displacement = 9 units.



8.

At $t = \frac{T}{4}$ particle will be at point B and at $t = \frac{T}{2}$ it will be at point A.

9. Acceleration of platform is maximum at extreme ($\therefore g = a\omega^2$)

10. At time t_1 , velocity of the particle is negative i.e. going towards $-X_m$. From the graph, at time t_1 , its speed is decreasing. Therefore particle lies in between $-X_m$ and 0. At time t_2 , velocity is positive and its magnitude is less than maximum i.e. it has yet not crossed O

It lies in between $-X_m$ and 0

Phase of particle at time t_1 is $(180 + \theta_1)$

Phase of particle at time t_2 is $(270 + \theta_2)$

Phase difference is $90 + (\theta_2 - \theta_1)$

$\theta_2 - \theta_1$ can be negative making $\Delta\phi < 90^\circ$ but can not be more than 90°

$$11. \quad u = 5x^2 - 20x \Rightarrow F = \frac{-du}{dx} = -10(x - 2)$$

M.P. at $x = 2m$

12. Let the deformation in each spring be x_1, x_2 and x_3 respectively as shown

Let the block be displaced by x .

Then $x = x_1 + 2x_2 + x_3, \dots (i)$

$$F = 2Kx_3; 2F = Kx_2; 2F = 2Kx_1; \dots (ii)$$

Also $F = K_{eq}x, \dots (iii)$

$\therefore$ From equation (i), (ii) and (iii)

$$\Rightarrow K_{eq} = \frac{2K}{11}$$

$$\therefore T = 2\pi \sqrt{\frac{11m}{2K}}$$

$$F = K_{eq}x = Kx_1$$

$$\frac{2K}{11}x = Kx_1$$

$$x_1 = \frac{2}{11}x$$

$$F = K_{eq}x = \frac{Kx_2}{2}$$

$$x_2 = \frac{4}{11}x$$

$$F = K_{eq}x = 2Kx_3$$

$$\frac{2K}{11}x = 2Kx_3$$

$$x_3 = \frac{1}{11}x$$

$$13. \quad mg = KX_o \quad X_o = \frac{10}{500} = 2cm$$

$$\omega = \sqrt{\frac{500}{1}} = 10\sqrt{5} \frac{rad}{sec}$$

$$\therefore \text{Maximum velocity} = A\omega = 3 \times 10^{-2} \times 10\sqrt{5}$$

$$= 30\sqrt{5} \text{ cm/sec.} \Rightarrow \text{Max}^m \text{Acc}^n = A\omega^2 = 15 \text{ m/s}^2$$

14. Phase by $\omega = \frac{\omega \times 2\pi}{3\omega} = \frac{2\pi}{3}$

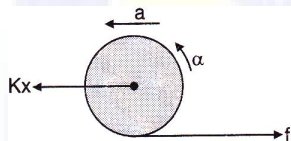
Phase by $2\omega = \phi + 2\omega \times \frac{2\pi}{3\omega}$, $\phi + \frac{4\pi}{3} - \frac{2\pi}{3} = 0, 2\pi, 4\pi$

$$\phi = \frac{2\pi}{3}, \frac{4\pi}{3}, \dots$$

15. The forces on the disc at displacement x of the spring are shown in figure

For pure rolling to take place, we have

$$\Rightarrow \frac{kx - f}{m} = \frac{R(fR)}{\frac{1}{2}mR^2} = \frac{2f}{m}$$



$$\Rightarrow f = \left(\frac{k}{3}\right)x$$

The cylinder is starting from $x = A$, so the $x-t$ equation would be

$$x = A \cos(\omega t)$$

$$\Rightarrow f = \frac{kA}{3} \cos(\omega t)$$

$\Rightarrow f-t$ curve will be a cosine curve.

16. Energy of oscillation $E = \frac{1}{2}kA^2 = \frac{1}{2} \times 10 \times 4 = 20 \text{ J}$

At mean position this is totally kinetic energy (translational + rotational). In case of pure rolling ratio of rotational and translational kinetic energy is $\frac{1}{2}$ in case of disc

$$\text{Hence, } K_r = \frac{1}{2}mV^2 = \frac{2}{3} \times 20 = \frac{40}{3}$$

$$\Rightarrow V^2 = \frac{80}{3 \times 2} = \frac{40}{3} \text{ m}^2 \text{ s}^{-2}$$

$$\Rightarrow V = \sqrt{\frac{40}{3}} \text{ ms}^{-1}$$

17. $F = -4x + 8$

Let us write $X = (x + 2)$

This is the equation of SHM.

Hence, mean position is $x = 2 \text{ m}$

18. $F = -4X$

$$\Rightarrow k = 4 \text{ Nm}^{-1}$$

$$A = \sqrt{\frac{2E}{k}} = \sqrt{\frac{2 \times 18}{4}} = 3 \text{ m}$$

Mean position of the particle is $x = 2 \text{ m}$

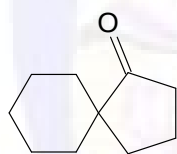
Hence, the extreme points are 5 m and -1 m

CHEMISTRY

19. Conceptual

20. It is free radical substitution at allylic position

21. d) Its pinacol-pinacolone type reaction and product is formed after formation of carbocation and its rearrangement by ring expansion.



22. Conceptual

23. Electrophilic addition reaction

24. Conceptual

25. Conceptual

26. Conceptual

27. Conceptual

28. Conceptual

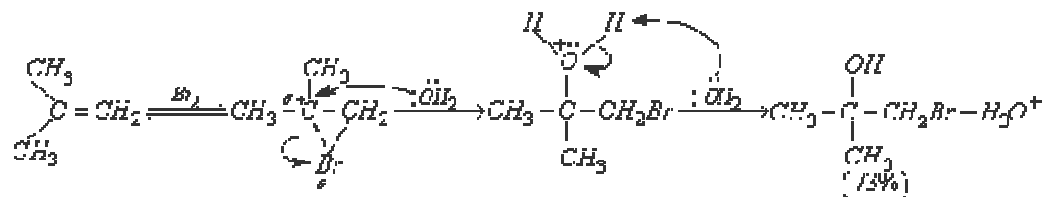
29. Conceptual

30. Conceptual

31. Conceptual

32. Conceptual

33.



34. Conceptual

35. Conceptual

36. Conceptual

MATHS37. Greatest side $= \sqrt{1 + \sin \alpha \cdot \cos \alpha}$, apply cosine rule.38. $2b = a + c$, use $AM \geq GM$ 39. Use $\tan\left(\frac{A-C}{2}\right) = \frac{a-c}{a+c} \cot \frac{B}{2}$ 40. Let $A(0,0) B(0,\beta) C(\alpha,0)$

$$Q\left(\frac{2\alpha}{3}, \frac{\beta}{3}\right), P\left(\frac{\alpha}{3}, \frac{2\beta}{3}\right)$$

$$AQ^2 = 16, AP^2 = 9$$

$$4\alpha^2 + \beta^2 = 144, 4\alpha^2 + 16\beta^2 = 324$$

$$15\beta^2 = 80$$

$$\alpha^2 + \beta^2 = 45$$

41. Let $A = K, B = 2K, C = 3K$

Use sine rule

42. Conceptual

43. Conceptual

44. $C^2 = 2 - \sqrt{2}$ then use sine rule45. $2b = a + c$

$$\tan^2 \frac{\theta_1}{2} + \tan^2 \frac{\theta_3}{2} = \frac{1 - \cos \theta_1}{1 + \cos \theta_1} + \frac{1 - \cos \theta_2}{1 + \cos \theta_2}$$

$$46. AE^2 = 16 + \frac{25}{9} - 2 \times 4 \times \frac{5}{3} \times \frac{4}{5}$$

47. In $\triangle ABD$

$$\frac{AB}{\sin \angle ADB} = \frac{BD}{\sin A} = 2 \Rightarrow \angle ADB = 30^\circ$$

$$\sin A = \frac{BD}{2} \Rightarrow \angle A = 60^\circ \text{ Or } 120^\circ$$

48. Using cosine rule

$$x^2 = (x+1)^2 + b^2 - 2(x+1)b \cos \frac{\pi}{3}$$

$$b^2 - (x+1)b + 2x+1 = 0$$

Since b is real we have

$$(x+1)^2 - 4(2x+1) \geq 0$$

$$x \geq 3 + \sqrt{12}$$

Least integral value is 7

$$49. \quad \cos \frac{B-C}{2} = 2 \sin \frac{A}{2}$$

$$50. \quad a^2 = b^2 + c^2 - 2bc \cos A$$

$$\Rightarrow c^2 - 2b \cos A c + b^2 - a^2 = 0$$

$$c_1 + c_2 = 2b \cos A, c_1 c_2 = b^2 - a^2$$

$$51. \quad a + c = 2b$$

$$52.$$

$$53. \quad \tan \left(\frac{A-B}{2} \right) = \frac{1}{3}$$

$$\frac{a-b}{a+b} \cot \frac{C}{2} = \frac{1}{3} \Rightarrow \cot \frac{C}{2} = 1$$

Success



JEE ADVANCED WEEKEND TEST



Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr. Super60-I

Time: 07:30 AM to 10:30 AM

Date: 16-12-17

Max. Marks: 180

Name of the Student: _____

H.T. NO:

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2013_PAPER-I

IMPORTANT INSTRUCTIONS:

- 1) This booklet is your Question Paper.
- 2) Use the Optical Response Sheet (ORS) provided separately for answering the questions
- 3) Blank spaces are provided within this booklet for rough work.
- 4) Write your name, roll number and sign in the space provided on the back cover of this booklet.
- 5) You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET:

- 6) Darken the appropriate bubbles on the ORS by applying sufficient pressure. This will leave an impression at the corresponding place on the Candidate's sheet.
- 7) The ORS will be collected by the invigilator at the end of the examination.
- 8) Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
- 9) Write your name, roll number and code of the examination center, and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else** on the ORS. Darken the appropriate bubble under each digit of your roll number.

DARKENING THE BUBBLES ON THE ORS

- 10) Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
- 11) Darken the bubble **COMPLETELY**.
- 12) The correct way of darkening a bubble is as:
- 13) The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- 14) Darken the bubbles **ONLY** IF you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.

16-12-17_Jr.Super60-I_Jee-Adv_WAT-29_WEEKEND SYLLABUS

MATHS: SYLLABUS COVERED FROM_11-12-17 TO 15-12-17

PHYSICS: SYLLABUS COVERD FROM 11-12-17 TO 15-12-2017

CHEMISTRY: Alkynes, Preparations reactions of benzene and alkyl benzenes, Electrophilic aromatic substitution, mechanism of halogenations, nitration, sulphonation, nitrosation, alkylation and acylation

JEE-ADVANCE-2013-P1-Model**Time: 03:00hour's****IMPORTANT INSTRUCTIONS****Max Marks: 180****PHYSICS:**

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 10)	Questions with Single Correct Choice	2	0	10	20
Sec – II(Q.N : 11 – 15)	Questions with Multiple Correct Choice	4	-1	5	20
Sec – III(Q.N : 16 – 20)	Questions with Integer Answer Type	4	-1	5	20
Total				20	60

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 21 – 30)	Questions with Single Correct Choice	2	0	10	20
Sec – II(Q.N : 31 – 35)	Questions with Multiple Correct Choice	4	-1	5	20
Sec – III(Q.N : 36 – 40)	Questions with Integer Answer Type	4	-1	5	20
Total				20	60

MATHEMATICS:

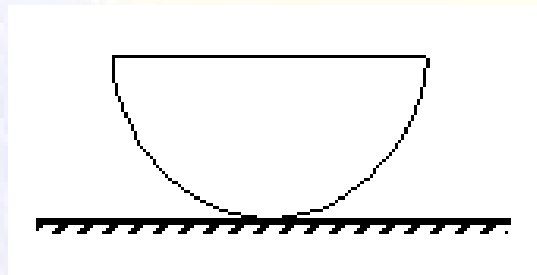
Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 41 – 50)	Questions with Single Correct Choice	2	0	10	20
Sec – II(Q.N : 51 – 55)	Questions with Multiple Correct Choice	4	-1	5	20
Sec – III(Q.N : 56 – 60)	Questions with Integer Answer Type	4	-1	5	20
Total				20	60

Sec: Jr. Super60-I**space for rough work****Page 2**

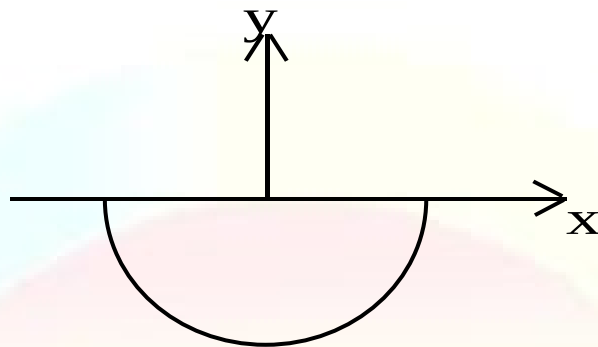
PHYSICS:**Max. Marks: 60****SECTION - I****SINGLE CORRECT ANSWER TYPE**

This section contains 10 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. A hemisphere is lying on a surface as shown. For small oscillations of this without slip, the time period is (centre of mass lies at a distance of $\frac{3R}{8}$ from centre of curvature where R is radius)



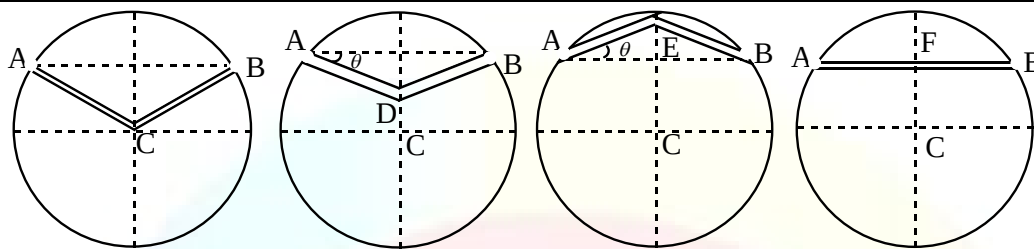
- A) $2\pi\sqrt{\frac{26R}{15g}}$ B) $2\pi\sqrt{\frac{13R}{8g}}$ C) $2\pi\sqrt{\frac{26R}{25g}}$ D) 2π
2. A semicircular uniform wire of radius $R = \pi$ can rotate freely about x-axis. The centre of curvature is at origin as shown. The acceleration due to gravity is given by $\vec{g} = -x^2 \hat{j}$ units. For small oscillations of the wire its time period is given by



- A) $\sqrt{3}\pi$ B) 2π C) $\sqrt{6}\pi$ D) π

3. In the figure are shown four tunnels ACB, ADB, AEB and AFB, which are supposed to be bored through the earth between two places A and B. The tunnel AFB runs straight from A to B and the point C is the center of the earth.

A particle released from place A reaches the place B in time intervals T_C , T_D , T_E and T_F respectively. Assume that the particle does not collide with the walls of the tunnel anywhere and passes the corner smoothly. Which of the following relations do these time intervals bear?



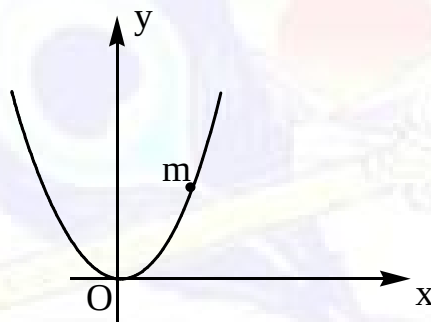
A) $T_C = T_D = T_E = T_F$

B) $T_C = T_F > T_D = T_E$

C) $T_C = T_F < T_D = T_E$

D) $T_E > T_C = T_F > T_D$

4. If a smooth bead of mass m oscillates on the parabolic surface $Y = kx^2$, which is in a vertical plane, its time period of small oscillations is :



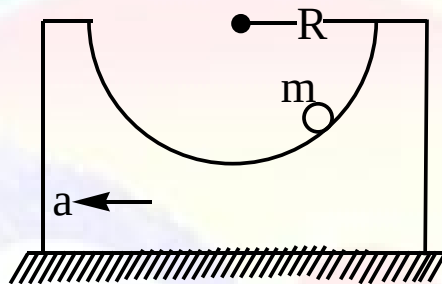
A) $\pi \sqrt{\frac{2}{kg}}$

B) $2\pi \sqrt{\frac{k}{g}}$

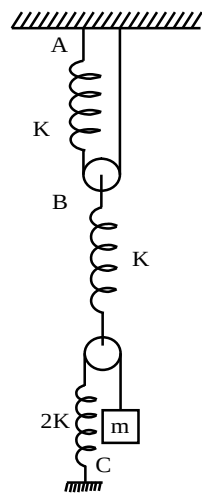
C) $2\pi \sqrt{\frac{2k}{m}}$

D) $2\pi \sqrt{\frac{k}{m}}$

5. A smooth wedge is having a cylindrical cut of radius R . A bead of mass m is placed inside the wedge. If the wedge moves with an acceleration ' a '. Find the frequency of small oscillation of the bead about the position of stable equilibrium of the bead relative to the wedge



- A) $\frac{1}{2\pi} \sqrt{\frac{g^2 + a^2}{R}}$ B) $\frac{1}{2\pi} \sqrt{\frac{g}{R}}$ C) $\frac{1}{2\pi} \sqrt{\frac{g+a}{R}}$ D) $\frac{1}{2\pi} \sqrt{\frac{g-a}{R}}$
6. In the given figure, the block is attached with a system of three ideal springs A,B,C. The block is displaced by a small distance x from its equilibrium position vertically downwards and released. T represents the time period of small vertical oscillations of the block. Then (Pulleys are ideal)



A) $T = 2\pi \sqrt{\frac{11m}{2K}}$

B) The deformation of the spring A is $(2/7)$ times the displacement of the block

C) The deformation of the spring C is $(1/7)$ times the displacement of the block

D) The deformation of the spring B is $(4/7)$ times the displacement of the block

7. The potential energy of a particle of mass 1 kg in motion along the x-axis is given by $U = (10 - 10 \cos 6x)$ J. The period of small oscillations is $(\pi^2 \approx 10)$

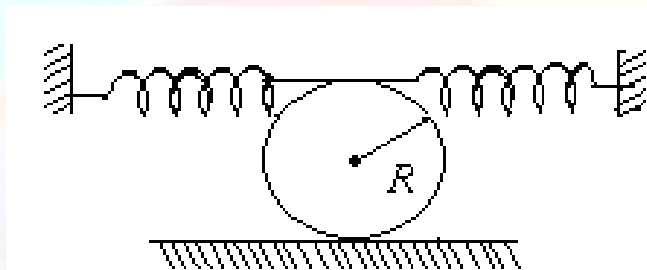
A) $\frac{1}{3}$ s

B) $\frac{3}{2}$ s

C) $\frac{2}{3}$ s

D) $\frac{1}{4}$ s

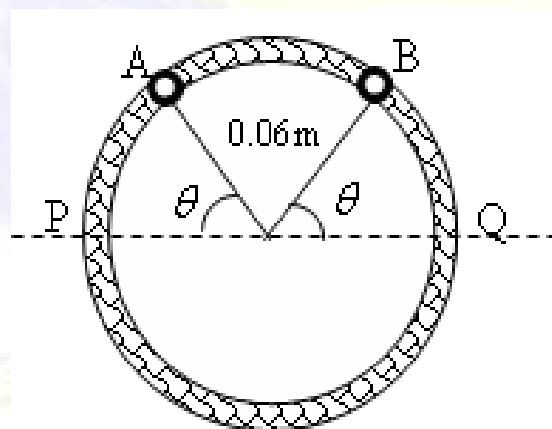
8. a solid uniform cylinder of mass M performs small oscillations in horizontal plane if slightly displaced from its mean position as shown in figure. Initially springs have its natural lengths and cylinder does not slip on ground during oscillations. If force constant of each spring be K , find time-period of oscillations



- A) $\frac{\pi}{2} \sqrt{\frac{3m}{K}}$ B) $\pi \sqrt{\frac{m}{3K}}$ C) $2\pi \sqrt{\frac{2m}{3K}}$ D) $2\pi \sqrt{\frac{3m}{K}}$
9. In a uni-dimensional potential field the potential energy of a particle depends on the co-ordinate x as $U(x) = U_0(2 - \cos ax)$, U_0 and a are constants. Find the time period of small oscillations that the particle performs about the equilibrium position.

- A) $2\pi \sqrt{\frac{m}{U_0 a^2}}$ B) $\frac{\pi}{2a} \sqrt{\frac{m}{U_0}}$ C) $2\pi \sqrt{\frac{2m}{U_0 a^2}}$ D) $\pi \sqrt{\frac{m}{U_0 a^2}}$

10. Two identical balls A and B, each of mass 0.1kg , are attached to two identical massless springs. The spring mass system is constrained to move inside a rigid smooth pipe bent in the form of circle as shown in figure. The pipe is fixed in a horizontal plane. The centres of the balls can move in a circle of radius 0.06m . Each spring has a natural length of $0.06\pi\text{ m}$ and spring constant 0.1N/m . Initially, both the balls are displaced by an angle $\theta = \frac{\pi}{6}$ rad with respect to the diameter PQ of the circle (as shown in fig.) and released from rest. Then the frequency of oscillation of ball B is



A) $\frac{2}{\pi}$

B) $\frac{1}{\pi}$

C) $\frac{3}{\pi}$

D) $\frac{4}{\pi}$

SECTION - II

MULTIPLE CORRECT ANSWER(S) TYPE

This section contains 5 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which ONE or MORE are correct.

11. A simple pendulum has a bob of mass m and swings with an angular amplitude ϕ .

The tension in the thread is T . At a certain time, the string makes an angle θ with the vertical ($\theta \leq \phi$)

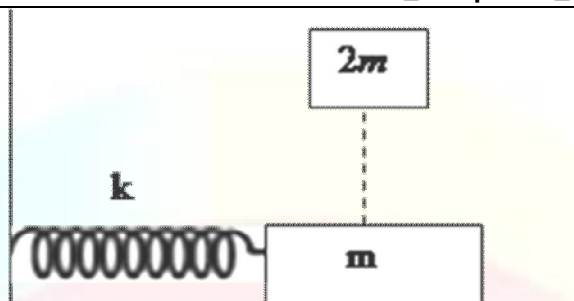
A) $T = mg \cos \theta$, for all values of θ

B) $T = mg \cos \theta$, only for $\theta = \phi$

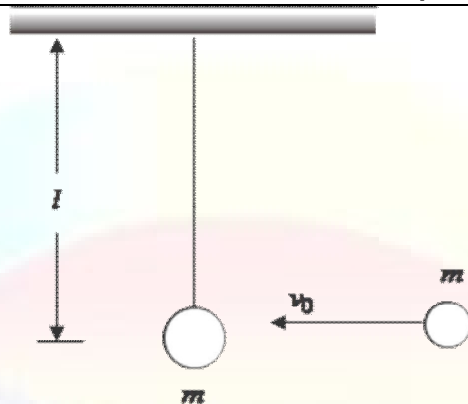
C) $T = mg$, for $\theta = \cos^{-1} \left[\frac{1}{3}(2 \cos \phi + 1) \right]$

D) T will be larger for smaller values of θ

12. An object of mass m is performing simple harmonic motion on a smooth horizontal surface as shown in the figure. Just as the oscillating object reaches its extreme position, another object of mass $2m$ is dropped on to oscillating object, which sticks to it. For this situation mark out the correct statement (s)

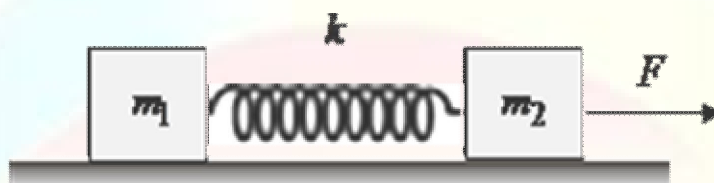


- A) Amplitude of oscillation remains unchanged
- B) Time period of oscillations changed
- C) The total mechanical energy of the system does not change
- D) The maximum speed of the oscillating object changes
13. A simple pendulum consists of a bob of mass m and a light string of length l as shown in the given figure. Another identical ball moving with the small velocity v_0 collides with the pendulum bob and sticks to it at $t = 0$. For this new pendulum of mass $2m$, mark out the correct statement (s) (θ is angle made with vertical)

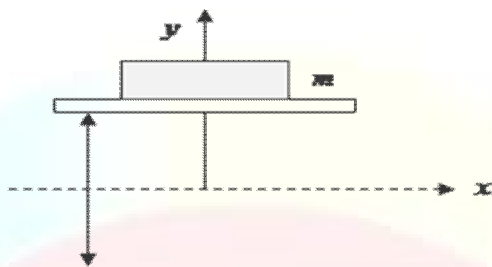


- A) Time period of the pendulum is $2\pi\sqrt{\frac{l}{g}}$.
- B) The equation of motion for this pendulum is $\theta = \frac{v_0}{2\sqrt{gl}} \sin\left[\sqrt{\frac{g}{l}}t\right]$.
- C) The equation of motion for this pendulum is $\theta = \frac{v_0}{2\sqrt{gl}} \cos\left[\sqrt{\frac{g}{l}}t\right]$.
- D) Time period of the pendulum is $2\pi\sqrt{\frac{2l}{g}}$.

14. Two blocks connected by a spring rest on a smooth horizontal plane as shown in the given figure. A constant force F starts acting on block m_2 as shown in the figure. Which of the following statements are not correct ?



- A) Length of the spring increases continuously if $m_1 > m_2$.
 B) Blocks start performing SHM about centre of mass of the system, which moves rectilinearly with constant acceleration.
 C) Blocks start performing oscillations about centre of mass of the system with increasing amplitude
 D) Acceleration of ' m_2 ' is maximum at initial moment of time only
15. A horizontal platform with a mass ' m ' placed on it is executing SHM along y-axis. If the amplitude of oscillation is 2.5 cm, the minimum period of the motion for the mass not to be detached from the platform is $(g = 10 \text{ m/s}^2)$



A) $\frac{10}{\pi} s$

B) $\frac{\pi}{10} s$

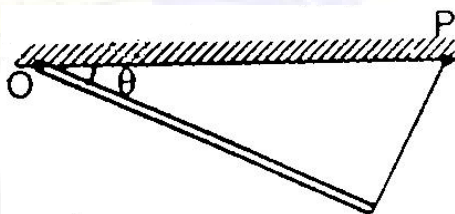
C) $\frac{\pi}{\sqrt{10}} s$

D) $\frac{2\pi}{\sqrt{10}} s$

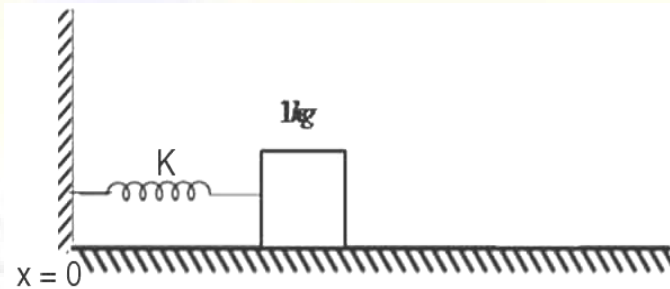
SECTION - III
INTEGER ANSWER TYPE

This section contains 5 questions. The answer to each question is single digit integer, ranging from 0 to 9 (both inclusive).

16. A uniform rod of mass m and length L is hinged at one of its end with the ceiling and other end of the rod is attached with a thread which is attached with horizontal ceiling at point P . If one end of the rod is slightly displaced horizontally and perpendicular to the rod and released. If the time period of small oscillation is $2\pi\sqrt{\frac{2\ell\sin\theta}{xg}}$. Find x .



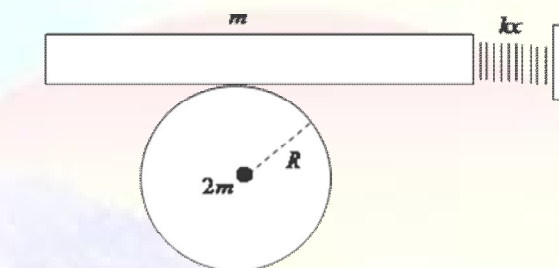
17. One end of an ideal spring is fixed to a wall at origin O and axis of spring is parallel to the x -axis. A block of mass $m = 1$ kg is attached to the free end of the spring and it is performing SHM. Equation of position of the block in coordinate system shown in figure is $x = 10 + 3\sin(10t)$, where 't' is in seconds and x in centimetres. The force constant of spring is $4n^2$. Where $n = \dots$



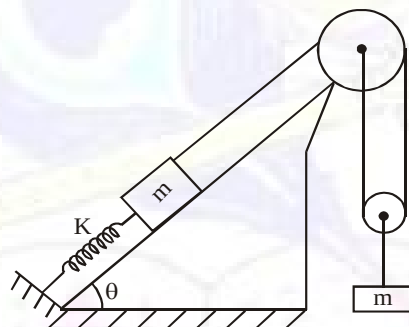
18. A uniform plank of mass m , free to move in the horizontal direction only, is placed at the top of a solid cylinder of mass $2m$ and radius R . The plank is attached to a fixed wall by means of a light spring of spring constant k . There is no slipping between the

cylinder and the plank, and between the cylinder and the ground. Find the time period

of small oscillations of the system is $\pi\sqrt{\frac{xm}{k}}$ where $x = \dots$



19. In the given diagram the threads, spring and pulleys are mass less, no friction anywhere



Find the angular frequency of the system (In rad/sec)

[given $K = 20 \text{ Nm}^{-1}$ and $m = 4 \text{ kg}$]

20. Consider a hypothetical field in which force on any particle at position 'x' varies

according to the relation $\vec{F} = F_0 \left(1 - \frac{x}{4}\right) \hat{i}$ where F is in newton, x is in meter and F_0 is a

positive constant. A particle of mass m is released from rest from $x = 12\text{m}$ at $t = 0$ in

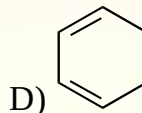
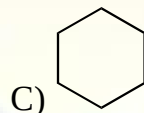
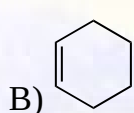
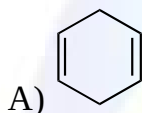
this field. If $2\pi\sqrt{\frac{m}{F_0}} = 6$ second and then find value of 't' in second when the particle

reaches the origin first time.

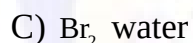
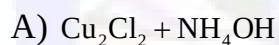
CHEMISTRY:**Max. Marks: 60****SECTION - I****SINGLE CORRECT ANSWER TYPE**

This section contains 10 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

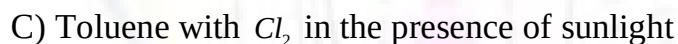
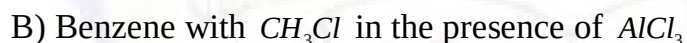
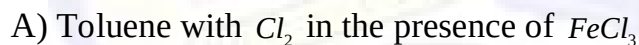
21.  + H₂ $\xrightarrow[\text{high pressure}]{\text{Ni, high temp.}}$ (A). Which of the following can be isolated as the product of this reaction.



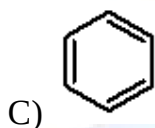
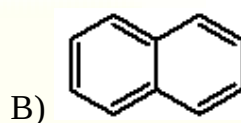
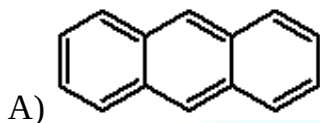
22. Which of the following reaction can be used to differentiate $\text{CH}_2 = \text{CH} - \text{CH} = \text{CH}_2$ & $\text{CH} \equiv \text{C} - \text{CH}_2 - \text{CH}_3$?



23. Benzyl chloride can be obtained as one of the products prepared by reacting

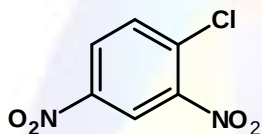


24. Which of the following will have maximum resonance energy per ring?



D) All have equal Resonance energy per ring

25.



Which IUPAC name is correct for given compound?

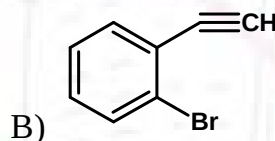
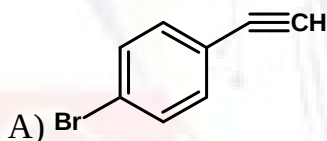
A) 1-chloro-2,4-dinitrobenzene

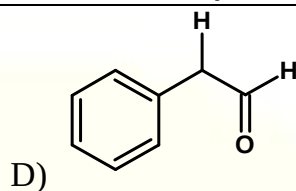
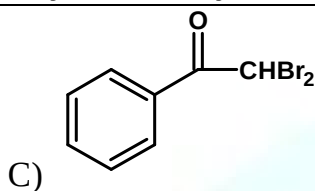
B) 4-chloro-1,3-dinitrobenzene

C) 2,4-dinitrochlorobenzene

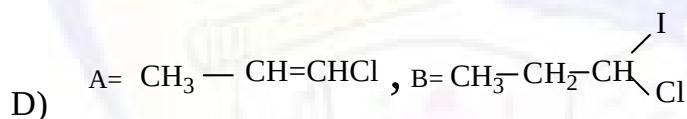
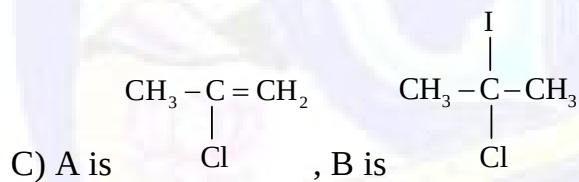
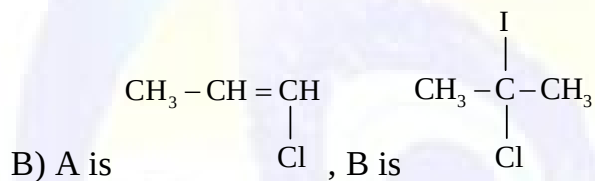
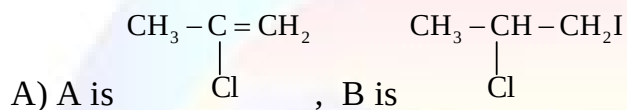
D) 2-chloro-1,5-dinitrobenzene

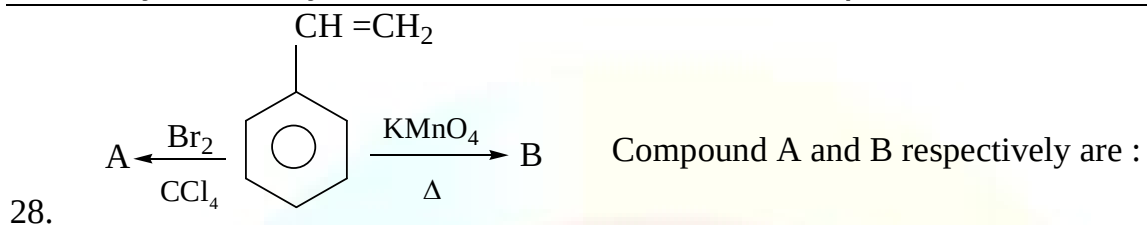
26. $Ph-C \equiv CH \xrightarrow[2eq]{Br_2/H_2O}$ Product is?



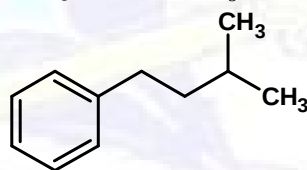
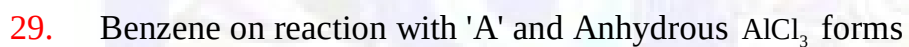


27. $CH_3 - C \equiv CH \xrightarrow[(\text{leq})]{HCl} A \xrightarrow[(\text{major})]{HI} B$; then find A and B in the reaction sequence?

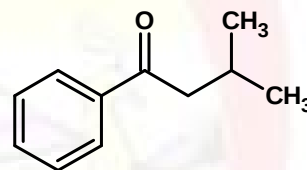




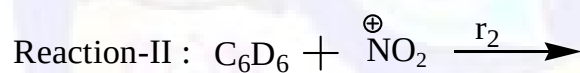
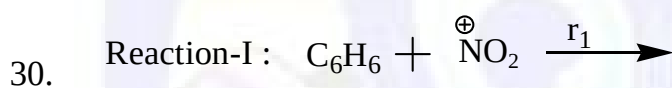
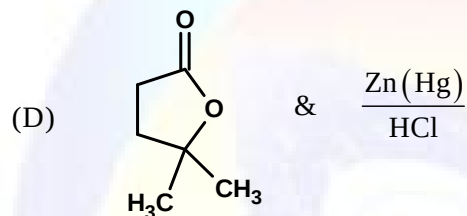
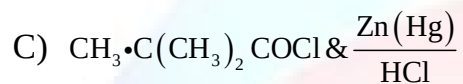
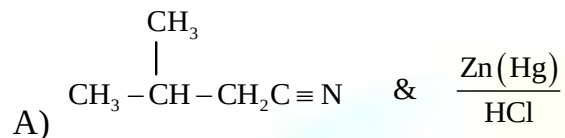
- A) *o*-bromo styrene, benzoic acid
 B) *p*-bromo styrene, benzaldehyde
 C) *m*-bromo styrene, benzaldehyde
 D) styrene dibromide, benzoic acid



which on reaction with 'B' forms



'A' and 'B' are:



A) rate $r_2 > r_1$

B) rate $r_1 > r_2$

C) rate $r_1 \approx r_2$

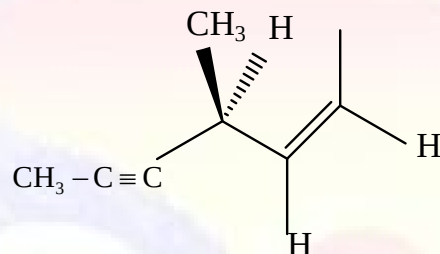
D) r_1 & r_2 are not comparable

SECTION - II

MULTIPLE CORRECT ANSWER (S) TYPE

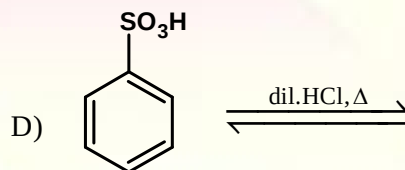
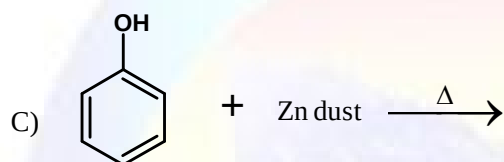
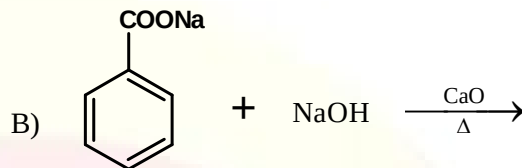
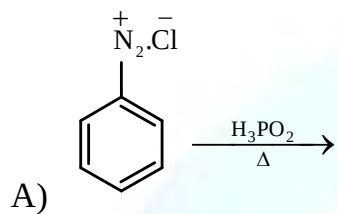
This section contains 5 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which ONE or MORE are correct.

31. Hydrogenation of the following compound in the presence of poisoned palladium catalyst does NOT give:

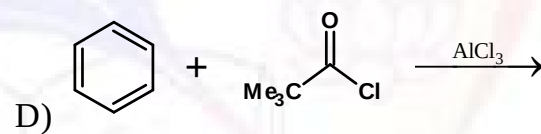
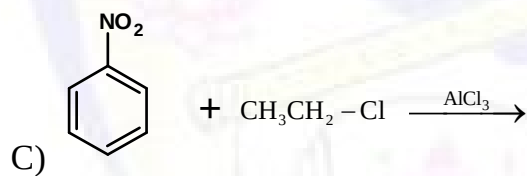
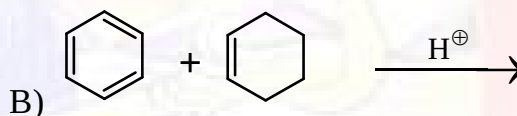
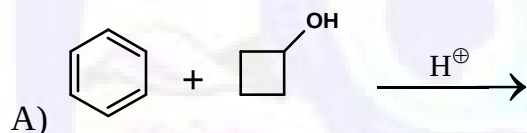


- A) an optically active compound
- B) an optically inactive compound
- C) a racemic mixture
- D) a diastereomeric mixture

32. Which of the following will form Benzene as final product?

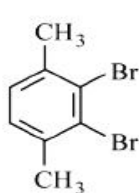


33. Which of the following reaction(s) give alkylation product:

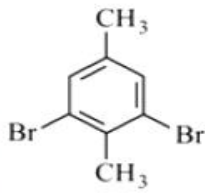


34. Out of the following four structures given, which structures when given IUPAC name will have bromine atoms located at same IUPAC numbering in their respective IUPAC names.

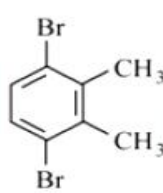
(Hint: Suppose in one structure = 1,2-dibromo, then in other structure also it should be 1,2-dibromo)



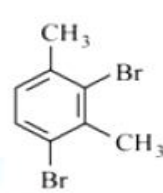
(K)



(L)



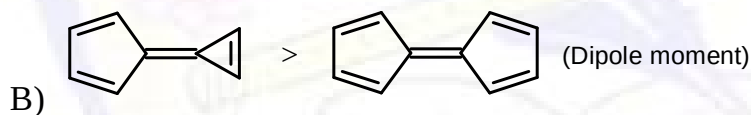
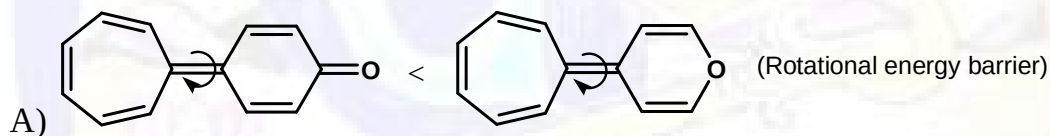
(M)



(N)

- A) K and L B) M and N C) L and M D) L and N

35. Identify the true statements from following?



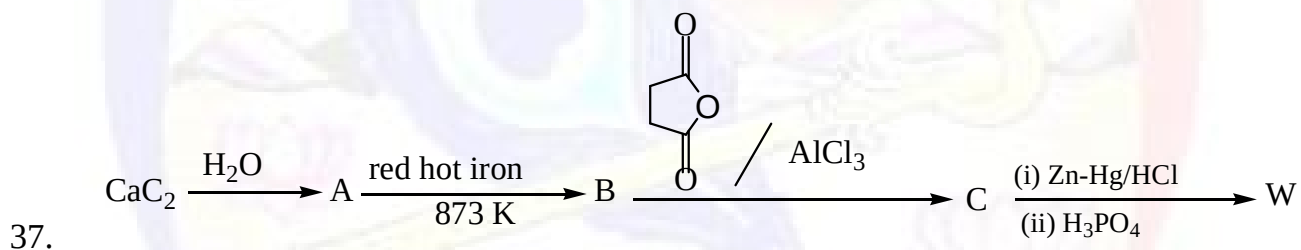
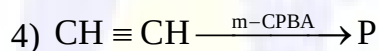
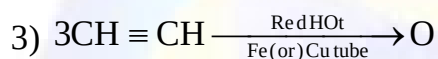
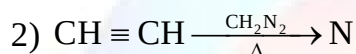
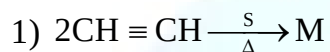
- C) Friedel crafts reaction fails for nitrobenzene

- D) $C_6T_6 < C_6D_6 < C_6H_6$ (rate towards iodination)

SECTION - III
INTEGER ANSWER TYPE

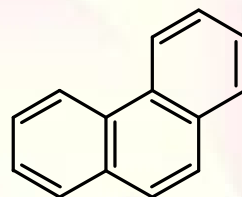
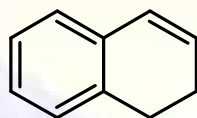
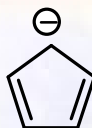
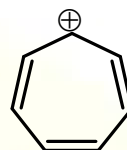
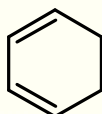
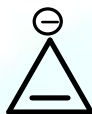
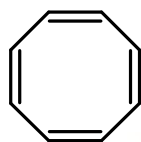
This section contains 5 questions. The answer to each question is single digit integer, ranging from 0 to 9 (both inclusive).

36. How many out of the following reactions, can give aromatic product?

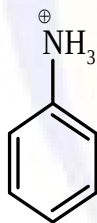


Find the number of oxygen atoms in W.

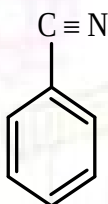
38. Among the following, the number of aromatic compound(s) is



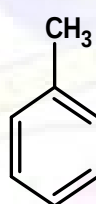
39. Examine the structural formulas shown below and find out how many compounds can not give Friedel Crafts reaction.



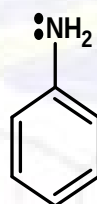
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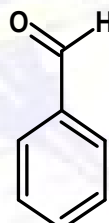
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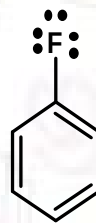
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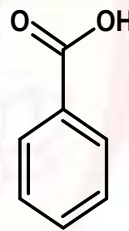
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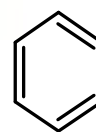
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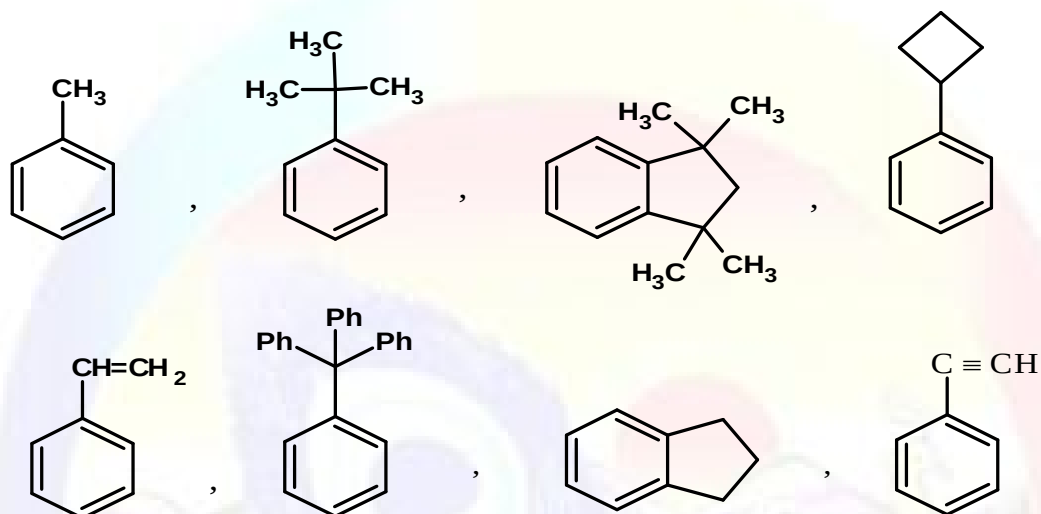
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40. Examine the structural formulas shown below and find out how many compounds will show side chain oxidation reaction with acidic KMnO_4



MATHS:**Max. Marks: 60****SECTION - I****SINGLE CORRECT ANSWER TYPE**

This section contains 10 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

41. In a triangle if $r_1 > r_2 > r_3$, then
 A) $a > b > c$ B) $a < b < c$ C) $a > b$ and $b < c$ D) $a < b$ and $b > c$
42. If in a triangle $\frac{r}{r_1} = \frac{r_2}{r_3}$, where r, r_1, r_2, r_3 have their usual meanings, then
 A) $A = 90^\circ$ B) $B = 90^\circ$ C) $C = 90^\circ$ D) None of these
43. If in a ΔABC $(r_2 - r_1)(r_3 - r_1) = 2r_2r_3$, then the ΔABC is
 A) Acute angled B) obtuse angled C) equilateral Δ D) right angled Δ
44. The product of the distances of the incentre from the angular points of a ΔABC is:
 A) $4R^2r$ B) $4Rr^2$ C) $\frac{(a+b+c)R}{s}$ D) $\frac{(a+b+c)s}{R}$
45. A circle of radius 4 unit is inscribed in an equilateral triangle ABC. Sides AB and AC of triangle are produced and a circle is drawn touching the sides BC and produced part of sides AB and AC. The area of the circle drawn is
 A) 16π sq. unit B) 64π sq. unit C) 144π sq. unit D) 256π sq. unit

46. The sides of a triangle ABC satisfy the relations $a + b - c = 2$ and $2ab - c^2 = 4$. Then the radius of the circle escribed opposite to the angle A is
- A) 1 B) $\sqrt{3}$ C) $\frac{\sqrt{3}}{2}$ D) $\frac{1}{\sqrt{3}}$
47. In triangle ABC, if $r_1 = 3, r_2 = 10, r_3 = 15$, then $c =$
- A) 5 B) 12 C) 13 D) $13/2$
48. In a $\Delta^{\text{le}}ABC, r_1 = 2r_2 = 3r_3$. D is mid point of BC and $\angle ADC = \theta$ then $\cos \theta =$
- A) $-7/25$ B) $3/25$ C) $\frac{1}{25}$ D) $\frac{-12}{25}$
49. Let H be the ortho centre of $\Delta^{\text{le}}ABC$, then angle subtended by the side BC at the centre of in circle of $\Delta^{\text{le}}BHC$ is
- A) $\frac{A}{2} + \frac{\pi}{2}$ B) $\frac{B+C}{2} + \frac{\pi}{2}$ C) $\frac{B-C}{2} + \frac{\pi}{2}$ D) $\frac{B+C}{2} + \frac{\pi}{4}$
50. Three Circles with radii a, b, c touch one another externally tangents at contact points meet at P then the distance of P to the contact point of any two circles is
- A) $\frac{abc}{a+b+c}$ B) $\sqrt{\frac{abc}{a+b+c}}$ C) $\sqrt{\frac{a+b+c}{abc}}$ D) $\frac{a+b+c}{abc}$

SECTION - II

MULTIPLE CORRECT ANSWER(S) TYPE

This section contains 5 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which ONE or MORE are correct.

51. Let A, A_1, A_2, A_3 be the areas of the inscribed and escribed circles of ΔABC , then:

A) $\sqrt{A_1} + \sqrt{A_2} + \sqrt{A_3} = \sqrt{\pi} (r_1 + r_2 + r_3)$ B) $\sqrt{A_1} + \sqrt{A_2} + \sqrt{A_3} = \sqrt{\pi} (4R + r)$

C) $\frac{1}{\sqrt{A_1}} + \frac{1}{\sqrt{A_2}} + \frac{1}{\sqrt{A_3}} = \frac{1}{\sqrt{A}}$ D) $\frac{1}{\sqrt{A_1}} + \frac{1}{\sqrt{A_2}} + \frac{1}{\sqrt{A_3}} = \frac{s^2}{\sqrt{\pi} r_1 r_2 r_3}$

52. Three equal circle each of radius r touch one another. The radius of the circle touching all the three given circles DEF is

A) $(2 + \sqrt{3})r$ B) $\frac{(2 + \sqrt{3})}{\sqrt{3}}r$ C) $\frac{(2 - \sqrt{3})}{\sqrt{3}}r$ D) $(2 - \sqrt{3})r$

53. If a right angled ΔABC of maximum area is inscribed within a circle of radius R , Δ represents area of triangle ABC and r, r_1, r_2, r_3 represent inradius and exradii, and s is the semi perimeter then

A) $\Delta = R^2$ B) $\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{\sqrt{2} + 1}{R}$

C) $r = (\sqrt{2} - 1)R$ D) $s = (1 + \sqrt{2})R$

54. In a triangle ABC, $\angle C = \frac{\pi}{2}$, then

A) $r_3 = r_2 + r_1 + r$

B) $(a^2 + b^2) \sin(A - B) = (a^2 - b^2) \sin(A + B)$

C) $\frac{2 \cos C}{c} + \frac{\cos B}{b} + \frac{2 \cos A}{a} = \frac{c}{ab} + \frac{b}{ac}$

D) $r + 2R = s$

55. In a ΔABC which of the following results are true.

A) $\frac{ab - r_1 r_2}{r_3} = r$

B) $\frac{bc - r_2 r_3}{r_1} = r$

C) $\frac{ac - r_1 r_3}{r_2} = r$

D) $\cos A + \cos B + \cos C = 1 + \frac{r}{R}$

SECTION - III INTEGER ANSWER TYPE

This section contains 5 questions. The answer to each question is single digit integer, ranging from 0 to 9 (both inclusive).

56. AD, BE and CF are the perpendiculars from the angular points of a ΔABC upon the opposite sides. The perimeters of the ΔDEF and ΔABC are in the ratio $\frac{kr}{R}$ then find the value of k.

57. With usual notation in a ΔABC , if $R = \frac{1}{k} \frac{(r_1 + r_2)(r_2 + r_3)(r_3 + r_1)}{r_1 r_2 + r_2 r_3 + r_3 r_1}$ where k has the value equal to ?
58. In a triangle ABC , $a : b : c = 4 : 5 : 6$ and the ratio of the radius of the circumcircle to that of the incircle is K , then the value of $16/K$ is
59. The value of $\frac{b-c}{r_1} + \frac{c-a}{r_2} + \frac{a-b}{r_3}$ is
60. Consider a triangle ABC and let a, b and c denote the lengths of the sides opposite to vertices A, B and C , respectively. Suppose $a=6$, $b=10$ and the area of triangle is $15\sqrt{3}$. if $\angle ACB$ is obtuse and if r denotes the radius of the incircle of the triangle, then find the value of r^2 .



Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

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2013-P1

Date: 16-12-17

Max.Marks:180

KEY SHEET

PHYSICS

1	C	2	C	3	D	4	A	5	A
6	A	7	A	8	A	9	A	10	B
11	BCD	12	ABCD	13	AB	14	ACD	15	B
16	3	17	5	18	7	19	2	20	4

CHEMISTRY

21	C	22	A	23	C	24	C	25	A
26	C	27	C	28	D	29	B	30	C
31	ACD	32	ABCD	33	ABD	34	D	35	ABCD
36	3	37	1	38	5	39	5	40	5

MATHS

41	A	42	C	43	D	44	B	45	C
46	B	47	C	48	A	49	B	50	B
51	ABCD	52	BC	53	ABCD	54	ABCD	55	ABCD
56	1	57	4	58	7	59	0	50	3

SOLUTIONS PHYSICS

1. In the position shown torque, $\tau = mgAC \sin \theta$
 $= mg(AC)\theta$

Where C is centre of mass and θ is angle between AC and AO.

For small oscillations $AC = R - OC$

$$= R - \frac{3R}{8} = \frac{5R}{8}$$

$$I_A = I_C + m\left(\frac{5R}{8}\right)^2 = \left[\frac{2}{5}MR^2 - M\left(\frac{3R}{8}\right)^2\right] + M\left(\frac{5R}{8}\right)^2$$

$$\tau = I\alpha \Rightarrow mg \frac{5R}{8} \theta = -I_A \alpha$$

$$\alpha = \frac{-mg 5R}{8I_A} \theta \quad \omega^2 = \frac{mg 5R}{8I_A}$$

2. Let $R d\theta$ be an element of wire located at an angular position of θ wrt -ve y axis.
 When slightly rotated by angle β torque due to gravity is

$$\vec{\tau} = \int d\vec{\tau} = \int d\vec{F} \times \vec{r}$$

$$= \int dm \vec{g} \times \vec{r} = - \int \left[R d\theta \frac{M}{\pi R} \right] x^2 (R \cos \theta) \sin \beta$$

$$\Rightarrow \tau = \int R \left(\frac{M d\theta}{\pi} x^2 R \cos \theta \right) \beta, \text{ for small } \beta$$

$$\text{as } x = R \sin \theta \quad \tau = \beta \int_{-\pi/2}^{\pi/2} \frac{MR \cos \theta}{\pi} R^2 \sin^2 \theta d\theta \text{ ----- (1)}$$

$$\text{But } \tau = I\alpha = MR^2 \alpha \text{ ----- (2)}$$

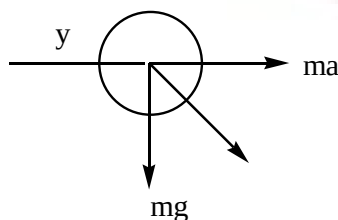
$$\text{Where } \alpha = \frac{d^2 \beta}{dt^2}$$

$$\alpha = -\omega^2 \beta \quad \Rightarrow \quad T = \frac{2\pi}{\omega}$$

3. Conceptual

4. Conceptual

- 5.



$$m\omega^2 y = m \left(\sqrt{g^2 + a^2} \right) \frac{y}{R}$$

6. Let the deformation in each spring be x_1, x_2 and x_3 respectively as shown

Let the block be displaced by x .

$$\text{Then } x = x_1 + 2x_2 + x_3 \dots (i)$$

$$F = 2Kx_3; 2F = Kx_2; 2F = 2Kx_1 \dots (ii)$$

$$\text{Also } F = K_{eq} x \dots (iii)$$

$\therefore$ From equation (i), (ii) and (iii)

$$\frac{F}{K_{eq}} = \frac{F}{K} + \frac{4F}{K} + \frac{R}{2K}$$

$$\Rightarrow K_{eq} = \frac{2K}{11}$$

$$\therefore T = 2\pi \sqrt{\frac{11m}{2K}}$$

$$F = K_{eq} x = Kx_1$$

$$\frac{2K}{11} x = Kx_1$$

$$x_1 = \frac{2}{11} x$$

$$F = K_{eq} x = \frac{Kx_2}{2}$$

$$\frac{2K}{11} x = \frac{Kx_2}{2}$$

$$x_2 = \frac{4}{11} x$$

$$F = K_{eq} x = 2Kx_3$$

$$\frac{2K}{11} x = 2Kx_3$$

$$x_3 = \frac{1}{11} x$$

7. $U = 10 - 10 \cos 6x$

$$\therefore F = -\frac{dU}{dx} = -60 \sin 6x \approx -360x$$

(for small x)

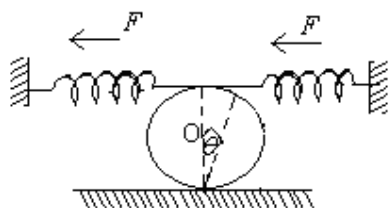
$$\therefore a = -\frac{360}{1} x \Rightarrow \omega^2 = 360$$

$$\therefore T = \frac{2\pi}{\omega} = \frac{1}{3} \text{ s}$$

$\therefore$ (A)

8. Net torque, $\tau = F \times 2R + F \times 2R$

$$\text{or, } \tau = 2K(2R\theta)(2R) = 8KR^2\theta$$



$$\tau = I \alpha = \frac{3}{2}MR^2 \frac{d^2\theta}{dt^2} = -8KR^2\theta$$

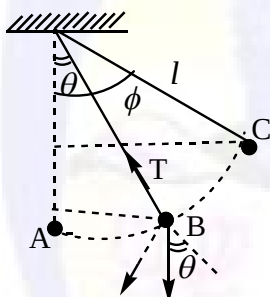
$$\omega = \sqrt{\frac{16K}{3m}} \text{ so, } T = 2\pi \sqrt{\frac{3m}{16K}} = \frac{\pi}{2} \sqrt{\frac{3m}{K}}$$

9. $F = -\frac{dU}{dx}$, so, $a = \frac{-U_0 c^2}{m} x = -\omega^2 x$

$$\omega^2 = \frac{U_0 c^2}{m}, T = 2\pi \sqrt{\frac{m}{U_0 c^2}}$$

10. Conceptual

11.



$$T - mg \cos \theta = \frac{mv^2}{l}$$

$$\frac{mv^2}{l} = 2mg(\cos \theta - \cos \phi)$$

$$T - mg \cos \theta = \frac{mv^2}{l} = 2mg(\cos \theta - \cos \phi)$$

$$T = mg(3\cos \theta - 2\cos \phi)$$

$$3\cos \theta = 2\cos \phi + 1$$

$$\theta = \cos^{-1} \left[\frac{1}{3}(2\cos \phi + 1) \right]$$

12. Period of oscillation changes as it depends on mass and becomes three times. The amplitude of oscillation does not change, because the new object is attached when the original object is at rest. Total energy does not change as at extreme position the energy is in the form of potential energy stored in spring which is independent of mass, and hence maximum; KE also does not change but as mass changes the maximum speed changes.

13. The time period of simple harmonic pendulum is independent of mass, so it would be same as that $T = 2\pi\sqrt{l/g}$. After collision, the combined mass acquires a velocity of $\frac{v_0}{2}$, as a result of this velocity, the mass (2m) moves up and at an angle θ_0 (say) with vertical, it stops, this is the extreme position of bob.

$$0 - \frac{2m}{2} \left(\frac{v_0}{2} \right)^2 = -2mgl(1 - \cos\theta_0)$$

$$\frac{v_0^2}{8gl} = 1 - \cos\theta = 2\sin^2 \frac{\theta_0}{2}$$

$$\sin \frac{\theta_0}{2} = \frac{v_0}{4\sqrt{gl}}$$

$$\text{If } \theta_0 \text{ is small, } \sin \frac{\theta_0}{2} = \frac{\theta_0}{2} \Rightarrow \theta_0 = \frac{v_0}{2\sqrt{gl}}$$

14. Conceptual

15. Acceleration $a = \omega^2 x$

$$\text{Maximum acceleration } a = \omega^2 A = \left(\frac{2\pi}{T} \right)^2 \frac{25}{100}$$

When block and platform are separated, $a = g = 10$

$$\frac{4\pi^2}{T^2} \cdot \frac{1}{40} \Rightarrow T^2 = \frac{\pi^2}{100} T = \frac{\pi}{10} \text{ sec}$$

16. Consider a small angular displacement β

Torque about the rotational axis

$$\tau = - \left(mg \frac{\ell}{2} \sin \theta \right) (\beta)$$

$$\tau = I \alpha$$

$$I = m \frac{(\ell \sin \theta)^2}{3} = m \frac{\ell^2 \sin^2 \theta}{3}$$

$$- \left(mg \frac{\ell}{2} \sin \theta \right) (\beta) = \frac{m \ell^2 \sin^2 \theta}{3} \alpha$$

$$\alpha = - \left(\frac{3g}{2\ell \sin \theta} \right) \beta$$

$$\therefore T = 2\pi \sqrt{\frac{2\ell \sin \theta}{3g}}$$

17. Conceptual

$$18. E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 + \frac{1}{2}I_P\omega^2$$

$$\Rightarrow \frac{1}{2}mv^2 + \frac{1}{2}kx^2 + \frac{3}{8}mv^2$$

$$E = v^2 + \frac{4k}{7m}x^2 = \text{constant}$$

Which gives $\omega^2 = \frac{4k}{7m}$ or $T = \pi \sqrt{\frac{7m}{k}}$

19. If mass m on inclined plane moves by a distance x , the hanging mass moves by distance $x/2$

Using Energy Equation

$$E = \frac{1}{2}k(x + x_0)^2 + \frac{1}{2}mv^2 + \frac{1}{2}m\left(\frac{v}{2}\right)^2 + mgx \sin \theta - mg\frac{x}{2} + U_0$$

At mean position $Kx_0 + mg \sin \theta = \frac{mg}{2}$

Differentiating with respect to time we get

$$\ddot{a} + \frac{4k}{5m}\ddot{x} = 0$$

$$\Rightarrow \omega = \sqrt{\frac{4k}{5m}} = \sqrt{\frac{4 \times 20}{5 \times 4}} = 2 \text{ rad/sec}$$

20. Mean position ($F = 0$) is at $x = 4 \text{ m}$.

At a position l with respect to mean position

$$x = (l + 4)$$

Acceleration, $\ddot{a} = \frac{F_0[1 - \frac{(l+4)}{4}]}{m} = -\frac{F_0 l}{4m}$

Therefore motion is SHM and $\omega^2 = \frac{F_0}{4m}$

or, time period, $T = 2\pi \sqrt{\frac{4m}{F_0}} = 2 \times 6 = 12 \text{ s}$.

$x = +12 \text{ m}$ is extreme position & $x = +4 \text{ m}$ is mean position.

Therefore amplitude of motion $A = (+12) - (+4) = 8 \text{ m}$.

Origin is at a distance $4 \text{ m} = \frac{A}{2}$ from mean position.

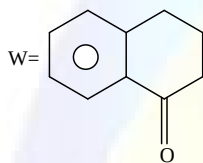
Minimum time taken to reach from $l = +A$ ($\phi_1 = 90^\circ$) to $l = -\frac{A}{2}$ ($\phi_2 = 210^\circ$) is

$$\Delta t = \frac{(210) - (90)}{360} \times 12 = 4 \text{ sec.}$$

CHEMISTRY

21. Ortho position is less crowded
22. 1 - ynes gives red precipitate with $\text{Cu}_2\text{Cl}_2 + \text{NH}_4\text{OH}$
23. In the presence of light free radical reaction taken place.
24. Maximum number of electrons participating in resonance per ring.
25. Electron density more on Anisole
26. H_2O attack at more substituted carbon.
27. Electrophilic addition taken place following markownikoff rule.

28. Conceptual
29. Directive influence of more activating group is deciding the product.
30. Isotopic effect is absent in nitration
31. syn addition takes place converting alkyne to cis-alkene
32. Conceptual
33. Friedel craft's alkylation
34. Conceptual
35. First ring is aromatic
36. $x=2$, $y=3$ so answer is 5



- 37.
38. Conceptual
39. Friedel craft's reaction fails for strongly deactivating groups, aniline.
40. Conceptual

MATHS

41. We have $r_1 > r_2 > r_3 \Rightarrow \frac{\Delta}{s-a} > \frac{\Delta}{s-b} > \frac{\Delta}{s-c}$
 Dividing each by Δ , then $\frac{1}{s-a} > \frac{1}{s-b} > \frac{1}{s-c}$
 $\Rightarrow s-a < s-b < s-c$ (subtracting s from each) $\Rightarrow -a < -b < -c \Rightarrow a > b > c$
42. Given that $\frac{r}{r_1} = \frac{r_2}{r_3}$
 $\Rightarrow \frac{\Delta(s-a)}{s\Delta} = \frac{\Delta(s-c)}{(s-b)\Delta}$
 $\Rightarrow (s-a)(s-b) = s(s-c)$
 $\Rightarrow ab = s(a+b-c)$
 $\Rightarrow a^2 + b^2 = c^2 \Rightarrow$ the triangle is right angled at C.
43. $\Rightarrow \left[\frac{\Delta}{s-b} - \frac{\Delta}{s-a} \right] \left[\frac{\Delta}{s-c} - \frac{\Delta}{s-a} \right] = 2 \frac{\Delta}{(s-b)} \cdot \frac{\Delta}{(s-c)}$
 $\Rightarrow \left[\frac{(s-a)}{(s-a)} - \frac{(s-b)}{(s-b)} \right] \left[\frac{(s-a)}{(s-c)} - \frac{(s-c)}{(s-a)} \right] = \frac{2}{(s-b)(s-c)}$
 $\Rightarrow \left[\frac{(b-a)}{(s-a)^2} \right] = 2$

$$\Rightarrow (b-a)(c-a) = 2(s-a)^2 \Rightarrow 2(b-a)(c-a) = [2s-2a]^2$$

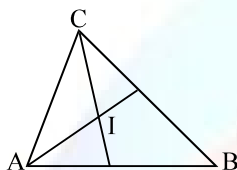
$$\Rightarrow (b-a)(c-a) = [b+c+a]^2$$

$$\Rightarrow 2(bc-ab-ac+a^2) = b^2+c^2+a^2+2bc-2ca-2ab$$

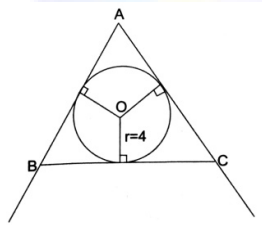
$$\Rightarrow a^2 = b^2 + c^2$$

$\therefore \Delta ABC$ is a rt. Δ

$$44. \quad 1. \quad \Pi IA = r^3 \Pi \operatorname{cosec} \frac{A}{2} = \frac{4r^3}{4 \Pi \sin \frac{A}{2}} = \frac{4r^3 \cdot R}{r} = 4r^2 R \Rightarrow \Rightarrow B$$



$$45. \quad \text{Here } r=4 \Rightarrow R=8$$



Trigonometry

$$A = 2s \sin A = 2(8) \sin 60^\circ = 16 \left(\frac{\sqrt{3}}{2} \right) = 8\sqrt{3}$$

$$\text{Now, } r_1 = s \tan \frac{A}{2} = 12\sqrt{3} \left(\frac{1}{\sqrt{3}} \right) = 12$$

$\therefore$ Area of drawn circle = 144π square units

$$46. \quad a+b-c=2 \text{ and } 2ab-c^2=4$$

$$\therefore a=b=c \text{ and } r_1 = \frac{\Delta}{s-a} = \frac{\left(\frac{\sqrt{3}}{4} 2^2 \right)}{3-2} = \sqrt{3}$$

$$47. \quad r = \frac{1}{\frac{1}{3} + \frac{1}{10} + \frac{1}{15}} = \frac{30}{10+3+2} = 2.$$

$$c = \sqrt{(15-2)(3+1)} = 13$$

$$r_1 = 2r_2 = 3r_3$$

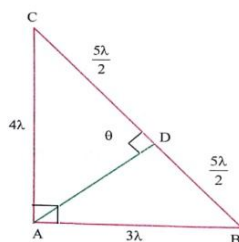
$$\frac{1}{s-a} = \frac{2}{s-b} = \frac{3}{s-c}; s-a = \frac{s-b}{2} = \frac{s-c}{3} = \lambda$$

$$a = s - \lambda \quad b = s - 2\lambda \quad c = s - 3\lambda$$

$$a + b + c = 3s - 6\lambda; 2S = 3s - 6\lambda; 6\lambda = S$$

$$\lambda = \frac{s}{6} \Rightarrow a = s - \frac{s}{6} = \frac{5s}{6}; \quad b = s - \frac{2s}{6} = \frac{4s}{6}$$

$$c = s - \frac{3s}{6} = \frac{3s}{6}$$



48.

By cosine rule

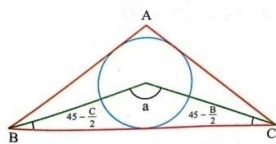
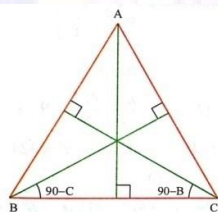
$$AD^2 = (3\lambda)^2 + \left(\frac{5\lambda}{2}\right)^2 - 2(3\lambda)\left(\frac{5\lambda}{2}\right)\cos\theta$$

$$= 9\lambda^2 + \frac{25\lambda^2}{4} - 9\lambda^2; \quad AD = \frac{5\lambda}{2}$$

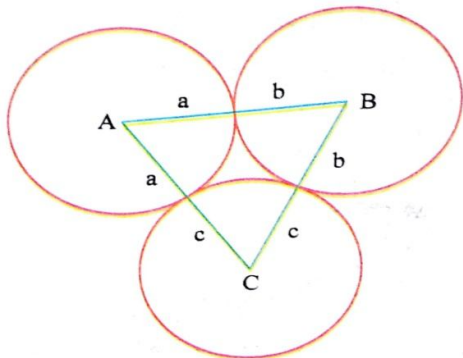
By cosine rule

$$\therefore \cos\theta = \frac{\left(\frac{5\lambda}{2}\right)^2 + \left(\frac{5\lambda}{2}\right)^2 - (4\lambda)^2}{2\left(\frac{25\lambda^2}{4}\right)} = \frac{-7\lambda^2}{\frac{25\lambda^2}{2}} = -\frac{7}{25}$$

49. H-Orthocentre



$$\alpha + 4S - \frac{C}{2} + 4S - \frac{B}{2} = 180; \alpha = \frac{\pi}{2} + \frac{B+C}{2}$$



50.

Required distance is distance between Incenter and side

51. (a,b,c,d) $A = \pi r^2; r = \Delta / s;$

$$A_1 = \pi r_1^2; r_1 = \frac{\Delta}{s-a}; A_2 = \pi r_2^2; r_2 = \frac{\Delta}{s-b};$$

$$A_3 = \pi r_3^2; r_3 = \frac{\Delta}{(s-a)}$$

$$\therefore \sqrt{A_1} + \sqrt{A_2} + \sqrt{A_3} = \sqrt{\pi} (r_1 + r_2 + r_3)$$

$\Rightarrow$ (a) is true

Also $r_1 + r_2 + r_3 = r + 4R$ is true

$$\Rightarrow \sqrt{A_1} + \sqrt{A_2} + \sqrt{A_3} = \sqrt{\pi} (4r + r)$$

$\Rightarrow$ (b) is also true

$$\text{Now, } \frac{1}{\sqrt{A_1}} + \frac{1}{\sqrt{A_2}} + \frac{1}{\sqrt{A_3}}$$

$$= \frac{1}{\sqrt{\pi}} \cdot \left(\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} \right) = \frac{1}{\sqrt{\pi}} \cdot \frac{1}{r} = \sqrt{\frac{1}{\pi r^2}} = \frac{1}{\sqrt{A}}$$

$\Rightarrow$ (c) is also true

$$\text{Also } \frac{1}{\sqrt{A_1}} + \frac{1}{\sqrt{A_2}} + \frac{1}{\sqrt{A_3}} = \frac{1}{\sqrt{\pi}} \cdot \frac{1}{r} = \frac{1}{\sqrt{\pi}} \left(\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} \right)$$

$$= \frac{1}{\sqrt{\pi}} \left(\frac{r_2 r_3 + r_1 r_3 + r_1 r_2}{r_1 r_2 r_3} \right)$$

$$= \frac{1}{\sqrt{\pi}} \left[\frac{\Delta^2}{(s-b)(s-c)} + \frac{\Delta^2}{(s-a)(s-c)} + \frac{\Delta^2}{(s-a)(s-b)} \right] \cdot \left(\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} \right)$$

$$= \frac{\Delta}{\sqrt{\pi}} \left[\frac{(s-a) + (s-b) + (s-c)}{(s-a)(s-b)(s-c)} \right] \cdot \frac{1}{r_1 r_2 r_3}$$

$$= \frac{\Delta}{\sqrt{\pi}} \left[\frac{s}{(s-a)(s-b)(s-c)} \right] \cdot \frac{1}{r_1 r_2 r_3} = \frac{1}{\sqrt{\pi}} \frac{s^2}{r_1 r_2 r_3}$$

$\Rightarrow$ (d) is correct

52. Conceptual

53. For a right angled triangle inscribed in a circle of radius R the length of the hypotenuse is $2R$. Then area is maximum when its is isosceles triangle With each side $= \sqrt{2}R$

$$\therefore S = \frac{1}{2} (2\sqrt{2} + 2)R = (\sqrt{2} + 1)R$$

$$\Delta = \frac{1}{2} \sqrt{2}R \cdot \sqrt{2}R = R^2$$

$$r = \frac{\Delta}{S (\sqrt{2} + 1)R} \Rightarrow r = (\sqrt{2} - 1)R$$

$$\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{1}{r} = \frac{1}{(\sqrt{2} - 1)R} = \frac{\sqrt{2} + 1}{R}$$

54. $r_3 = r_1 + r_2 + r$

$$\Rightarrow \frac{\Delta}{s-c} = \frac{\Delta}{s-a} + \frac{\Delta}{s-b} + \frac{\Delta}{s}$$

$$\Rightarrow \frac{\Delta}{s(s-c)} = \frac{2s-(a+b)}{(s-a)(s-b)} \quad (2s = a+b+c)$$

$$\Rightarrow s(s-c) = (s-a)(s-b)$$

$$\Rightarrow s^2 - (a+b)s + ab = s^2 - cs$$

$$\Rightarrow s(a+b-c) = ab$$

$$\Rightarrow (a+b+c)(a+b-c) - 2ab = 0$$

$$\Rightarrow (a+b)^2 - c^2 - 2ab = 0$$

$$\Rightarrow a^2 + b^2 = c^2 \Rightarrow \angle C = \pi/2$$

$$\angle C = \pi/2 \Rightarrow R = c/2 \Rightarrow c = 2R$$

$$r = (s-c) \tan(C/2) = s-c$$

$$\Rightarrow r = s - 2R \Rightarrow r + 2R = s$$

Multiplying both sides by abc in option C

$$2ab \cos C + ac \cos B + 2bc \cos A = c^2 + b^2$$

$$\Rightarrow b^2 + a^2 - c^2 + \frac{a^2 + c^2 - b^2}{2} + b^2 + c^2 - a^2 = c^2 + b^2$$

$$\Rightarrow a^2 + b^2 = c^2 \Rightarrow \angle C = \pi/2$$

$$a^2 + b^2 = c^2$$

$$\Rightarrow \sin^2 A + \sin^2 B - \sin^2 C = 0 \quad (a = 2R \sin A)$$

$$\Rightarrow \sin(A-B)(\sin^2 A + \sin^2 B - \sin^2 C) = 0$$

$$\Rightarrow (\sin^2 A + \sin^2 B) \sin(A-B) = \sin^2(A+B) \sin(A-B) \quad (A+B = \pi - C)$$

$$= \sin(A+B)(\sin^2 A - \sin^2 B)$$

$$\Rightarrow (a^2 + b^2) \sin(A-B) = (a^2 - b^2) \sin(A+B) \quad (\sin A = a/2R)$$

$$(A) \frac{ab - r_1 r_2}{r_3} = \frac{ab}{r_3} \left\{ 1 - \frac{\Delta^2}{ab(s-a)(s-b)} \right\}$$

$$= \frac{ab}{r_3} \left\{ 1 - \frac{s(s-c)}{ab} \right\}$$

$$= \frac{ab}{s \tan C/2} (1 - \cos^2 C/2)$$

55. $= \frac{2ab \sin C/2 \cos C/2}{2S}$

$$= \frac{ab \sin C}{2} \times \frac{1}{S} = \frac{\Delta}{S} = r$$

Similarly (B) and (C) are also correct.

$$(D) \cos A + \cos B + \cos C = 1 + 4 \sin A/2 + \sin B/2 \sin C/2$$

$$= 1 + \frac{4R \sin A/2 \sin B/2 \sin C/2}{R} = 1 + \frac{r}{R}$$

56. Note that ΔDEF is a pedal triangle whose sides are $R \sin 2A$, $R \sin 2B$ and $R \sin 2C$.

$$\Delta \text{ ratio} = \frac{R \sum \sin 2A}{a+b+c} = \frac{4R \Pi \sin A}{2R \sum \sin A} = \frac{2 \cdot 8 \Pi \sin \frac{A}{2} \Pi \cos \frac{A}{2}}{4 \Pi \cos \frac{A}{2}} = 4 \Delta \sin \frac{A}{2} = \frac{r}{R}$$

57. Numerator of RHS = $\frac{s^2 \cdot 4R}{k}$ and denominator of RHS = s^2

(using $r_1 = s \tan \frac{A}{2}$ etc.)

$$\Delta \frac{\text{numerator}}{\text{denominator}} \text{ of RHS} = \frac{4R}{k} = \text{LHS} = R \quad \Delta \quad k = 4$$

58. We have $R = \frac{abc}{4\Delta}$ and $r = \frac{\Delta}{s}$

$$\frac{R}{r} = \frac{abc}{4\Delta} \cdot \frac{s}{\Delta} = \frac{abc \cdot s}{4\Delta^2} = \frac{abc \cdot s}{4s(s-a)(s-b)(s-c)}$$

But $a : b : c = 4 : 5 : 6$ (given)

$$\Rightarrow \frac{a}{4} = \frac{b}{5} = \frac{c}{6} = k \text{ (let)}$$

$$a = 4k, b = 5k, c = 6k$$

$$\Rightarrow s = \frac{1}{2}(a+b+c) = \frac{1}{2}(4k+5k+6k) = \frac{15k}{2}$$

$$\therefore \frac{R}{r} = \frac{(4k)(5k)(6k)}{4\left(\frac{15k}{2}-4k\right)\left(\frac{15k}{2}-5k\right)\left(\frac{15k}{2}-6k\right)}$$

$$= \frac{30k^3}{k^3\left(\frac{15-8}{2}\right)\left(\frac{15-10}{2}\right)\left(\frac{15-12}{2}\right)} = \frac{30 \cdot 8}{7 \cdot 5 \cdot 3} = \frac{16}{7} = K$$

$$\Rightarrow \frac{16}{K} = \frac{16}{16/7} = 7$$

59. $\frac{b-c}{r_1} + \frac{c-a}{r_2} + \frac{a-b}{r_3} \Rightarrow (b-c)\left(\frac{s-a}{\Delta}\right) + (c-a)\left(\frac{s-b}{\Delta}\right) + (a-b)\left(\frac{s-c}{\Delta}\right)$

$$\Rightarrow \frac{(s-a)(b-c) + (s-b)(c-a) + (s-c)(a-b)}{\Delta}$$

$$= \frac{s(b-c+c-a+a-b) - [ab-ac+bc-ba+ac-bc]}{\Delta} = \frac{0}{\Delta} = 0$$

60. $\Delta = \frac{1}{2}ab \sin c$

$$\Rightarrow \sin C = \frac{2\Delta}{ab} = \frac{2 \times 15\sqrt{3}}{6 \times 10} = \frac{\sqrt{3}}{2} \Rightarrow C = 120^\circ$$

$$\Rightarrow \sqrt{a^2 + b^2 - 2abc \cos C}$$

Now, $r = \frac{\Delta}{s}$

$$\Rightarrow r^2 = \frac{225 \times 3}{\left(\frac{6+10+14}{2}\right)^2} = 3$$

Success



JEE ADVANCED WEEKEND TEST



Sri Chaitanya IIT Academy., India.

A.P., TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr. Super60-I

Time: 07:30 AM to 10:30 AM

Date: 30-12-17

Max.Marks:186

Name of the Student: _____

H.T. NO:

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2016_PAPER-II

IMPORTANT INSTRUCTIONS:

- 1) This booklet is your Question Paper.
- 2) Use the Optical Response Sheet (ORS) provided separately for answering the questions
- 3) Blank spaces are provided within this booklet for rough work.
- 4) Write your name, roll number and sign in the space provided on the back cover of this booklet.
- 5) You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET:

- 6) Darken the appropriate bubbles on the ORS by applying sufficient pressure. This will leave an impression at the corresponding place on the Candidate's sheet.
- 7) The ORS will be collected by the invigilator at the end of the examination.
- 8) Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
- 9) Write your name, roll number and code of the examination center, and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else** on the ORS. Darken the appropriate bubble under each digit of your roll number.

DARKENING THE BUBBLES ON THE ORS

- 10) Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
- 11) Darken the bubble **COMPLETELY**.
- 12) The correct way of darkening a bubble is as:
- 13) The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- 14) Darken the bubbles **ONLY** IF you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.

30-12-17_Jr.Super60-I_Jee Adv_WAT-30_Weekend Syllabus

MATHS: SYLLABUS COVERED FROM 25-12-17 to 29-12-2017

PHYSICS: SYLLABUS COVERD FROM 25-12-17 to 29-12-2017

CHEMISTRY: General EAS, directive influence of groups and activating –deactivating groups.
alkyl halides, Mechanism of Nucleophilic Substitution reactions (SN2, SOLVENT EFFECTi)

JEE-ADVANCE-2016-P2-Model**Time: 3:00 hours****IMPORTANT INSTRUCTIONS****Max Marks: 186****PHYSICS:**

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 6)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 7 – 14)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 15 – 18)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	3	0	4	12
Total				18	62

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 24)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 25 – 32)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 33 – 36)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	3	0	4	12
Total				18	62

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 42)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 43 – 50)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 51 – 54)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	3	0	4	12
Total				18	62

Sec: Jr. Super60-I**space for rough work****Page 2**

PHYSICS

Max Marks: 62

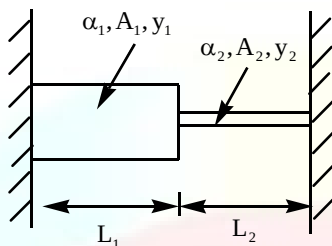
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.

1. Work done in converting 1g of ice at -10°C into steam at 100°C is (specific heats, latent heats are $s_{\text{ice}}=0.5 \text{ cal/g } ^{\circ}\text{C}$; $s_{\text{water}}=1 \text{ cal/g } ^{\circ}\text{C}$; $L_f = 80 \text{ cal/g}$; $L_v = 540 \text{ cal/g}$)
A) 3045 J B) 6056 J C) 721 J D) 6 J
2. The coefficient of linear expansion of an inhomogeneous rod changes linearly from α_1 to α_2 from one end to the other end of the rod. The effective coefficient of linear expansion of rod is
A) $\alpha_1 + \alpha_2$ B) $\frac{\alpha_1 + \alpha_2}{2}$ C) $\sqrt{\alpha_1 \alpha_2}$ D) $\alpha_1 - \alpha_2$
3. Two elastic rods are joined between fixed supports as shown in fig. Condition for no change in the lengths of individual rods with the increase of temperature
(α_1, α_2 = linear expansion coefficient, A_1, A_2 = area of rods, y_1, y_2 = Young's modulus) is

Sec: Jr. Super60-I**space for rough work****Page 3**



$$A) \frac{A_1}{A_2} = \frac{\alpha_1 y_1}{\alpha_2 y_2}$$

$$B) \frac{A_1}{A_2} = \frac{L_1 \alpha_1 y_1}{L_2 \alpha_2 y_2}$$

$$C) \frac{A_1}{A_2} = \frac{L_2 \alpha_2 y_2}{L_1 \alpha_1 y_1}$$

$$D) \frac{A_1}{A_2} = \frac{\alpha_2 y_2}{\alpha_1 y_1}$$

4. The coefficient of linear expansion of glass is α_g Per $^{\circ}\text{C}$, and the cubical expansion of mercury is γ_m per $^{\circ}\text{C}$. The volume of the bulb of a mercury thermometer at 0°C is V_0 and cross section of the capillary is A_0 . What is the length of mercury column in capillary at $T^{\circ}\text{C}$, if the mercury just fills the bulb at 0°C ?

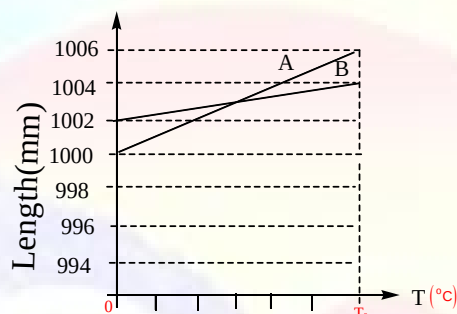
$$A) \frac{V_0 T (\gamma_m + 3\alpha_g)}{A_0 (1 + 2\alpha_g T)}$$

$$B) \frac{V_0 T (\gamma_m - 3\alpha_g)}{A_0 (1 + 2\alpha_g T)}$$

$$C) \frac{V_0 T (\gamma_m + 2\alpha_g)}{A_0 (1 + 3\alpha_g T)}$$

$$D) \frac{V_0 T (\gamma_m - 2\alpha_g)}{A_0 (1 + 3\alpha_g T)}$$

5. The variation of lengths of two metal rods A and B with change in temperature is shown in Fig. The coefficients of linear expansion α_A for the metal A and the temperature T will be



(given $\alpha_B = 9 \times 10^{-6} / ^\circ\text{C}$)

- A) $\alpha_A = 3 \times 10^{-6} / ^\circ\text{C}$, 500°C B) $\alpha_A = 3 \times 10^{-6} / ^\circ\text{C}$, 222.22°C
 C) $\alpha_A = 27 \times 10^{-6} / ^\circ\text{C}$, 500°C D) $\alpha_A = 27 \times 10^{-6} / ^\circ\text{C}$, 222.22°C
6. 1 g of steam at 100°C and an equal mass of ice at 0°C are mixed. The temperature of the mixture in steady state will be (Latent heat of steam = 540 cal/g Latent heat of ice = 80 cal/g , specific heat of water = $1 \text{ cal/g}^\circ\text{C}$)
- A) 50°C B) 100°C C) 67°C D) None of these

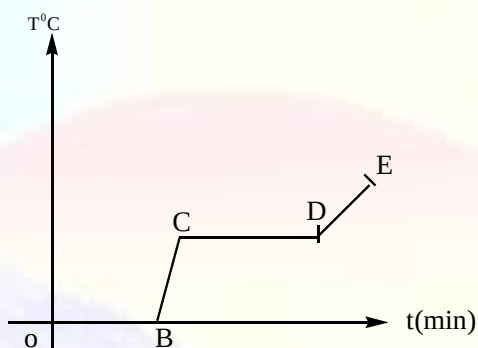
SECTION – II
(MULTIPLE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -2 in all other cases

7. An aluminium bush is to be fitted over a brass rod. At 20°C , inner diameter of aluminium ring is 5.00 cm and brass rod has a diameter of 5.05 cm. (Given $\alpha_{\text{Al}} = 24 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1}$, $\alpha_{\text{brass}} = 19 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1}$). To make the assembly, which of the following can be performed? (Melting point of Aluminium is approx. 700°C)
- A) Aluminium ring is heated above 437°C
 - B) Brass rod is cooled
 - C) Both aluminium ring and brass rod can be heated
 - D) Both aluminium ring and brass rod can be cooled using liquid nitrogen
8. Iron floats over mercury because density of mercury is more than the density of iron. Coefficient of volumetric expansion of iron and mercury are $35 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1}$ and $180 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1}$. Now consider an iron cube floating in a bowl of mercury at 0°C . When temperature is increased upto 25°C , then choose the correct options.
- A) Cube submerged more at 25°C
 - B) Cube submerged less at 25°C
 - C) Percentage change in the fraction of volume submerged is about 1%
 - D) Percentage change in the fraction of volume submerged is about 0.4%

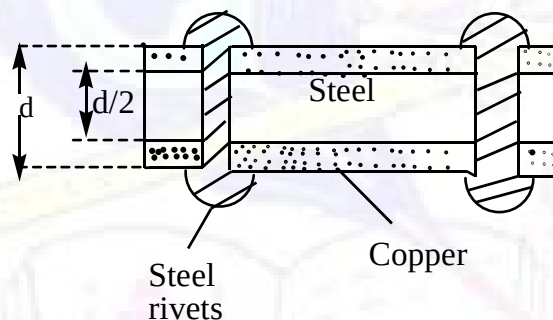
9. Refer to plot of temperature and time showing temperature change and state changes in ice on heating



Which of the following are correct?

- A) Region OB represents ice and water in thermal equilibrium
- B) At B water starts boiling
- C) At C all the water is converted into steam
- D) C to D represents water and steam in equilibrium at boiling point

10. Equal masses of three different liquids A, B and C are taken at 10°C , 15°C and 20°C respectively. The steady state temperature when A and B are mixed, is 13°C and when B and C are mixed, is 16°C . Which of the following statements is/are correct ?
- A) Ratio of specific heats of A and B is $\frac{2}{3}$
- B) Ratio of specific heats of A and C is $\frac{8}{3}$
- C) When A and C are mixed, temperature of mixture $\left(\frac{140}{11}\right)^{\circ}\text{C}$
- D) When all of three A, B and C are mixed, temperature of mixture is 15°C
11. A hollow copper tube is filled completely using a steel rod. The rod and tube are joined by two rivet pins as shown in the figure



Temperature of assembly is raised by ΔT after setting the rivets. Choose the correct options.

A) Stress in copper (σ_{cu}) tube = $-\frac{1}{3}$ stress in steel (σ_{steel}) tube

B) Stress in copper tube = – Stress in steel tube.

C) Strain in steel tube = $\alpha_{\text{steel}} \Delta T$

D) Strain in copper tube = $\frac{\sigma_{\text{cu}}}{Y_{\text{cu}}} + \alpha_{\text{cu}} \Delta T$

12. When a solid is heated (without phase change), which of the following occurs?

A) Intermolecular distance increases

B) Potential energy of molecules increases

C) Vibrational kinetic energy of molecules increases

D) Amplitude of vibrations of molecules increases

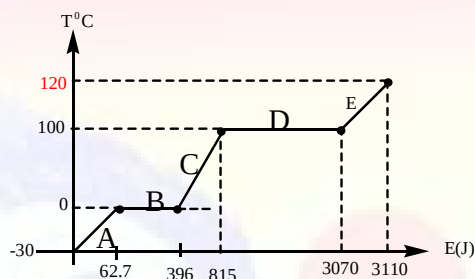
13. An ice cube of mass 0.1 kg at 0°C is placed in an isolated container at 227°C . Specific heat of container is given by $S=A+BT$, where
- $A = 100\text{ Cal Kg}^{-1}\text{K}^{-1}$ and $B = 2 \times 10^{-2}\text{ Cal Kg}^{-2}\text{K}^{-2}$. If final temperature after stabilization is 27°C , then which of these are correct ? (Take, $L_{\text{ice}} = 80\text{ cal g}^{-1}\text{K}^{-1}$ and $C_{\text{water}} = 10^3\text{ cal kg}^{-1}\text{K}^{-1}$)
- A) Ice receives a total of 10700 calories of heat
B) Container lost around 10800 calories of heat
C) Container has a mass of around 0.5 kg
D) Water equivalent of container is 300 g
14. A bimetallic strip is formed using two identical strips of copper and brass. On heating temperature of strip rises by ΔT and strip bends to form an arc of radius of curvature R . Then, R is
- A) Proportional to ΔT
B) Inversely proportional to ΔT
C) Proportional to $|\alpha_{\text{brass}} - \alpha_{\text{copper}}|$
D) Inversely proportional to $|\alpha_{\text{brass}} - \alpha_{\text{copper}}|$

SECTION – III (PARAGRAPH TYPE)

This section contains **2 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph for Questions 15 and 16

A heating curve for 1 g ice at -30°C is converted to steam, at 120°C is shown in



Note that X-axis shows energy added in joules and y-axis shows temperature in $^{\circ}\text{C}$

15. Ratio of slopes of portion A, portion C and portion E is
- A) 2090:4190:2010 B) 2010:4190:2090
- C) $\frac{1}{2090} : \frac{1}{4190} : \frac{1}{2010}$ D) $\frac{1}{2010} : \frac{1}{4190} : \frac{1}{2090}$
16. Ratio of latent heat of vaporization to latent heat of fusion of water is
- A) $\frac{2260}{360}$ B) $\frac{2260}{396}$ C) $\frac{2260}{3070}$ D) $\frac{2260}{333}$

Paragraph for Questions 17 and 18

An immersion heater, in an insulated vessel of negligible heat capacity brings 10g of water to the boiling point from 16°C in 7 min .Then

17. Power of heater is nearly

- A) 8.4×10^3 B) 84 W C) $8.4 \times 10^3 \text{ cal/s}$ D) 20W

18. The water is replaced by 200 g alcohol, which is heated from 16°C to the boiling point of 78°C in 6 min 12 s. The specific heat of alcohol is

- A) $0.6 \text{ J/kg}^{\circ}\text{C}$ B) $0.6 \text{ cal/g}^{\circ}\text{C}$ C) $0.6 \text{ cal/kg}^{\circ}\text{C}$ D) $6 \text{ J/Kg}^{\circ}\text{C}$

CHEMISTRY

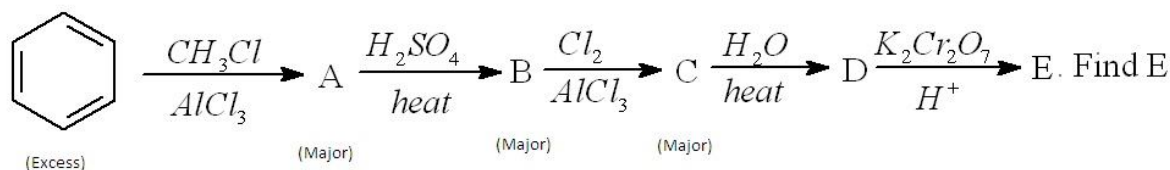
Max Marks: 62

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

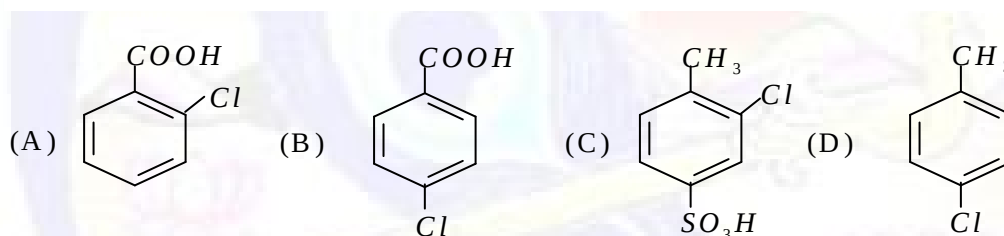
This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.

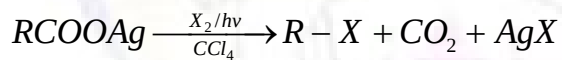
19.



Note: Assume all Products A, B, C are obtained after mono substitution



20. Yield of the reaction w.r.t to R-X here as shown below generally will follow the order:



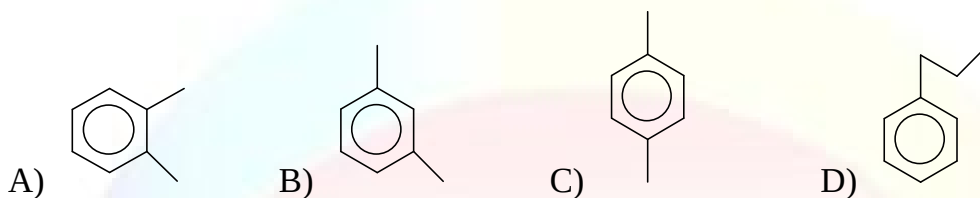
A) $1^\circ > 2^\circ > 3^\circ$ B) $3^\circ > 2^\circ > 1^\circ$ C) $2^\circ > 3^\circ > 1^\circ$ D) $3^\circ > 1^\circ > 2^\circ$

Sec: Jr. Super60-I

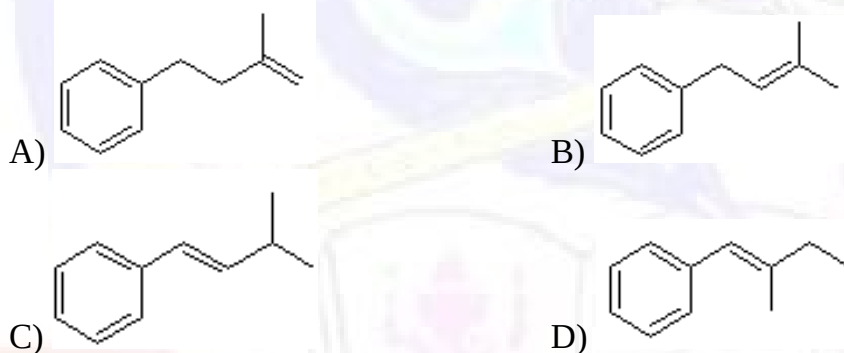
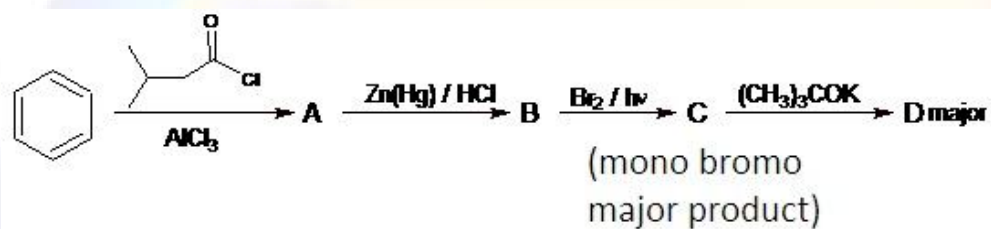
space for rough work

Page 13

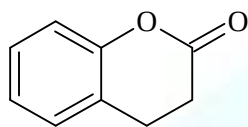
21. Ring nitration of dimethyl benzene results in the formation of only one nitro dimethyl benzene. The dimethyl benzene is



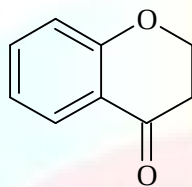
22. The major product (D) in the following reaction sequence is



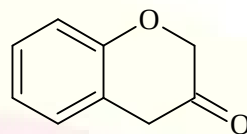
23.



(X)



(Y)



(Z)

The decreasing order of electrophilic aromatic substitution is

A) $X > Z > Y$ B) $Y > Z > X$ C) $Z > Y > X$ D) $Z > X > Y$

24. In which alkyl halide, dipole moment is maximum?

A) CH_3Cl B) CH_3F C) CH_3Br D) CH_3I

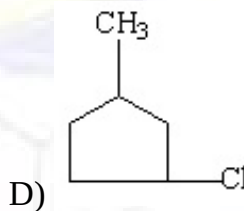
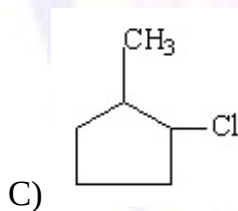
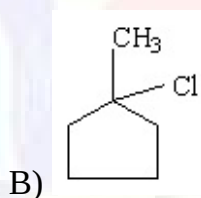
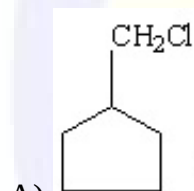
SECTION – II
(MULTIPLE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

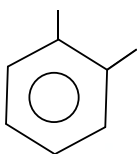
Marking scheme: +4 for correct answer, 0 if not attempted and -2 in all other cases



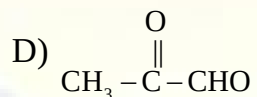
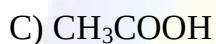
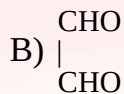
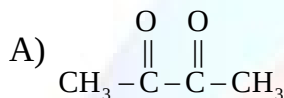
Possible monochloro substituted structural isomeric products (excluding stereo isomers) are



26. Which of the following products will be obtained if sample of



undergoes reductive ozonolysis?



27. Correct statement(s) regarding electrophilic aromatic substitution is

A) Among arylhalides, arylfluoride is the most deactivating.

B) Halogens on benzene ring direct the electrophile to ortho and para positions as they stabilize the arenium intermediates formed by ortho-para attack by their +R effect.

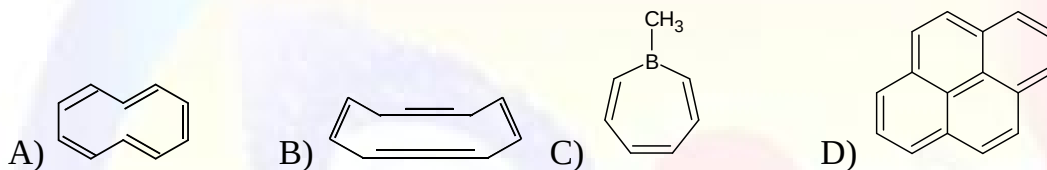
C) Halogens on benzene ring are activating groups as they are +R groups.

D) Halobenzenes react at a faster rate than benzene towards nitration.

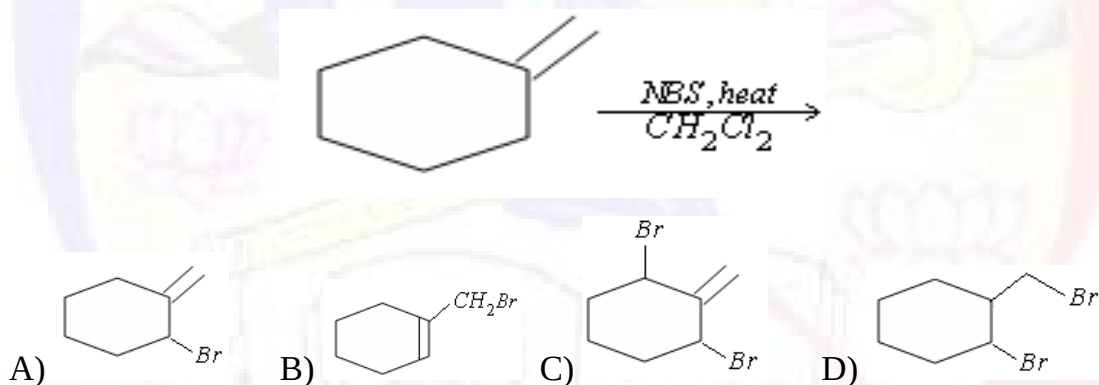
28. Which of the following is/are correct statement?

- A) Generally antiaromatic compounds are paramagnetic in nature
- B) The rate of nitration – $C_6H_6 > C_6D_6 > C_6T_6$
- C) The rate of sulphonation – $C_6H_6 > C_6D_6 > C_6T_6$
- D) Anthracene gives three monochloro products on chlorination

29. Which of the following is / are aromatic?



30. Predict the possible product(s) in the following reaction.



31. Which of the following is the weakest nucleophile in aprotic solvents?

- A) F^- B) Cl^- C) Br^- D) I^-

32. Boiling point order correct is:

- A) $CCl_4 > CHCl_3 > CH_2Cl_2 > CH_3Cl$
B) $CCl_4 > CHCl_3 > CH_3Cl > CH_2Cl_2$
C) $CHCl_3 > CCl_4 > CH_2Cl_2 > CH_3Cl$
D) $CH_3Cl > CH_2Cl_2 > CHCl_3 > CCl_4$

SECTION – III
(PARAGRAPH TYPE)

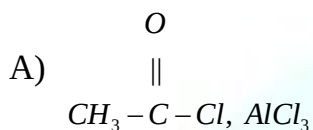
This section contains **2 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph for Questions 33 and 34

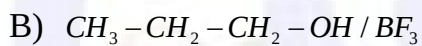
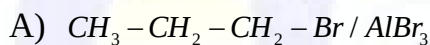
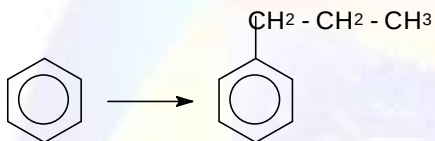
Alkylation and acylation of benzene is known as Friedel Craft's reaction. This reaction takes place in the presence of Lewis acid as a catalyst.

33. Benzene can be rapidly converted into acetophenone by which of these



D) All

34. The given conversion can be best achieved by which of these

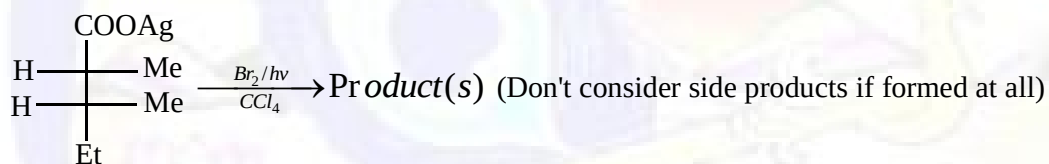
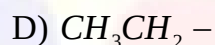
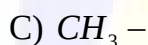
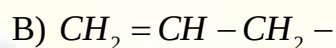
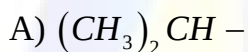


D) All

Paragraph for Questions 35 and 36

There are many ways of preparing alkyl halides from alcohols, acid salts. Alkyl halides are also prepared from alkenes, alkynes. Different alcohols may form different alkyl halides with different rates. Alkyl halides obtained through these methods may or may not be stereochemically active.

35. $ROH \xrightarrow[\text{Anhydrous } ZnCl_2]{\text{Conc. HCl}} R-Cl + H_2O$ in which case R-Cl formation shall be kinetically most favorable? Structurally 'R' is:



Relation between product(s) is/are:

A) Enantiomers

B) Diastereomers

C) Structural isomers

D) Homomers

MATHEMATICS**Max Marks: 62****SECTION – I
(SINGLE CORRECT ANSWER TYPE)**

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.

37. The number of points $(a+1, a)$, where a is an integer lying inside the region bounded by the circles $x^2 + y^2 - 2x - 1 = 0$ and $x^2 + y^2 - 2x - 17 = 0$, is
A) 2 B) 4 C) 1 D) 0
38. If a vertex of an equilateral triangle is $(2\sqrt{2}, -1)$ and its centroid is $(0, 0)$ then the radius of its incircle is
A) 3 B) $\frac{2}{3}$ C) $\frac{3}{2}$ D) $\frac{4}{3}$
39. The equation of a tangent to the circle $x^2 + y^2 - 6x + 4y - 3 = 0$ which is perpendicular to the line $y = 2x - 1$, is
A) $x + 2y + 1 = 0$ B) $x + 2y + (1 \pm 4\sqrt{5}) = 0$
C) $x + 2y \pm 4\sqrt{5} = 0$ D) $x + 2y + 3 = 0$

40. The chord of contact of the pair of tangents drawn from each point on the line $2x + y = 4$ to the circle $x^2 + y^2 = 1$ passes through the point
- A) $\left(\frac{1}{4}, \frac{1}{2}\right)$ B) $\left(4, \frac{1}{2}\right)$ C) $\left(\frac{1}{2}, 4\right)$ D) $\left(\frac{1}{2}, \frac{1}{4}\right)$
41. The area of a square circumscribing the circle $3(x^2 + y^2) - 6x + 8y = 0$ (in sq units)
- A) $\frac{91}{9}$ B) $\frac{100}{9}$ C) $\frac{89}{9}$ D) $\frac{121}{9}$
42. The line $2x - y + 1 = 0$ is tangent to the circle at the point $(2, 5)$ and the centre of the circle lies on $x - 2y = 4$. The radius of the circle is
- A) $3\sqrt{5}$ B) $5\sqrt{3}$ C) $2\sqrt{5}$ D) $5\sqrt{2}$

SECTION - II**(MULTIPLE CORRECT ANSWER TYPE)**

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -2 in all other cases

43. A regular hexagon is inscribed in the circle $x^2 + y^2 - 2(x + y + 1) = 0$ with one vertex at $(3, 1)$. The coordinates of the two vertices of the hexagon consecutive to the vertex $(3, 1)$ are
- A) $(2, 1 + \sqrt{3})$ B) $(1 + \sqrt{3}, 2)$ C) $(2, 1 - \sqrt{3})$ D) $(3, 2)$

44. P(1,1) is a point on a circle $x^2 + y^2 = 2$. If P travels on the circle and reaches Q such that $\text{arc PQ} = \frac{1}{12}(\text{circumference of the circle})$, then the coordinates of Q is/are
- A) $\left(\frac{\sqrt{3}-1}{2}, \frac{\sqrt{3}+1}{2}\right)$ B) $\left(\frac{3-\sqrt{3}}{2}, \frac{3+\sqrt{3}}{2}\right)$
C) $\left(\frac{\sqrt{3}+1}{2}, \frac{\sqrt{3}-1}{2}\right)$ D) $\left(\frac{3+\sqrt{3}}{2}, \frac{3-\sqrt{3}}{2}\right)$
45. Let x, y be variables satisfying the equation $x^2 + y^2 + 8x - 10y - 40 = 0$.
Let $a = \max\{(x+2)^2 + (y-3)^2\}$ and $b = \min\{(x+2)^2 + (y-3)^2\}$, then
A) $a + b = 18$ B) $a + b = \sqrt{2}$ C) $a - b = 4\sqrt{2}$ D) $ab = 73$
46. If α, β, γ are the parameters of points A, B, C on circle $x^2 + y^2 = a^2$ and if the ΔABC be equilateral, then
A) $\sum \cos \alpha = 0$ B) $\sum \sin \alpha = 0$ C) $\sum \tan \alpha = 0$ D) $\sum \cot \alpha = 0$
47. Equations of tangent (s) to the circle $x^2 + y^2 + 4x - 4y + 4 = 0$ which makes equal intercepts on the coordinates axes is/are
A) $x + y = 2\sqrt{2}$ B) $x - y = 2\sqrt{3}$ C) $x + y + 2\sqrt{2} = 0$ D) $x - y + 2\sqrt{3} = 0$

48. If the lines $3x - 4y + 4 = 0$ and $6x - 8y - 7 = 0$ are tangents to the same circle then
- A) radius of the circle is $\frac{3}{4}$
- B) radius of the circle is $\frac{3}{2}$
- C) centre of circle lies on $12x - 16y + 1 = 0$
- D) centre of circle lies on $12x - 16y + 31 = 0$
49. Circle (S) touching the x-axis at a distance of 3 units from the origin and having an intercept of length $2\sqrt{7}$ on the y-axis is (are)
- A) $x^2 + y^2 - 6x + 8y + 9 = 0$ B) $x^2 + y^2 - 6x + 7y + 9 = 0$
- C) $x^2 + y^2 - 6x - 8y + 9 = 0$ D) $x^2 + y^2 - 6x - 7y + 9 = 0$
50. Let RS be the diameter of the circle $x^2 + y^2 = 1$, where $S = (1, 0)$. Let P be a variable point (other than R and S) on the circle and tangents to the circle at 'S' and 'P' meet at the point 'Q'. The normal to the circle at 'P' intersects a line drawn through 'Q' parallel to 'RS' at a point E. Then the locus of 'E' passes through the point(S)
- A) $\left(\frac{1}{3}, \frac{1}{\sqrt{3}}\right)$ B) $\left(\frac{1}{4}, \frac{1}{2}\right)$ C) $\left(\frac{1}{3}, \frac{-1}{\sqrt{3}}\right)$ D) $\left(\frac{1}{4}, \frac{-1}{2}\right)$

SECTION – III
(PARAGRAPH TYPE)

This section contains **2 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph for Questions 51 and 52

Consider the two circles $c_1 : x^2 + y^2 = r_1^2$ and $c_2 : x^2 + y^2 = r_2^2$ ($r_2 < r_1$). Let $A(r_1, 0)$ be a fixed point on c_1 and B be a variable point on c_2 . The line BA meet the circle c_2 again at C .

Then

51. Maximum value of BC^2 is

A) $4r_1^2$

B) $4r_2^2$

C) $4(r_2 + r_1)(r_2 - r_1)$

D) $(r_2 + r_1)(r_2 - r_1)$

52. Locus of mid point of AB is

A) $\left(x + \frac{r_2}{2}\right)^2 + y^2 = \frac{r_1^2}{8}$

B) $\left(x - \frac{r_1}{2}\right)^2 + y^2 = \frac{r_2^2}{4}$

C) $\left(x - \frac{r_2}{2}\right)^2 + y^2 = \frac{r_1^2}{2}$

D) $(x - r_1)^2 + y^2 = 4r_2^2$

Paragraph for Questions 53 and 54

The circle 'C' of radius 1, touching the pair of lines $12x^2 - 25xy + 12y^2 = 0$ ($x > 0$)

53. The length of tangent from origin to circle 'C' is

- A) $\frac{1}{7}$ B) 9 C) $\sqrt{50}$ D) 7

54. The equation of circle 'C' is

A) $x^2 + y^2 + 10x + 10y + 49 = 0$

B) $x^2 + y^2 - 10(x + y) + 49 = 0$

C) $x^2 + y^2 + 10(x - y) + 29 = 0$

D) $x^2 + y^2 - 7(x + y) + 3 = 0$



Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr. Super60-I

Time: 07:30AM to 10:30AM

WAT-30

2016_P2

Date: 30-12-17

Max. Marks:186

KEY SHEET

PHYSICS

1	A	2	B	3	D	4	B	5	D
6	B	7	A	8	AD	9	AD	10	ABC
11	AD	12	ABCD	13	AC	14	BD	15	C
16	D	17	B	18	B				

CHEMISTRY

19	A	20	A	21	C	22	C	23	D
24	A	25	ABCD	26	ABD	27	B	28	ACD
29	CD	30	ABC	31	D	32	A	33	A
34	C	35	B	36	B				

MATHS

37	A	38	C	39	B	40	D	41	B
42	A	43	AC	44	AC	45	ACD	46	AB
47	AC	48	AC	49	AC	50	AC	51	B
52	B	53	B	54	B				

SOLUTIONS

PHYSICS

1. Work done in converting 1 g of ice at -10°C to steam at 100°C

= Heat supplied to raise temperature of 1g of ice from -10°C to 0°C ($m \times c_{\text{ice}} \times \Delta T$)

+ Heat supplied to convert 1 g ice into water at 0°C ($m \times L_{\text{ice}}$)

+ Heat supplied to raise temperature of 1 g of water from 0°C to 100°C ($m \times c_{\text{water}} \times \Delta T$)

+ Heat supplied to convert 1 g water into steam at 100°C [$m \times L_{\text{vapour}}$]

$$= [m \times c_{\text{ice}} \times \Delta T] + [m \times L_{\text{ice}}] + [m \times c_{\text{water}} \times \Delta T] + [m \times L_{\text{vapour}}]$$

$$[1 \times 0.5 \times 10] + [1 \times 80] + [1 \times 1 \times 100] + [1 \times 540] = 725 \text{ cal} = 725 \times 4.2 = 3045 \text{ J}$$

2. The effective value of α at a distance x from the left end is

$$\Delta L = \int_0^L \alpha_x dx \Delta t$$

$$\alpha_x = \alpha_1 + \left(\frac{\alpha_2 - \alpha_1}{L} \right) x \quad L = \left(\frac{\alpha_1 + \alpha_2}{2} \right) L \Delta T$$

$$\alpha_{\text{eff}} = \frac{\alpha_1 + \alpha_2}{2}$$

3. Since tension in the two rods will be same

$$A_1 Y_1 \alpha_1 \theta = A_2 Y_2 \alpha_2 \theta$$

$$A_1 Y_1 \alpha_1 = A_2 Y_2 \alpha_2$$

4. Expansion in mercury = $V_0 \gamma_m T$

$$\text{Expansion in glass bulb} = V_0 (3\alpha) T$$

$$\text{Apparent expansion in mercury} = V_0 \gamma_m T - V_0 (3\alpha) T$$

$$\therefore \Delta l = \frac{V_0 \gamma_m T - V_0 3\alpha_g T}{A_0 (1 + 2\alpha_g T)}$$

Length of mercury column in capillary is

$$h = \Delta l = \frac{V_0 T (\gamma_m - 3\alpha_g)}{A_0 (1 + 2\alpha_g T)}$$

5. Slope of line A = $\frac{(1006 - 100) \text{ mm}}{T^{\circ}\text{C}} = \frac{\Delta L}{\Delta T} = L \alpha_A$

$$\text{i.e., } \frac{6}{T} \text{ mm}^{\circ}\text{C} = (1000 \text{ mm}) \alpha_A \quad (\text{i})$$

Similarly, for line B,

$$\frac{2}{T} \text{ mm / } ^\circ\text{C} = (1002 \text{ mm}) \alpha_B \quad (\text{ii})$$

Dividing Eq.(i) by Eq.(ii),

$$3 = \frac{100\alpha_A}{1002\alpha_B} \approx \alpha_A = 3\alpha_B \quad (\text{iii})$$

From E.q. (iii), $\alpha_A = 3 \times 9 \times 10^{-6} = 27 \times 10^{-6} / ^\circ\text{C}$

$$\begin{aligned} \text{From E.q. (i). } T &= \frac{6}{1000\alpha_A} = \frac{6 \times 10^6}{1000 \times 27} \\ &= 222.22^\circ\text{C} \end{aligned}$$

6. Heat required by ice to convert in 100°C water will be $Q_i = 1 \times 80 + 1 \times 100 = 180 \text{ cal}$. Steam will supply 180 cal and convert in water of 100°C . Temperature of mixture will be 100°C . Whole of the steam will not be condensed.

7. $L_2 = L_1(1 + \alpha \Delta T) \quad 5.050 = 5(1 + 24 \times 10^{-6} \times \Delta T) \Rightarrow 437^\circ\text{C}$

For options (b) and (d), temperature below -273°C is required which is unattainable.

For option (c), temperature of 3000°C is required which is above melting point of aluminium.

8. As $\gamma_{\text{Hg}} > \gamma_{\text{Fe}}$, density of mercury decreased more on heating so iron cube will sink more on heating. Also, fraction of volume submerged $= \frac{V_{\text{out}}}{V_{\text{cube}}}$

Fractional change on heating is given by

$$\frac{(V_0 / V_0)_{25^\circ\text{C}} - (V_0 / V_0)_{0^\circ\text{C}}}{(V_0 / V_c)_{0^\circ\text{C}}} = \frac{(1 + 180 \times 10^{-6})}{(1 + 35 \times 10^{-6}) \times 25} = 3.6 \times 10^{-3}$$

%Change %Change = $0.36\% \approx 0.40\%$

9. OB shows fusion and CD shows vaporisation

10. When A and B are mixed $m_A(13 - 10) = m_B(15 - 13)$

$$3s_A = 2s_B \Rightarrow \frac{s_A}{s_B} = \frac{2}{3}$$

When B and C are mixed, $s_B \times 1 = s_C \times 4$

$$\Rightarrow \frac{s_B}{s_C} = 4 \Rightarrow \frac{3s_A}{4s_C} = 4 \Rightarrow \frac{s_A}{s_B} = \frac{8}{3}$$

When A and C are mixed,

$$s_A (\theta - 10)$$

$$= s_B (20 - \theta) \Rightarrow \theta = \frac{140}{11} ^\circ \text{C}$$

11. Force on steel = Force on copper

$$\sigma_{\text{steel}} A_{\text{steel}} + \sigma_{\text{cu}} A_{\text{cu}} = 0$$

$$\sigma_{\text{steel}} \cdot \frac{\pi}{4} d_{\text{steel}}^2 = -\sigma_{\text{cu}} \frac{\pi}{4} (d_2^2 - d_1^2)$$

$$\sigma_{\text{cu}} = \frac{-\left(\frac{d}{2}\right)^2}{d^2 - \left(\frac{d}{2}\right)^2} \cdot \sigma_{\text{steel}} \Rightarrow \sigma_{\text{cu}} = \frac{-\left(\frac{1}{4}\right)}{1 - \left(\frac{1}{4}\right)} \cdot \sigma_{\text{steel}}$$

$$\sigma_{\text{cu}} = -\frac{1}{3} \sigma_{\text{steel}}$$

$$\text{Also, strain in copper tube} = \frac{\sigma_{\text{cu}}}{\gamma_{\text{cu}}} + \alpha_{\text{cu}} \Delta T$$

12. conceptual

13. Heat received by ice = $mL + mC\Delta T = 10700 \text{ cal}$

Heat given by container, then

$$= -\int_{500}^{300} m_c (A + BT) dT = -m_c \left[AT + \frac{BT^2}{2} \right]_{500}^{300} = 21600 m_c$$

Heat lost = heat gained

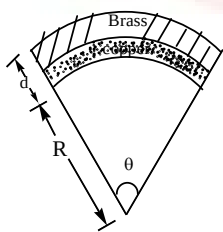
$$\therefore 10700 = 21600 m_c$$

$$\Rightarrow m_c = \frac{10700}{21600} \approx 0.495 \approx 0.5 \text{ kg}$$

Also heat lost by container $\approx 0.5 \times 21600 \approx 10800 \text{ cal}$

14. Let L_0 is length of each strip before heating

Length after heating will be



$$L_{\text{brass}} = L_0 (1 + \alpha_{\text{brass}} \Delta T)$$

And

$$L_{\text{copper}} = L_0(1 + \alpha_{\text{copper}} \Delta T) = RQ$$

$$\Rightarrow \frac{R+d}{R} = \frac{1 + \alpha_{\text{brass}} \Delta T}{1 + \alpha_{\text{copper}} \Delta T}$$

$$\Rightarrow R = \frac{d}{(\alpha_{\text{brass}} - \alpha_{\text{copper}}) \Delta T}$$

$$\therefore R \propto \frac{1}{|\alpha_{\text{brass}} - \alpha_{\text{copper}}|}$$

15. In portion A, C or E heat absorbed $Q = mC\Delta T$

Where C = specific heat

$$\Rightarrow \Delta T = \frac{Q}{mC} \quad \text{Which is comparable to } y = mx$$

$$\text{Hence, slope of line} = \frac{1}{mC} = \frac{1}{\text{Heat capacity}}$$

16. Ratio of L_v and $L_1 = \frac{mL_1}{mL_2} = \frac{2260}{333}$

17. In 7 min, temperature of 100 g of water is raised by $(1000 - 16^\circ\text{C}) = 84^\circ\text{C}$. The amount of

the heat provided by heater $Q_w = C_w m_w \Delta T = (1 \text{ cal/g}^\circ\text{C})(100 \text{ g})(84^\circ\text{C})$

$$= 8.4 \times 10^3 \text{ cal} = (8.4 \times 10^3 \times 4.186) \text{ J}$$

$$= 3.5 \times 10^4 \text{ J}$$

$$= \frac{Q_w}{t_1}$$

$$\text{Power of heater} = \frac{8.4 \times 10^3 \text{ cal}}{(7 \times 60) \text{ s}}$$

$$= (20 \times 4.18) \text{ J/s} = 83.6 \text{ W} \approx 84 \text{ W}$$

18. With 200g of alcohol in the vessel, the temperature rises from 16°C to 78°C in 6 min 12 s. Quantity of heat absorbed by alcohol

$$QA_a = C_a m_a (\theta_f - \theta_i)$$

$$= C_a \times (200 \text{ g})(78 - 16)^\circ\text{C}$$

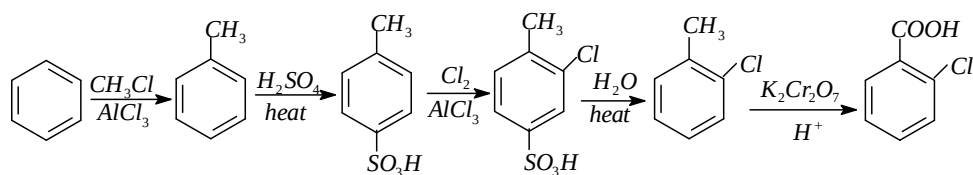
Specific heat of alcohol

$$C_a = \frac{20 \times 372}{200 \times 62} = 0.6 \text{ cal.g}^\circ\text{C}$$

$$= \frac{0.6 \times 4.186 \text{ J}}{10^{-3} \text{ kg}^\circ\text{C}} = 2.5 \times 10^3 \text{ J/kg}^\circ\text{C}$$

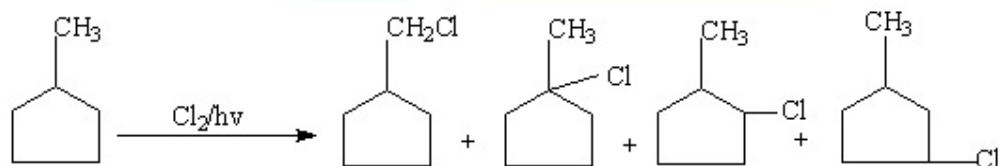
$$= 2.5 \text{ K/g}^\circ\text{C}$$

CHEMISTRY

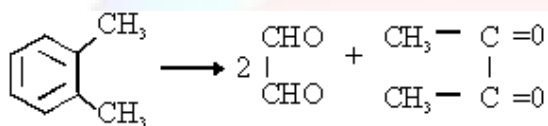


19.

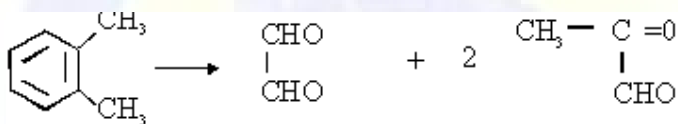
22. 1,4 gives one nitro product



25.



28.



31.

33. Acylium carbocation is formed.

34. Friedel – Craft's acylation and Clemmenson's reduction.

MATHS

37. Sol: $(a+1)^2 + a^2 - 2(a+1) - 1 > 0 \rightarrow a < -1 \text{ or } a > 1 \text{ --- (1)}$

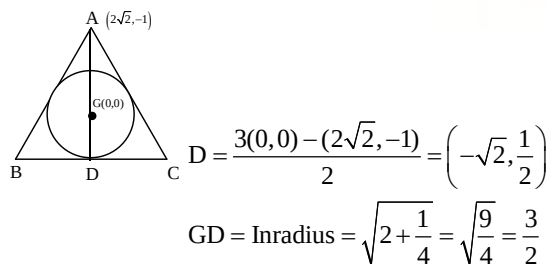
$$(a+1)^2 + a^2 - 2(a+1) - 17 < 0 \rightarrow a^2 - 9 < 0$$

$$\rightarrow a \in (-3, 3) \text{ --- (2)}$$

$$\text{From (1), (2)} \Rightarrow a \in (-3, -1) \cup (1, 3)$$

$$\Rightarrow a = -2, 2$$

38. Sol:



39. Sol: $x + 2y + k = 0$ is tangent to circle $C = (3, -2)$

$$r = \sqrt{9 + 4 + 3} = \sqrt{16} = 4$$

$$4 = \left| \frac{3 - 4 + K}{\sqrt{5}} \right| \Rightarrow |K - 1| = 4\sqrt{5}$$

$$\Rightarrow k = (1 \pm 4\sqrt{5})$$

$$\therefore \text{Reqd equation is } x + 2y + (1 \pm 4\sqrt{5}) = 0$$

40. Sol: Let $(\alpha, 4 - 2\alpha)$ be a point on the line

$$\text{Equation of chord of contact is } x(\alpha) + y(4 - 2\alpha) = 1$$

$$\Rightarrow \alpha(x - 2y) + (4y - 1) = 0$$

$$\Rightarrow \text{P O I is } x - 2y = 0, 4y - 1 = 0$$

$$y = \frac{1}{4}, x = 2y = 2\left(\frac{1}{4}\right) = \frac{1}{2}$$

$$\Rightarrow (x, y) = \left(\frac{1}{2}, \frac{1}{4}\right)$$

41. Sol: $\text{rad} = \frac{5}{3}$, diameter = side of square

$$\text{Area} = \left(\frac{10}{3}\right)^2 = \frac{100}{9}$$

42. Sol: Eqn of diameter is $x - 2y = 4 - (1)$

$$\text{Eqn of another diameter is } x + 2y = 2 + 2(5)$$

$$x + 2y = 10 - (2)$$

$$\text{P O I of } (1), (2) \text{ is centre } = (2, 2)$$

$$\text{rad} = \sqrt{(8 - 2)^2 + (2 - 5)^2} = \sqrt{36 + 9} = \sqrt{45} = 3\sqrt{5}$$

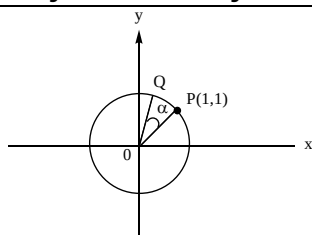
43. Sol: Centre = $(1, 1)$, vertex, $A_1 = (3, 1)$, $\text{rad} = 2$

CA_1 is parallel to x-axis

$$A_2, A_3 = (1 + 2\cos 60^\circ, 1 \pm 2\sin 60^\circ)$$

$$= \left(1 + 2\left(\frac{1}{2}\right), 1 \pm 2\left(\frac{\sqrt{3}}{2}\right) \right) = (1 + 1, 1 \pm \sqrt{3})$$

$$= (2, 1 \pm \sqrt{3})$$



44.

$$\text{Sol: } PQ = l = \frac{1}{12} (2\pi\sqrt{2}) = \frac{\pi\sqrt{2}}{6}$$

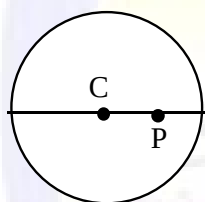
$$l = r\alpha \Rightarrow \alpha = \frac{1}{\sqrt{2}} \left(\frac{\pi\sqrt{2}}{6} \right) = \frac{\pi}{6}$$

$$\angle XOQ = (45^\circ \pm 30^\circ) = 75^\circ \text{ or } 15^\circ$$

$$Q = [\sqrt{2}\cos 75^\circ, \sqrt{2}\sin 75^\circ] \text{ or } [\sqrt{2}\cos 15^\circ, \sqrt{2}\sin 15^\circ]$$

45.

Sol:



$$c = (-4, 5) \quad cp = \sqrt{4+4} = 2\sqrt{2}$$

$$p = (-2, 3) \quad a = cp + r = 2\sqrt{2} + 9$$

$$r = 9 \quad b = -cp + r = 9 - 2\sqrt{2}$$

$$46. \text{ sol: } A = (a\cos\alpha, a\sin\alpha)$$

$$B = (a\cos\beta, a\sin\beta)$$

$$C = (a\cos\gamma, a\sin\gamma)$$

$$S = G = \left[\frac{a}{3} (\cos\alpha + \cos\beta + \sin\gamma), \frac{a}{3} (\sin\alpha + \sin\beta + \sin\gamma) \right] = (0, 0)$$

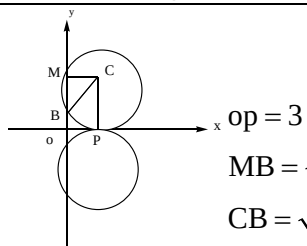
$$\sum \cos\alpha = 0, \sum \sin\alpha = 0$$

$$47. \text{ Sol: } x+y=a \text{ is eqn of tangent } r=a \Rightarrow a = \pm 2\sqrt{2}$$

$$48. \text{ sol: diameter of circle } \frac{15}{10} = \frac{3}{2}$$

$$\text{Line equidistant to given is } 6x - 8y + \left(\frac{8-7}{2} \right) = 0$$

$$12x - 16y + 1 = 0$$



49. sol:

$$OP = 3$$

$$MB = \sqrt{7}$$

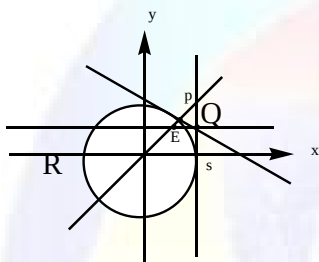
$$CB = \sqrt{9+7} = 4$$

$$C = (3, \pm 4)$$

$$r = 4$$

$$(x-3)^2 + (y \pm 4)^2 = 4^2$$

50. sol:



$$P = (\cos \theta, \sin \theta)$$

$$\rightarrow \text{Tangent at 'p' is } x \cos \theta + y \sin \theta = 1$$

$$\rightarrow Q = \left(1, \frac{1 - \cos \theta}{\sin \theta}\right)$$

$$\rightarrow \text{op Eqn is } y = (\tan \theta)x$$

$$E = \left(\frac{1 - \cos \theta}{\sin \theta \tan \theta}, \frac{1 - \cos \theta}{\sin \theta}\right)$$

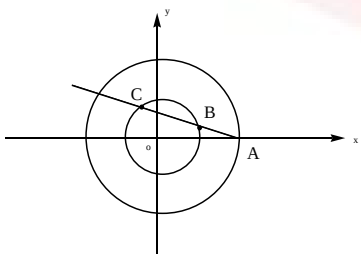
$$= (h, k) = \text{locus} \Rightarrow h = \frac{\tan \frac{\theta}{2}}{\tan \theta}, k = \tan \theta \rightarrow 2xy = y(1 - y^2)$$

Sec-3

$$\text{Sol: } BC^2 = (AC - AB)^2 = (AC + AB)^2 - 2AC \cdot AB$$

$$\max(BC)^2 = 4r_2^2 \Rightarrow BC \text{ becomes diameter of } c_2.$$

51,52. sol:



$$A = (r_1, \theta)$$

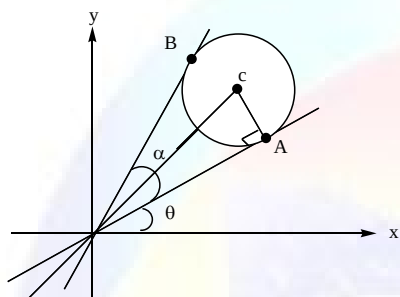
$$B = (h, k) \Rightarrow \text{mid point of } AB = \left(\frac{h+r_1}{2}, \frac{k}{2} \right) = (x, y)$$

$$h = 2x - r_1, k = 2y$$

$$h^2 + k^2 = r_2^2 \Rightarrow (2x - r_1)^2 + (2y)^2 = r_2^2$$

$$\Rightarrow \left(x - \frac{r_1}{2} \right)^2 + y^2 = \frac{r_2^2}{4}$$

53,54.sol:



α = angle b/w pair of lines

$$\tan \alpha = \frac{2\sqrt{\frac{625}{4} - 144}}{24} = \frac{7}{24} \Rightarrow \cos \alpha = \frac{24}{25}$$

Pair of lines are $(3x - 4y)(4x - 3y) = 0$ Slopes of OA, OB are $\frac{3}{4}, \frac{4}{3}$

$$= \frac{1}{7}$$

$$\tan \frac{\alpha}{2} = \sqrt{\frac{1 + \cos \alpha}{1 - \cos \alpha}} = \sqrt{\frac{1 - \frac{24}{25}}{1 + \frac{24}{25}}} = \frac{\frac{3}{4} + \frac{1}{7}}{1 - \frac{3}{4} - \frac{1}{7}} = 1$$

$$\angle XOC = 45^\circ$$

$$\tan \frac{\alpha}{2} = \frac{AC}{OA} = \frac{1}{OA} = \frac{1}{7}$$

OA = 7 = length of tan gent from (0,0) to circle

$$\text{centre} = \left[0 + \sqrt{50} \cos \frac{\pi}{4}, 0 + \sqrt{50} \sin \frac{\pi}{4} \right]$$

$$= [5, 5]$$

$$\text{rad} = 1$$

Eqn of circle is

$$(x - 5)^2 + (y - 5)^2 = 1$$

Success



JEE ADVANCED WEEKEND TEST



Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr. Super60-I

Time: 07:30 AM to 10:30 AM

Date 06-01-18

Max. Marks: 240

Name of the Student: _____

H.T. NO:

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2011_PAPER-I

IMPORTANT INSTRUCTIONS:

- 1) This booklet is your Question Paper.
- 2) Use the Optical Response Sheet (ORS) provided separately for answering the questions
- 3) Blank spaces are provided within this booklet for rough work.
- 4) Write your name, roll number and sign in the space provided on the back cover of this booklet.
- 5) You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET:

- 6) Darken the appropriate bubbles on the ORS by applying sufficient pressure. This will leave an impression at the corresponding place on the Candidate's sheet.
- 7) The ORS will be collected by the invigilator at the end of the examination.
- 8) Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
- 9) Write your name, roll number and code of the examination center, and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else** on the ORS. Darken the appropriate bubble under each digit of your roll number.

DARKENING THE BUBBLES ON THE ORS

- 10) Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
- 11) Darken the bubble **COMPLETELY**.
- 12) The correct way of darkening a bubble is as:
- 13) The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- 14) Darken the bubbles **ONLY** IF you are sure of the answer. There is **NO WAY** to erase or

"un-darken" a darkened bubble.

06-01-18_Jr.Super60-I_Jee Adv_WAT-31_Weekend Syllabus

MATHS: SYLLABUS COVERED FROM 01-01-18 to 05-01-18

PHYSICS: SYLLABUS COVERED FROM 01-01-18 to 05-01-2018

CHEMISTRY: Preparation and Reactions of alkyl halides, Mechanism of Nucleophilic Substitution reactions (SN1, SN2, SNi), Aryl halides

JEE-ADVANCE-2011-P1-Model

Time: 3:00 Hour's

IMPORTANT INSTRUCTIONS

Max Marks: 240

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 7)	Questions with Single Correct Choice	3	-1	7	21
Sec – II(Q.N : 8 – 11)	Questions with Multiple Correct Choice	4	0	4	16
Sec – III(Q.N : 12 – 16)	Questions with Comprehension Type (2 Comprehensions – 2 + 3 = 5Q)	3	-1	5	15
Sec – IV(Q.N : 17 – 23)	Questions with Integer Answer Type	4	0	7	28
Total				23	80

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 24 – 30)	Questions with Single Correct Choice	3	-1	7	21
Sec – II(Q.N : 31 – 34)	Questions with Multiple Correct Choice	4	0	4	16
Sec – III(Q.N : 35 – 39)	Questions with Comprehension Type (2 Comprehensions – 2 + 3 = 5Q)	3	-1	5	15
Sec – IV(Q.N : 40 – 46)	Questions with Integer Answer Type	4	0	7	28
Total				23	80

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 47 – 53)	Questions with Single Correct Choice	3	-1	7	21
Sec – II(Q.N : 54 – 57)	Questions with Multiple Correct Choice	4	0	4	16
Sec – III(Q.N : 58 – 62)	Questions with Comprehension Type (2 Comprehensions – 2 + 3 = 5Q)	3	-1	5	15
Sec – IV(Q.N : 63 – 69)	Questions with Integer Answer Type	4	0	7	28
Total				23	80

Sec: Jr. Super60-I

space for rough work

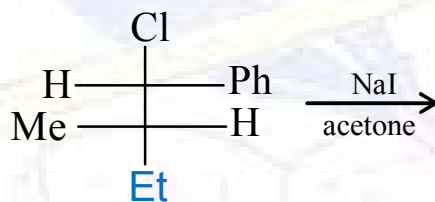
Page 2

SECTION – I**(SINGLE CORRECT CHOICE TYPE)**

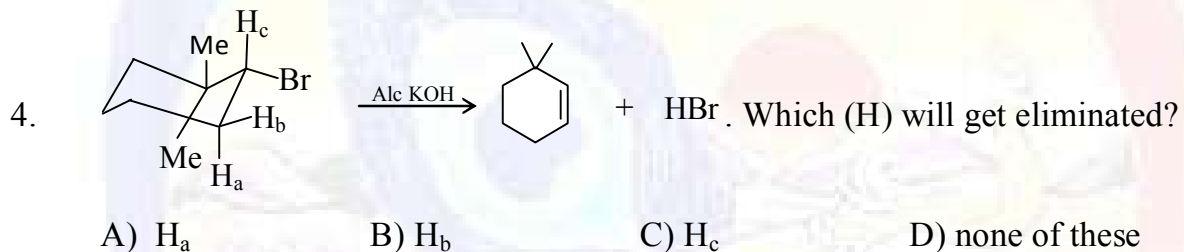
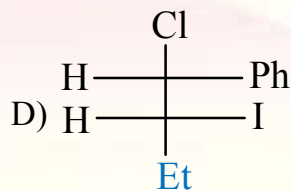
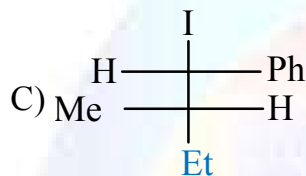
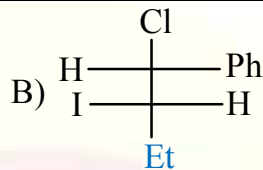
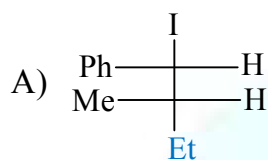
This section contains 7 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which ONLY ONE is correct

1. Which statement best explains the partial racemization that occurs in SN^1 reactions.
- A) The alkyl halide reacts as an intimate ion pair or a solvent-separated ion pair, and the carbocation does not become fully planar but re-attains much of its original configuration.
- B) The partial racemization is the result of contribution by a competing E^2 reaction
- C) The alkyl halide reacts as an intimate ion pair or a solvent-separated ion pair, and one face of the planar carbocation is partially blocked by the leaving group.
- D) The SN^1 reaction yields complete racemization, but afterwards one of the enantiomers is interconverted into the other more stable enantiomer.

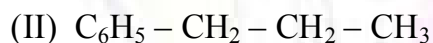
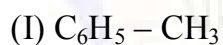
2. How does a change to a more polar solvent affect the reaction coordinate diagram for an SN^2 reaction of an alkyl halide with a negatively charged nucleophile?
- A) The energy of the reactants is increased, and the energy of the transition state remains about the same.
- B) The energy of the reactants is decreased slightly, and the energy of transition state is decreased significantly more.
- C) The energy of reactants is decreased, and the energy of the transition state is decreased significantly less.
- D) The energy of the reactants and the energy of the transition state increase by about the same amount.
3. In the following reaction,

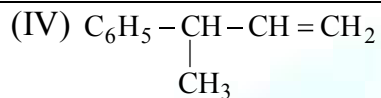


The structure of chief product is



5. Arrange the following compounds in the decreasing order of reactivity with NBS (N-bromosuccinimide).





Select the correct answer from the following

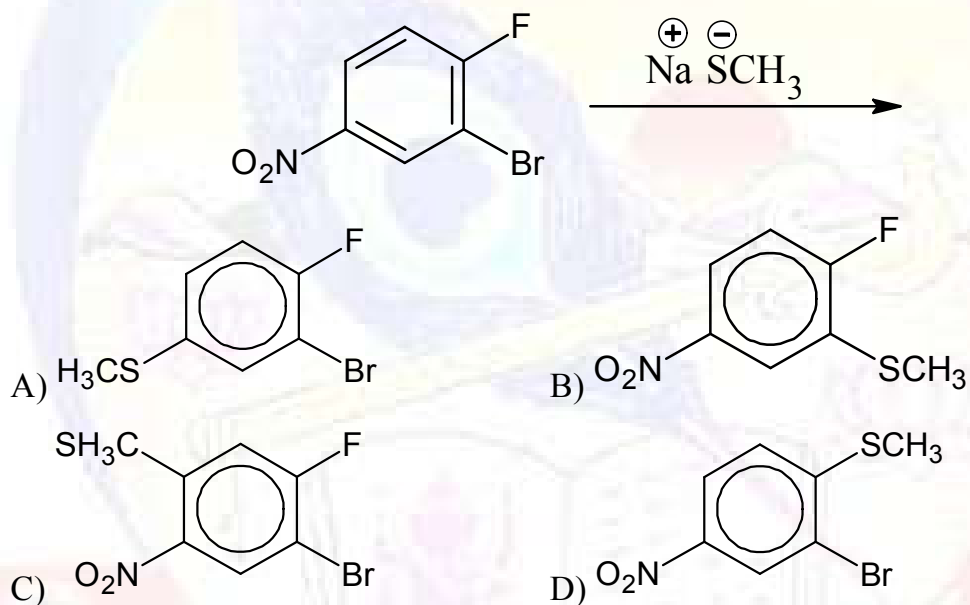
A) (IV) > (III) > (I) > (II)

B) (IV) > (III) > (II) > (I)

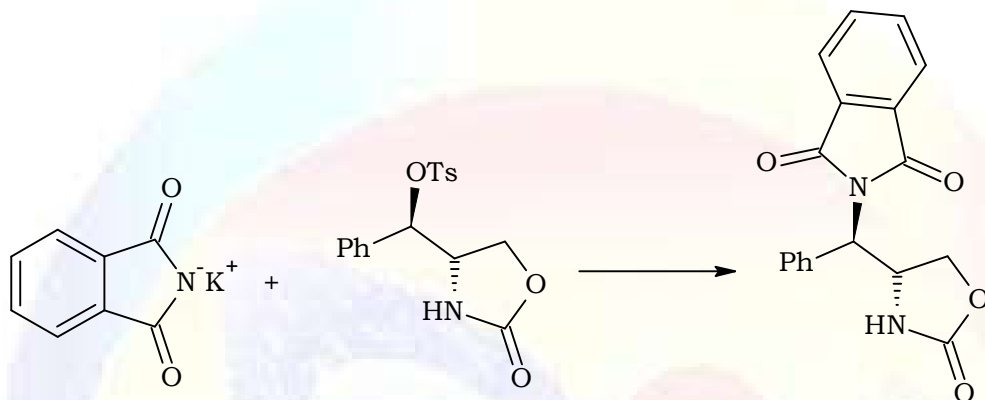
C) (I) > (II) > (III) > (IV)

D) (I) > (III) > (II) > (IV)

6. Identify the principal organic product of the following reaction.



7. Reason for retention of configuration in the following reaction is

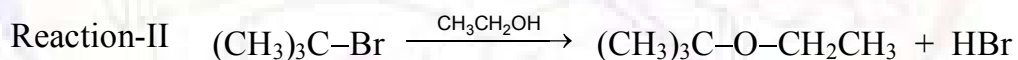
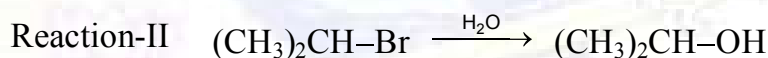
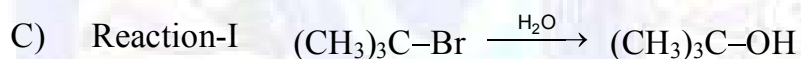
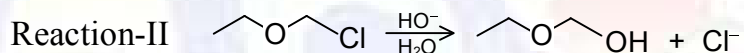
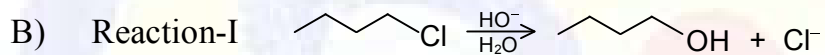
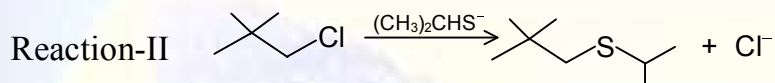
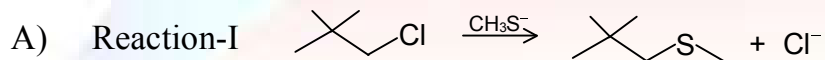


- A) OTs is an excellent leaving group.
- B) Neighboring group participation by Ph group.
- C) Neighboring group participation by NH group.
- D) The reaction involves S_Ni mechanism.

SECTION – II**(MULTIPLE CORRECT CHOICE TYPE)**

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which ONE OR MORE is/ are correct

8. Which reaction in each of the following pairs, **Reaction-I** will take place more rapidly than **Reaction-II**?



9. The rate law for the substitution reaction of 2-bromobutane and OH^- in

75% ethanol – 25% water at 30°C is

$$\text{Rate} = 3.2 \times 10^{-5} [\text{2-bromobutane}] [\text{OH}^-] + 1.5 \times 10^{-6} [\text{2-bromobutane}]$$

Which of the following statement(s) is/ are true?

A) The percent of the reaction takes place by the $\text{S}_{\text{N}}2$ mechanism when

$[\text{OH}^-] = 1.0 \text{ M}$ is 95.5 approximately

B) The percent of the reaction takes place by the $\text{S}_{\text{N}}1$ mechanism when

$[\text{OH}^-] = 1.0 \text{ M}$ is 95.5 approximately

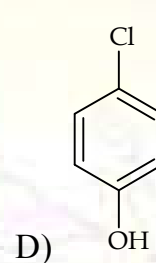
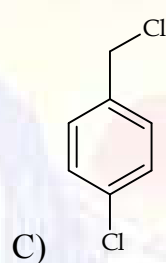
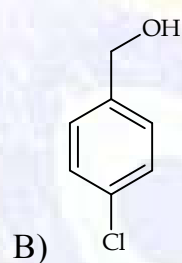
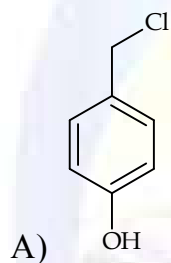
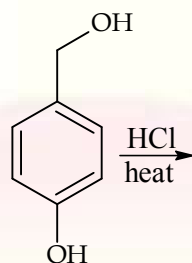
C) The percent of the reaction takes place by the $\text{S}_{\text{N}}1$ mechanism when

$[\text{OH}^-] = 0.001 \text{ M}$ is 98 approximately

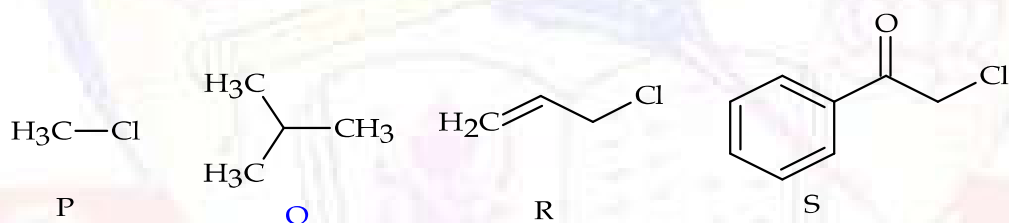
D) The percent of the reaction takes place by the $\text{S}_{\text{N}}2$ mechanism when

$[\text{OH}^-] = 0.001 \text{ M}$ is 98 approximately

10. Identify correct product of the given reaction:



11. KI in acetone, undergoes $\text{S}_\text{N}2$ reaction with each of P, Q, R and S. The rate reaction vary as



A) $\text{P} > \text{Q} > \text{R} > \text{S}$

B) $\text{S} > \text{P} > \text{R} > \text{Q}$

C) $\text{P} > \text{R} > \text{Q} > \text{S}$

D) $\text{R} > \text{P} > \text{S} > \text{Q}$

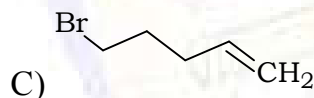
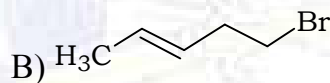
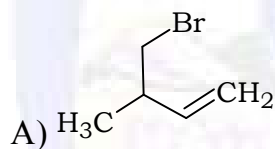
SECTION – III
(COMPREHENSION TYPE)

This section contains 2 groups of questions. One group has 3 questions and another group has 2 multiple choice questions based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which ONLY ONE is correct.

Paragraph for Questions 12 and 14

An organic compound **A** has molecular formula C_5H_9Br . **A** decolourises brown colour of bromine water but does not rotate plane polarized light. **A** on treatment with $HBr/(PhCO)_2O_2$ from $C_5H_{10}Br_2$ which on further treatment with Na/R_2O gives a cyclic compound which on dibromination gives seven isomeric products.

12. The starting compound **A** is most likely to be



13. Compound **A** on treatment with HBr will produce

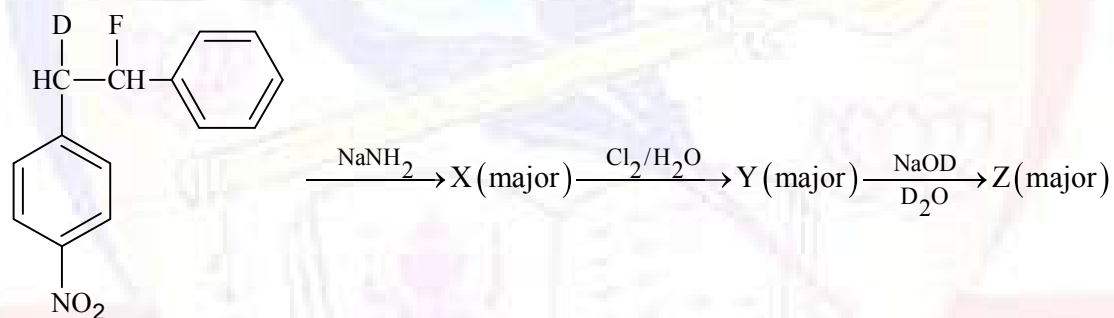
- A) An achiral dibromide B) A racemic mixture
C) A single pure enantiomer D) A meso dibromide

14. The true statement regarding the monochloro derivative product formed in the reaction below is

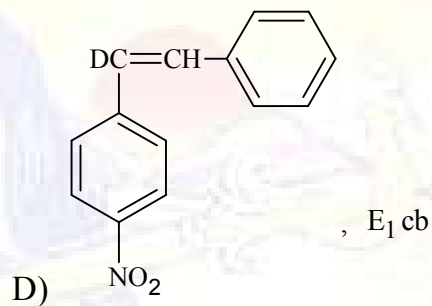
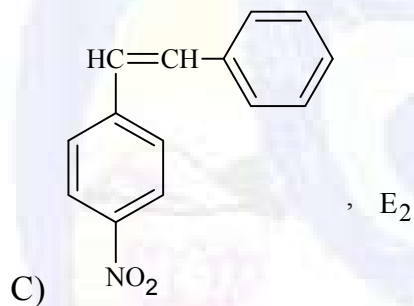
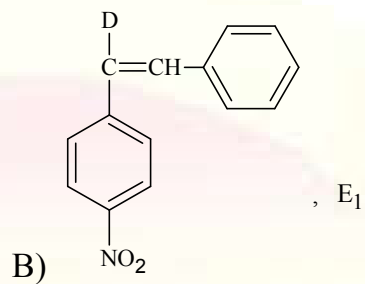
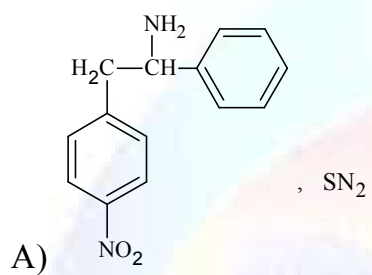


- A) Three position isomers are formed
 B) Total five isomers are produced
 C) Two pair of enantiomers are formed
 D) A pair of diastereomers are formed

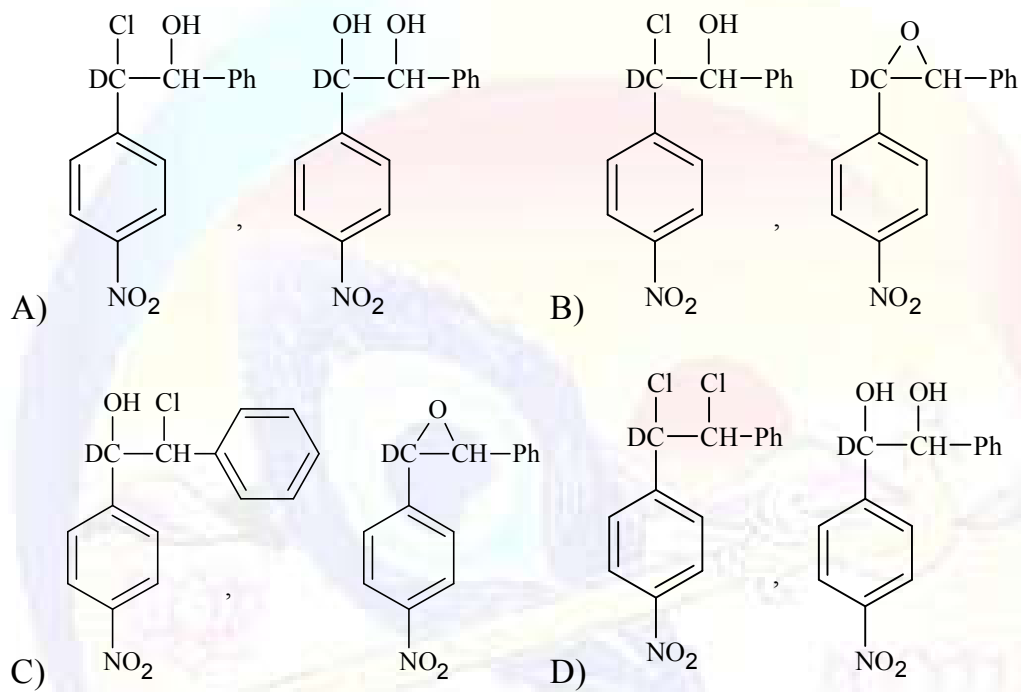
Paragraph for Questions 15 and 16



15. The product (X) and the most possible mechanism of its formation are



16. Product (Y) and (Z) are

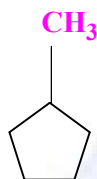


SECTION –IV**(INTEGER ANSWER TYPE)**

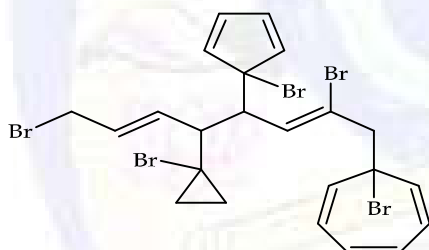
This section contains 7 questions. The answer to each of the questions is a single digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened.

17. In how many of the following pairs the former is more reactive than later with sodium methoxide in methanol at 50°C?
- (i) Chlorobenzene and o-chloronitrobenzene
 - (ii) p-chloronitrobenzene and m-chloronitrobenzene
 - (iii) 4-chloro-3-nitroacetophenone and 4-chloro-3-nitrotoluene
 - (iv) 2-fluoro-1,3-dinitrobenzene and 2-chloro-1,3-dinitrobenzene
 - (v) 4-chloronitrosobenzene and 4-chloroanisole
 - (vi) 2-iodo-1-(trifluoromethyl)benzene and 2-fluoro-1-(trifluoromethyl)benzene
18. Number of possible monochloro substituted products obtained from the reaction, of $(CH_3)_2CH-CH(CH_3)_2$ is,
19. A chloroderivative (X) on treatment with Zn-Cu couple in ethanol gives a hydrocarbon (Y). When (X) is dissolved in ether and treated with sodium, 2, 2, 5, 5 - tetramethylhexane is obtained. Identify how many different types of hydrogen atoms are present in Y.

20. When 3,3 dimethyl 2- butanol is treated with HBr the number of 3^0 carbons present in the major product formed is.



21. If mono chlorinated through free radical substitution reaction, the total number of possible isomers (Including stereo isomers) that can be formed for the above compound is (X) which on fractional distillation gives (y) fractions. Find (x-y).
22. How many moles of AgNO_3 will react with one mole of given compound?

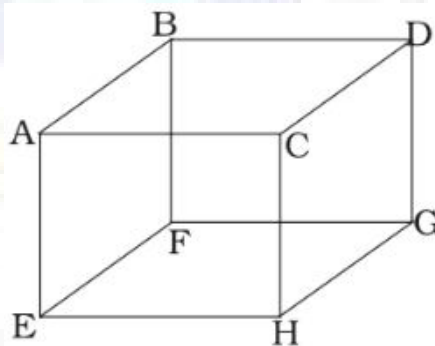


23. How many of the following are less reactive than $\text{CH}_3\text{CH}_2\text{I}$ in a S_N^2 reaction?
- | | | |
|-------------------------------|---|---|
| (i) MeI | (ii) $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ | (iii) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl}$ |
| (iv) Me_2CHI | (v) $\text{Me}_2\text{CHCH}_2\text{I}$ | (vi) $\text{CH}_2 = \text{CH}-\text{CH}_2\text{I}$ |
| (vii) PhCH_2I | | |

SECTION – I**(SINGLE CORRECT CHOICE TYPE)**

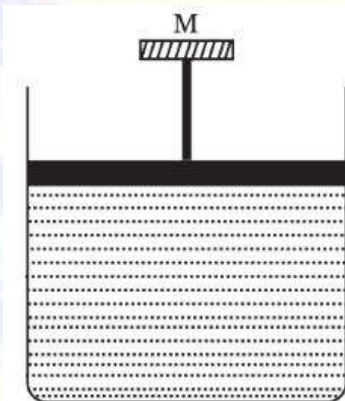
This section contains 7 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

24. 50 calories of heat is required to raise the temperature of 2 mole of an ideal gas at constant pressure from 30°C to 35°C . The amount of heat required to raise the temperature of the same gas through the same range at constant volume is
A) 30 calories B) 50 calories C) 70 calories D) 90 calories
25. 1 mole of an ideal gas is contained in a cubical volume V , ABCDEFGH at 300 K. One face of the cube (EFGH) is made up of a material which totally adsorbs any gas molecule incident on it at an instant. At this given instant of time,



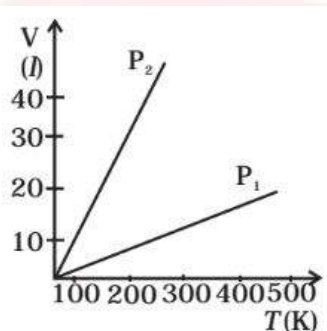
- A) The pressure on EFGH would be zero.
- B) the pressure on all the faces will be equal.
- C) the pressure of EFGH would be double the pressure on ABCD.
- D) the pressure on EFGH would be half that on ABCD.

26. A cylinder containing an ideal gas is in vertical position and has a piston of mass M that is able to move up or down without friction. If the temperature is increased slowly



- A) both p and V of the gas will change.
- B) only p will increase according to Charles's law.
- C) V will change but not p .
- D) p will change but not V .

27. Volume versus temperature graphs for a given mass of an ideal gas are shown in figure at two different values of constant pressure. What can be inferred about relation between P_1 & P_2 ?



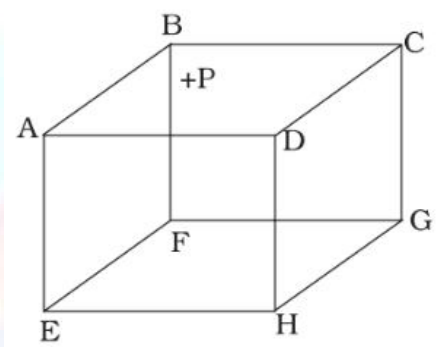
- A) $P_1 > P_2$ B) $P_1 = P_2$
C) $P_1 < P_2$ D) data is insufficient.
28. 1 mole of H_2 gas is contained in a box of volume $V = 1.00 \text{ m}^3$ at $T = 300\text{K}$. The gas is heated to a temperature of $T = 3000\text{K}$ and the gas gets converted to a gas of hydrogen atoms. The final pressure would be (considering all gases to be ideal)
- A) same as the pressure initially. B) 2 times the pressure initially.
C) 10 times the pressure initially. D) 20 times the pressure initially.

29. An inflated rubber balloon contains one mole of an ideal gas, has a pressure p , volume V and temperature T . If the temperature rises to $1.1 T$, and the volume is increase to $1.05 V$, the final pressure will be
- A) $1.1 p$ B) p
- C) less than p D) between p and $1.1p$
30. Boyle's law is applicable for an
- A) adiabatic process. B) isothermal process.
- C) isobaric process. D) isochoric process

(MULTIPLE CORRECT CHOICE TYPE)

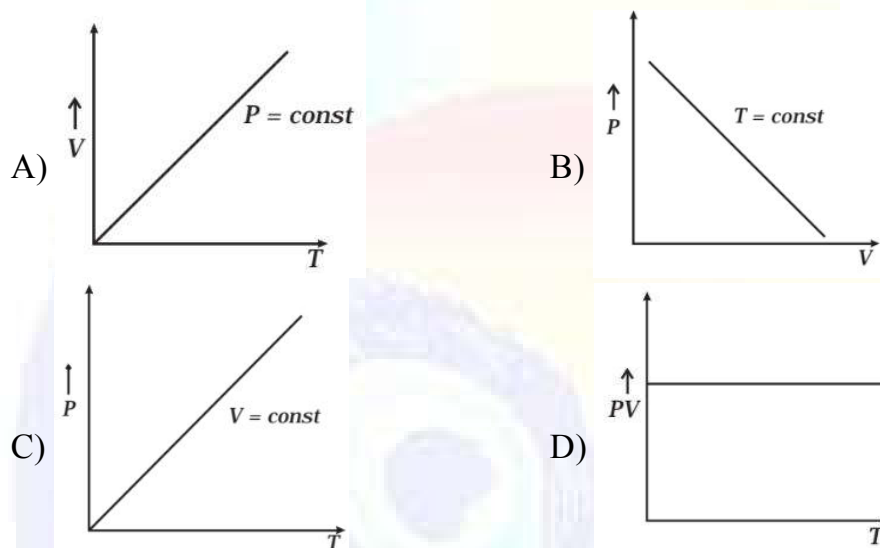
This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which ONE OR MORE is/ are correct

31. In the figure shown, the cube contains ionized hydrogen. The usual kinetic theory expression for pressure



- A) will be valid.
- B) will not be valid since the ions would experience forces other than due to collisions with the walls.
- C) will not be valid since collisions with walls would not be elastic.
- D) will not be valid because isotropy is lost.
32. In a diatomic molecule, the rotational energy at a given temperature
- A) obeys Maxwell's distribution.
- B) have the same value for all molecules.
- C) equals the translational kinetic energy for each molecule.
- D) is $(2/3)$ rd the translational kinetic energy for each molecule.

33. Which of the following diagrams depicts ideal gas behaviour?



34. Diatomic molecules like hydrogen have energies due to both translational as well as rotational motion. From the equation in kinetic theory $pV = \frac{2}{3}E$, E is

- A) the total energy per unit volume.
- B) only the translational part of energy because rotational energy is very small
- C) only the translational part of the energy because during collisions with the wall pressure relates to change in linear momentum.
- D) the translational part of the energy because rotational energies of molecules can be of either sign and its average over all the molecules is zero.

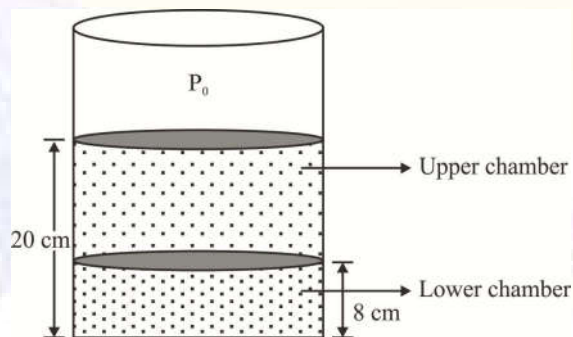
SECTION – III

(PARAGRAPH TYPE)

This section contains 2 paragraphs. Based upon one of the paragraphs 3 multiple choice questions based on the 3 multiple choice questions have to be answered. Each of these questions has 4 choices (A), (B), (C) and (D) for its answer, out of which ONLY ONE is correct.

Paragraph for Questions 35 and 36

In a cylindrical container of sufficiently large height, two easily moving pistons enclose certain amount of same ideal gas in two chambers as shown in the figure.



The upper piston is at a height 20 cm from the bottom and lower piston is at a height 8 cm from the bottom. The mass of each piston is m kg and cross sectional area of each piston is A m², where $\frac{mg}{A} = P_0$ and P_0 is the atmospheric pressure $= 1 \times 10^5$ N / m².

The cylindrical container and pistons are made of conducting material. Initially the temperature of gas is 27°C and whole system is in equilibrium. Now if the upper

piston is slowly lifted by 16 cm and held in that position with the help of some external force. As a result, the lower piston rises slowly by l cm.

35. The value of l is:
A) 2 cm B) 4 cm C) 8 cm D) 6 cm
36. Find the ratio of volume of gas in upper chamber to that of in lower chamber in final state:
A) 2: 1 B) 1: 2 C) 4: 1 D) 1: 4
37. Find the pressure of gas in lower chamber in final state:
A) $1.0 \times 10^5 \text{ N/m}^2$ B) $2.0 \times 10^5 \text{ N/m}^2$
C) $3.0 \times 10^5 \text{ N/m}^2$ D) $4.0 \times 10^5 \text{ N/m}^2$

Paragraph for Questions 37 and 39

An ideal gas undergoes a process in which $T = T_0 + aV^3$, where T_0 and "a" are positive constants and V is molar volume.

38. The volume for which pressure will be minimum is:
A) $\left(\frac{T_0}{2a}\right)^{1/3}$ B) $\left(\frac{T_0}{3a}\right)^{1/3}$ C) $\left(\frac{a}{2T_0}\right)^{2/3}$ D) $\left(\frac{a}{3T_0}\right)^{2/3}$

39. In above processes, minimum pressure attainable is:

A) $\frac{3}{4}(a^{5/3}R^{2/3}T_0^{2/3})2^{1/3}$

B) $\frac{3}{2}(a^{2/3}RT_0^{2/3})3^{1/2}$

C) $\frac{3}{2}(a^{1/2}R^{2/3}T_0^{3/4})4^{1/3}$

D) $\frac{3}{2}(a^{1/3}R^{2/3}T_0^{2/3})2^{1/3}$

SECTION –IV

(INTEGER ANSWER TYPE)

This section contains 7 questions. The answer to each of the questions is a single digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened.

40. A column of mercury 10 cm long is contained in the middle of a narrow, horizontal 1 m long tube which is closed at both ends. Both the halves of the tube contain air at a pressure of 76 cm of mercury. By what distance in centimeter (approx.) will the column of mercury be displaced if the tube is held vertically?
41. A glass tube sealed at one end and containing a quantity of air is immersed in mercury until the sealed end is 10 cm from the surface of mercury in the vessel. At 0°C the level of mercury in the tube is 5 cm above the level of mercury in the vessel. The length of the tube is 15 cm. Temperature of the air in the tube will be raised by $221 \times x$ Kelvin as to fill the tube completely? The atmospheric pressure is 75 cm of Hg. Neglect the expansion of mercury and glass. Value of 'x' is

42. Both limbs of a 'U' tube are of equal length. One of the limbs is sealed and the tube contains a column of 28 cm of air at atmospheric pressure. The air is separated from the atmosphere by mercury. Height of air in the sealed limb, will be approximately $3 \times x$ centimeter if the other limb is now filled to the top with mercury? Atmospheric pressure is 76 cm of mercury. Value of 'x' is
43. The diameter of a gas bubble formed at the bottom of a pond is $d = 4.0 \mu\text{m}$. When the bubble rises to the surface its diameter increases to $n = 1.1$ times. Depth of pond is 'x' meter. The nearest integer to 'x' is
(The atmospheric pressure is taken 10 m of water column, the gas expansion is assumed to be isothermal).
44. Two glass spheres of equal volume are connected by a small horizontal tube containing a small amount of mercury. The spheres are sealed at 20°C with exactly 1 litre of air in each side. Cross sectional area of tube is $5 \times 10^{-4} \text{ m}^2$. The mercury will be displaced by 34×10^{-x} centimeter if the temperature of one sphere is raised by 0.1°C while the other is maintained at 20°C . Value of 'x' is (neglect the expansion of glass spheres and tube)

45. Two adiabatic cylinders A and B, fitted with insulated pistons, contain equal amounts of an ideal diatomic gas at 300 K. The piston of A is free to move, while that of B is held fixed. The same amount of heat is given slowly to the gas in each cylinder. If the rise in temperature of the gas in cylinder A is 30K then rise in temperature of the gas in cylinder B is $7 \times x$ Kelvin. Value of 'x' is
46. The average degrees of freedom per molecule of a gas are 6. The gas performs 25 J of work in a process when it is expanded at constant pressure. Amount of heat absorbed by the gas is $25 \times x$. Value of 'x' is

MATHS**Max.Marks:80****SECTION – I****(SINGLE CORRECT CHOICE TYPE)**

This section contains 7 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which ONLY ONE is correct

47. The locus of the centre of the circle which touches the y-axis and also touches the circle $(x+1)^2 + y^2 = 1$ externally is

A) $\{(x, y) | x^2 = 4y\} \cup \{(0, y) | y \leq 0\}$

B) $\{(x, y) | y^2 = 4x\} \cup \{(x, 0) | x \leq 0\}$

C) $\{(x, y) | x^2 + 4y = 0\} \cup \{(0, y) | y \geq 0\}$

D) $\{(x, y) | y^2 + 4x = 0\} \cup \{(x, 0) | x \geq 0\}$

48. The circle $x^2 + y^2 = 1$ cuts the x-axis at P and Q. Another circle with centre at Q and variable radius intersects to first circle at R above the X-axis and the line segment PQ at S. The maximum area of the triangle QSR is

A) $\frac{2}{9}$

B) $\frac{5\sqrt{2}}{7}$

C) $\frac{4\sqrt{3}}{9}$

D) $\frac{\sqrt{2}}{13}$

49. Equation of a straight line meeting the circle $x^2 + y^2 = 100$ in two points, each point is at a distance of 4 units from the point (8, 6) on the circle, is
- A) $4x + 3y - 50 = 0$ B) $4x + 3y - 100 = 0$
C) $4x + 3y - 46 = 0$ D) $4x + 3y - 16 = 0$
50. CD is the common chord of the two circles of equal radii touching a line L at A and B. C is closer to L than D. The ratio of the circum radii of the triangles ACB and ADB is
- A) less than 1 B) more than 1
C) equal to 1 D) can't say
51. If the circle $x^2 + y^2 + 4x + 22y + c = 0$ bisects the circumference of the circle $x^2 + y^2 - 2x + 8y - d = 0$ then $c + d =$
- A) 50 B) 25 C) 60 D) 30
52. Tangents drawn from P(1, 8) to the circle $x^2 + y^2 - 6x - 4y - 11 = 0$ touches the circle at the points A and B, respectively. The radius of the circle which passes through the points of intersection of circles $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 2x - 6y + 6 = 0$ and intersects the circumcircle of the $\triangle PAB$ orthogonally is equal to
- A) $\frac{\sqrt{73}}{4}$ B) $\frac{\sqrt{71}}{2}$ C) 3 D) 2

53. Consider a family of circles passing through the point of intersection of the lines $\sqrt{3}(y-1)=x-1$ and $y-1=\sqrt{3}(x-1)$ and having its centre on the acute angle bisector of the given lines. Then the common chords of each member of the family and the circle $x^2+y^2+4x-6y+5=0$ are concurrent at
- A) $\left(\frac{1}{2}, \frac{1}{2}\right)$ B) $\left(\frac{1}{2}, \frac{3}{2}\right)$ C) $\left(\frac{3}{2}, \frac{3}{2}\right)$ D) $\left(-\frac{1}{2}, -\frac{1}{2}\right)$

SECTION – II**(MULTIPLE CORRECT CHOICE TYPE)**

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which ONE OR MORE is/ are correct

54. If two circles $x^2+y^2-6x-12y+1=0$ and $x^2+y^2-4x-2y-11=0$ cut a third circle orthogonally then the radical axis of the two circles passes through
- A) (1,1)
B) (0,6)
C) centre of the third circle
D) mid-point of the line joining the centres of the given circles.

55. Consider the circles $C_1 \equiv x^2 + y^2 - 2x - 4y - 4 = 0$ and $C_2 \equiv x^2 + y^2 + 2x + 4y + 4 = 0$ and the line $L \equiv x + 2y + 2 = 0$, then
- A) L is the radical axis of C_1 and C_2
 - B) L is the common tangent of C_1 and C_2
 - C) L is the common chord of C_1 and C_2
 - D) L is perpendicular to the joining centers of C_1 and C_2
56. If C_1, C_2 are two circle touching x-axis and y-axis. They both pass through point P and are orthogonal, then coordinates of P can be
- A) $(2, 4 + 2\sqrt{3})$
 - B) $(3, 6 + 3\sqrt{3})$
 - C) $(2 - \sqrt{3}, 1)$
 - D) $(1 + \sqrt{2}, 2)$
57. Circle(s) touching the x-axis at a distance of 3 units from the origin and having an intercept of length $2\sqrt{7}$ on the y-axis is (are)
- A) $x^2 + y^2 - 6x + 8y + 9 = 0$
 - B) $x^2 + y^2 - 6x + 7y + 9 = 0$
 - C) $x^2 + y^2 - 6x - 8y + 9 = 0$
 - D) $x^2 + y^2 - 6x - 7y + 9 = 0$

SECTION – III

(COMPREHENSION TYPE)

This section contains 2 paragraphs. Based upon one of the paragraphs 3 multiple choice questions based on the 3 multiple choice questions have to be answered. Each of these questions has 4 choices (A), (B), (C) and (D) for its answer, out of which ONLY ONE is correct.

Paragraph for Questions 58 and 60

Given equation of two intersecting circle's $S_1 = 0$ & $S_2 = 0$

Equation of family of circles passing through the intersection point's of $S_1 = 0$ &

$S_2 = 0$ is $S_1 + \lambda S_2 = 0$, (where $\lambda \neq -1$) Equation of common chord is $S_1 - S_2 = 0$

Equation of chord of contact for circle $x^2 + y^2 + 2gx + 2fy + c = 0$ with respect to external

point (x_1, y_1) is $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$

58. The equation of the circle described on the common chord of the circles

$x^2 + y^2 + 2x = 0$ and $x^2 + y^2 + 2y = 0$ as diameter is

A) $x^2 + y^2 + x - y = 0$

B) $x^2 + y^2 - x - y = 0$

C) $x^2 + y^2 - x + y = 0$

D) $x^2 + y^2 + x + y = 0$

59. The common chord of the circle $x^2 + y^2 + 6x + 8y - 7 = 0$ and a circle passing through the origin, and touching the line $y = x$, always passes through the point

A) $\left(-\frac{1}{2}, \frac{1}{2}\right)$

B) $(1, 1)$

C) $\left(\frac{1}{2}, \frac{1}{2}\right)$

D) none of these

Paragraph for Questions 61 and 62

A circle C_1 of radius r units rolls outside the circle $C_2: x^2 + y^2 + 2rx = 0$ touching it externally. The line of centers has an inclination 60° . Then

60. The point of contact of C_1 & C_2 is

A) $(-r, r\sqrt{3})$

B) $(-r, -r\sqrt{3})$

C) $\left(\frac{-r}{2}, \frac{-r\sqrt{3}}{2}\right)$

D) $\left(\frac{-r}{2}, \frac{r\sqrt{3}}{2}\right)$

61. The equation of direct common tangents are

A) $\sqrt{3}x - y + r(2 \pm \sqrt{3}) = 0$

B) $\sqrt{3}x - y + r(\sqrt{3} \pm 2) = 0$

C) $\sqrt{3}x - y + 2r(2 \pm \sqrt{3}) = 0$

D) $\sqrt{3}x - y + 2r(\sqrt{3} \pm 2) = 0$

62. The transverse common tangent is

A) $x + \sqrt{3}y + r = 0$

B) $x + \sqrt{3}y + 2r = 0$

C) $x + \sqrt{3}y - r = 0$

D) $x + \sqrt{3}y - 2r = 0$

SECTION –IV**(INTEGER ANSWER TYPE)**

This section contains 7 questions. The answer to each of the questions is a single digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened.

63. If the curves $\frac{x^2}{4} + y^2 = 1$ and $\frac{x^2}{a^2} + y^2 = 1$ for suitable value of “a” cut on four concyclic points, then find the radius of the smallest circle passing through these 4 points
64. Let $S_1 \equiv x^2 + y^2 - 4x - 8y + 4 = 0$ and S_2 its image in the line $y = x$. The radius of the circle touching $y = x$ at $(1, 1)$ and orthogonal to S_2 is $\frac{3}{\sqrt{\lambda}}$, then $\lambda^2 + 2 =$
65. A variable line $ax + by + c = 0$, where a, b, c (taken in that order) are in A.P. P is normal to a circle $(x - \alpha)^2 + (y - \beta)^2 = \gamma$, which is orthogonal to circle $x^2 + y^2 - 4x - 4y = 1$. Then the value of $(\alpha + \beta + \gamma)$ is
66. Through a given point $P(5, 2)$ secants are drawn to cut the circle $x^2 + y^2 = 25$ at points $A_1(B_1), A_2(B_2), A_3(B_3), A_4(B_4)$ such that $PA_1 + PB_1 = 5, PA_2 + PB_2 = 5, PA_3 + PB_3 = 7, PA_4 + PB_4 = 8$ and $PA_5 + PB_5 = 9$ then the value of $\frac{1}{51} \left[\sum_{i=1}^5 PA_i^2 + \sum_{i=1}^5 PB_i^2 \right]$ is

67. Let C , C_1 and C_2 be circles of radii 5, 3 and 2 respectively. C_1 and C_2 touch each other externally and C internally. A circle C_3 touches C_1 and C_2 externally and C internally. If the radius of C_3 is r then the value of $[4r]$ is, (where $[.]$ denotes the greatest integer function).....
68. Point A lies on circle $(x-20)^2 + y^2 = 4$ and points B lies on circle $x^2 + y^2 = 36$. The mid point of AB is M such that all possible positions of M form a region. If area of the region is $2\lambda\pi$. Then λ is equal to
69. If eight distinct points can be found on the curve $|x|+|y|=1$ such that from each point two mutually perpendicular tangents can be drawn to the circle $x^2 + y^2 = a^2$ then the range of $|a|$ is (α, β) then $\left[\frac{1}{\alpha} + \frac{1}{\beta}\right], ([.] \text{ g.i.f.}) =$



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A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr. Super60-I

WAT-31

Date: 06-01-18

Time: 07:30 AM – 10:30 AM

2011_P1

Max.Marks:240

KEY SHEET

CHEMISTRY

1	C	2	C	3	A	4	B	5	B
6	D	7	C	8	ACD	9	AC	10	A
11	B	12	C	13	B	14	A	15	D
16	B	17	4	18	3	19	1	20	2
21	4	22	2	23	4				

PHYSICS

24	A	25	D	26	C	27	A	28	D
29	D	30	B	31	BD	32	AD	33	AC
34	C	35	B	36	A	37	B	38	A
39	D	40	3	41	3	42	7	43	3
44	3	45	6	46	4				

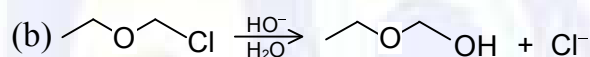
MATHS

47	D	48	C	49	C	50	C	51	A
52	A	53	B	54	AC	55	AD	56	ABC
57	AC	58	D	59	C	60	D	61	B
62	C	63	1	64	6	65	7	66	4
67	6	68	6	69	3				

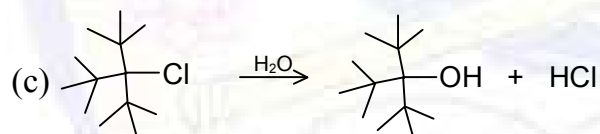
SOLUTIONS CHEMISTRY

1. CONCEPTUAL
2. CONCEPTUAL
3. CONCEPTUAL
4. Solution (b) Elimination is favoured when departing groups are anti or trans. Here H_b is anti to Br
5. Solution (b) The correct order is decided by the stability of the free radical formed as intermediate.
6. snar
7. conceptual
8. sol: (a) 

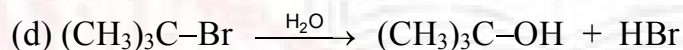
The nucleophile CH₃S⁻ is less sterically hindered than (CH₃)₂CHS⁻.



The electron-withdrawing oxygen increases the electrophilicity of the carbon that the nucleophile attacks.



Steric strain is decreased when Cl⁻ dissociates to form the carbocation in the rate-limiting step since the hybridization of the carbon atom changes from sp³ to sp², allowing the bond angle between the bulky groups to increase from 109.5° to 120°.



Because the reactants are neutral, the reaction will be faster in the more polar solvent.

9. sol: (a) Rate = [2-bromobutane] (3.2 × 10⁻⁵ + 1.5 × 10⁻⁶)
= 33.5 × 10⁻⁶ [2-bromobutane]

$$\% \text{ of } S_N1 \text{ reaction} = \frac{1.5 \times 10^{-6} [\text{2-bromobutane}]}{33.5 \times 10^{-6} [\text{2-bromobutane}]} \times 100 = 4.5\%$$

$$\% \text{ of } S_N2 \text{ reaction} = \frac{3.2 \times 10^{-5} [\text{2-bromobutane}]}{33.5 \times 10^{-6} [\text{2-bromobutane}]} \times 100 = 95.5\%$$

(b) Rate = [2-bromobutane] (3.2 × 10⁻⁸ + 1.5 × 10⁻⁶)

$$= 1.532 \times 10^{-6} [2\text{-bromobutane}]$$

$$\% \text{ of } S_N1 \text{ reaction} = \frac{1.5 \times 10^{-6} [2\text{-bromobutane}]}{1.532 \times 10^{-6} [2\text{-bromobutane}]} \times 100 \approx 98\%$$

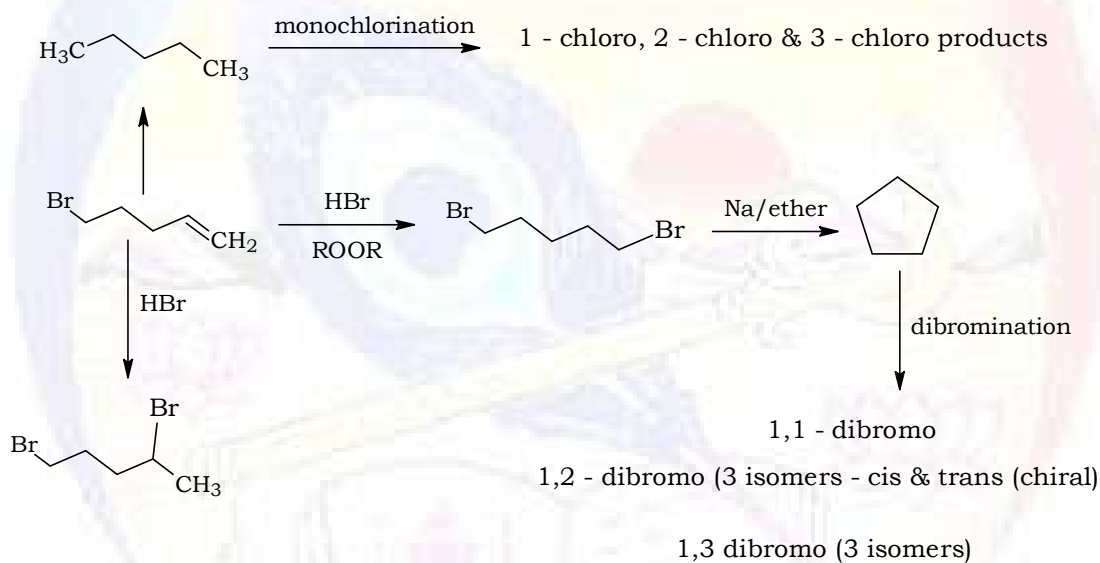
$$\% \text{ of } S_N2 \text{ reaction} = \frac{0.032 \times 10^{-6} [2\text{-bromobutane}]}{1.532 \times 10^{-6} [2\text{-bromobutane}]} \times 100 \approx 2\%$$

10. CONCEPTUAL

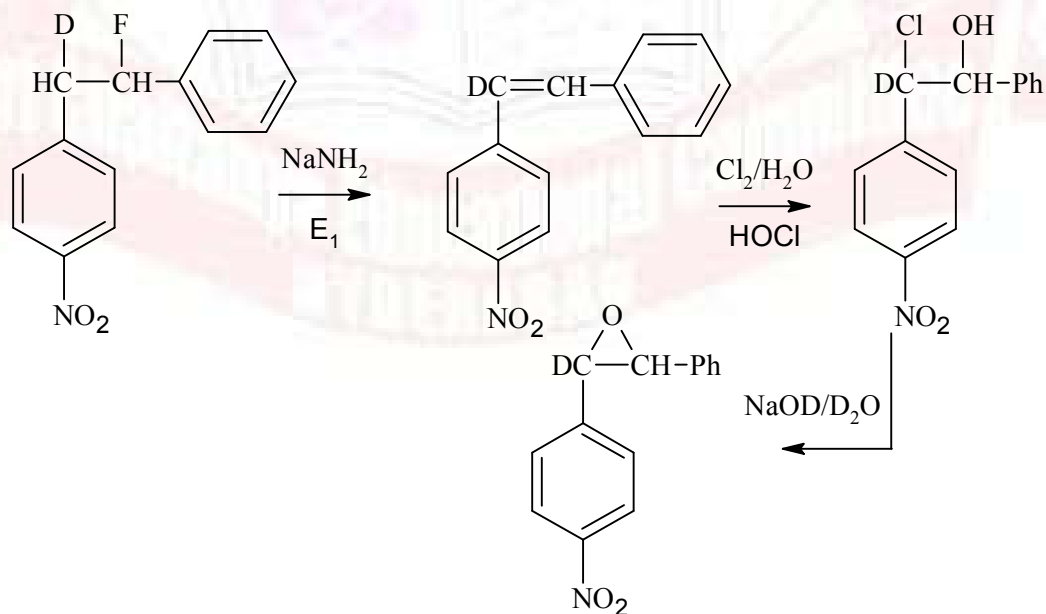
11. CONCEPTUAL

11. sol:- All the four are correct choices. The first three reactions involve benzyne intermediate

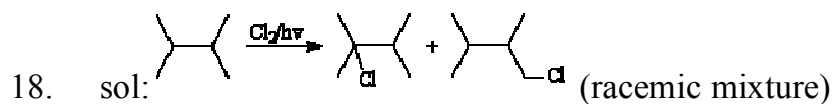
12, 13, 14 Sol:



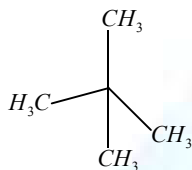
15, 16, SOL:



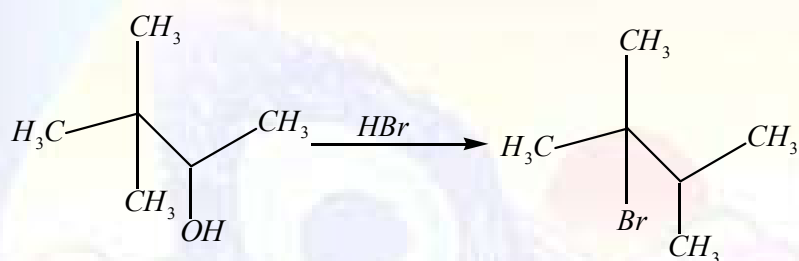
17. SOL: In (ii), (iii), (iv) and (v) first compound is more reactive than the second compound.



19. y is



20. ans: 2



Sol:

21. sol: (x) = 10 , (y) = 4
22. CONCEPTUAL
23. sol: (ii),(iii),(iv),(v) are less reactive than $\text{CH}_3\text{CH}_2\text{I}$ in a S_N^2 reaction

PHYSICS

24. $\Delta Q = \Delta U + W$
25. Comment for discussion: In the ideal case that we normally consider, each collision transfers twice the magnitude of its normal momentum. On the face EFGH, it transfers only half of that.
26. This is a constant pressure ($p = \text{Mg}/A$) arrangement.
28. Comment for discussion: The usual statement for the perfect gas law somehow emphasizes molecules. If a gas exists in atomic form (perfectly possible) or a combination of atomic and molecular form, the law is not clearly stated.
29. Comment for discussion: In this chapter, one has discussed constant pressure and constant volume situations but in real life there are many situations where both change. If the surfaces were rigid, p would rise to $1.1 p$. However, as the pressure rises, V also rises such that pV finally is $1.1 RT$ with $p_{\text{final}} > p$ and $V_{\text{final}} > V$.

32. Comment : The equation $\langle \text{K.E. of translation} \rangle = (3/2) RT$, $\langle \text{Rotational energy} \rangle = RT$ is taught. The fact that the distribution of the two is independent of each other is not emphasized. They are independently Maxwellian.

35-37. Let P_1 and P_2 be the initial pressure in lower chamber of gas and upper chamber of gas.

$$P_2 + P_0 + \frac{mg}{A} = 2P_0, V_2 = A \times 12 \times 10^{-2} \text{ m}^3$$

If P'_2 and V'_2 are final pressure and volume in upper chamber

$$V'_2 = A \times (28 - l) \times 10^{-2} \text{ m}^3$$

$$P_2 V_2 = P'_2 V'_2 \Rightarrow P'_2 = \frac{P_2 V_2}{V'_2} = \frac{24P_0}{28 - l}$$

Now consider lower chamber

$$P_1 = P_0 + \frac{2mg}{A} = 3P_0$$

$$\text{and } V_1 = A \times 8 \times 10^{-2} \text{ m}^3$$

$$P'_1 = P'_2 + \frac{mg}{A} = P_0 \left[\frac{52 - l}{28 - l} \right]$$

$$\text{and } V'_1 = A \times (8 + l) \times 10^{-2} \text{ m}^3$$

$$P_1 V_1 = P'_1 V'_1$$

$$3P_0 A \times 8 \times 10^{-2} = P_0 \left[\frac{52 - l}{28 - l} \right] \times A \times (8 + l) \times 10^{-2}$$

$$24 = \left[\frac{52 - l}{28 - l} \right] \times (8 + l)$$

Solving we get $l = 4 \text{ cm}$

$$\Rightarrow P'_1 = 2P_0 = 2 \times 10^5 \text{ N/m}^2$$

$$\Rightarrow P'_2 = \frac{24P_0}{28 - l} = P_0 = 1 \times 10^5 \text{ N/m}^2$$

$$\frac{V'_2}{V'_1} = \frac{28 - l}{8 + l} = \frac{24}{12} = 2$$

38. From ideal gas equation,

$$pV = nRT$$

$$p = \frac{nRT}{V} = \frac{nR(T_0 + aV^3)}{V}$$

$$p = \frac{nRT_0}{V} + nRaV^2$$

$$\frac{dp}{dV} = 0$$

$$\frac{-nRT_0}{V^2} + 2anRV = 0$$

$$\frac{T_0}{V^2} = 2aV$$

$$V^3 = \frac{T_0}{2a}$$

$$V = \left(\frac{T_0}{2a} \right)^{1/3}$$

39. From ideal gas equation,

$$p = \frac{nRT}{V}$$

$$p = \frac{nR(T_0 + aV^3)}{V}$$

$$0 = \frac{nRT_0}{V} + anRV^2$$

$$p = \frac{nRT_0}{T_0^{1/3}} (2a)^{1/3} + anR \cdot \frac{T_0^{2/3}}{(2a)^{2/3}}$$

for $n = 1$

$$\Rightarrow p = RT_0^{2/3} (2)^{1/3} (a)^{1/3}$$

$$+ \frac{a^{1/3} \cdot RT_0^{2/3}}{2^{2/3}}$$

$$\Rightarrow p = \frac{3RT_0^{2/3} a^{1/3}}{2^{2/3}} \Rightarrow p = \frac{3}{2} (a^{1/3} RT_0^{2/3}) (2)^{1/3}$$

40. by 3 cm downwards

41. 663°C

42. 21.77 cm

43. 3.3 m

44. 0.034cm

45. 42 K

46. 100 J

MATHS

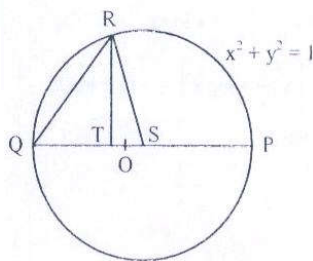
47. Key: D

Sol. Let $P(x_1, y_1)$ be the centre of the touching $(x+1)^2 + y^2 = 1$ externally and touching y-axis $\therefore 1 - x_1 = (x_1 + 1)^2 + y_1^2 \Rightarrow y_1^2 + 4x_1 = 0$ Also every circle with centre on positive x-axis and touching y-axis at origin satisfy the condition.

48. Key: C

Sol: Q is $(-1, 0)$ The circle with centre at Q and variable radius r has the equation $(x+1)^2 + y^2 = r^2$ This circle meets the line segment QP at S where $QS = r$

It meets the circle $x^2 + y^2 = 1$ at $R\left(\frac{r^2-2}{2}, \frac{r}{2}\sqrt{4-r^2}\right)$ found by solving the equations of the two circles simultaneously. A = area of the triangle QSR



$$= \frac{1}{2} QS \times RT$$

$$= \frac{1}{2} r \left(\frac{r}{2} \sqrt{4-r^2} \right) \text{ since } RT \text{ is the } y \text{ coordinate of } R$$

$$\frac{dA}{dr} = \frac{1}{4} \left\{ 2r\sqrt{4-r^2} + \frac{r^2(-r)}{\sqrt{4-r^2}} \right\} = \frac{\{2r(4-r^2) - r^3\}}{4\sqrt{4-r^2}} = \frac{8r-3r^3}{4\sqrt{4-r^2}}$$

$$\frac{dA}{dr} = 0 \text{ when } r(8-3r^2) = 0 \text{ giving } r = \sqrt{\frac{8}{3}}$$

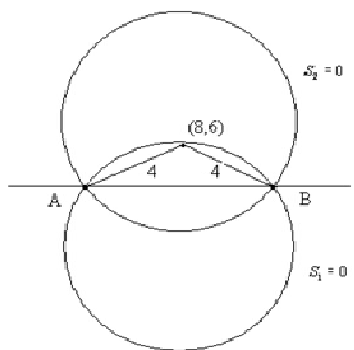
$$\frac{d^2A}{dr^2} = \frac{4\sqrt{4-r^2}(8-9r^2) - (8r-3r^3)\frac{(-r)}{\sqrt{4-r^2}}}{16(4-r^2)}, \text{ where, } r = \sqrt{\frac{8}{3}}, \frac{d^2A}{dr^2} < 0$$

Hence A is maximum when $r = \sqrt{\frac{8}{3}}$ and the maximum area =

$$\frac{8}{4 \times 3} \sqrt{4 - \frac{8}{3}} = \frac{16}{12\sqrt{3}} = \frac{4}{3\sqrt{3}} = \frac{4\sqrt{3}}{9}$$

49. Key. C

$$\text{SOL. } S_1 = x^2 + y^2 = 100$$



equation of circle centred at $(8,6)$ & radius 4 units is $(x-8)^2 + (y-6)^2 = 16$

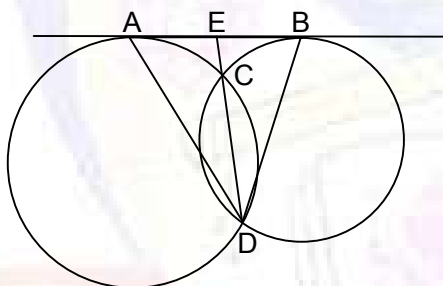
required line AB is the common chord of $S_1 = 0$ & $S_2 = 0$, is $S_1 - S_2 = 0$

$$4x + 3y - 46 = 0$$

50. C

Let R_1 be the circum radii of $\triangle ACB$

$$\Rightarrow R_1 = \frac{AB}{2 \sin \angle ADB}$$



Now $AE = EB$

(common chord bisects the common tangent)

Also $\angle EAC = \angle ADE = \theta$

$$\Rightarrow \angle ACE = 90 - \theta \Rightarrow \angle ACB = 180 - \angle ADB$$

$$\Rightarrow R_1 = R_2$$

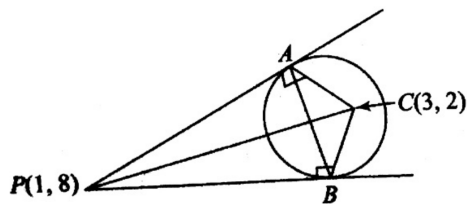
51. A

The common chord of the two circles passes through $(1, -4)$

$$\Rightarrow 6(1) + 14(-4) + c + d = 0 \Rightarrow c + d = 50$$

52. A

Equations of circle circumscribing $\triangle PAB$ is



$$(x-1)(x-3) + (y-8)(y-2) = 0$$

$$\Rightarrow x^2 + y^2 - 4x - 10y + 19 = 0$$

Equation of circle passing through points of intersection of circles

$$x^2 + y^2 - 2x - 6y + 6 = 0 \text{ and } x^2 + y^2 + 2x - 6y + 6 = 0 \text{ is given by}$$

$$(x^2 + y^2 - 2x - 6y + 6) + \lambda(x^2 + y^2 + 2x - 6y + 6) = 0$$

$$\Rightarrow x^2 + y^2 + \frac{(2\lambda - 2)}{\lambda + 1}x - 6y + 6 = 0$$

As circle (ii) is orthogonal to circle (i), we have

$$-2\left(\frac{2\lambda - 2}{\lambda + 1}\right) - 5(-6) = 19 + 6$$

$$\Rightarrow 4\lambda - 4 = 5\lambda + 5$$

$$\Rightarrow \lambda = -9$$

Hence, required equation of circle is

$$x^2 + y^2 + \frac{5}{2}x - 6y + 6 = 0$$

$$\therefore \text{Radius of circle} = \sqrt{\frac{25}{16} + 9 - 6} = \frac{\sqrt{73}}{4}$$

53. B

$$\text{The given lines are } \sqrt{3}(y-1) = x-1$$

$$y-1 = \sqrt{3}(x-1)$$

These lines intersect at the point (1, 1)

Also lines make an angle 30° and 60° with the axes so acute angle bisector makes an angle 45° with the axes. Therefore, for acute angle bisector is $x=y$

Since centre lies on this line, let it be (a, a)

$\therefore$ Equation of the circle is

$$(x-a)^2 + (y-a)^2 = (1-a)^2 + (1-a)^2$$

$$\text{Or } x^2 + y^2 - 2ax - 2ay + 4a - 2 = 0$$

The common chord of this variable circle with the given circle

$$x^2 + y^2 + 4x - 6y + 5 = 0 \text{ is}$$

$$(4 + 2a)x + (2a - 6)y + (7 - 4a) = 0$$

$$\text{i.e. } (4x - 6y + 7) + 2a(x + y - 2) = 0$$

Which represents a family of straightlines passing through the point of intersection of

the lines $4x - 6y + 7 = 0$ And $x + y - 2 = 0$ Which is $\left(\frac{1}{2}, \frac{3}{2}\right)$

54. Key. A, C

Sol. Radical axis of the given circles is $x + 5y - 6 = 0$ which passes through (1, 1)

Let the given circles intersect the circle

$$x^2 + y^2 + 2gx + 2fy + c = 0 \text{ Orthogonally then } 2g(-3) + 2f(-6) = c + 1$$

$$2g(-2) + 2f(-1) = c - 11 \Rightarrow 2g(-1) + 2f(-5) = 12 \Rightarrow -g - 5f - 6 = 0$$

$\Rightarrow$ the radical axis passes through the centre $(-g, -f)$ of the third circle.

55. Key. A, C, D

$$\text{Sol. } C_1 \equiv x^2 + y^2 - 2x - 4y - 4 = 0 \dots\dots\dots(i)$$

$$\text{And } C_2 \equiv x^2 + y^2 + 2x + 4y + 4 = 0 \dots\dots\dots(ii)$$

$$\therefore \text{ Radical axis is } C_1 - C_2 = 0$$

$$\Rightarrow -4x - 8y - 8 = 0$$

$$\text{Or } x + 2y + 2 = 0 \text{ which is } L = 0$$

(a) option is correct

centre and radius of $C_1 = 0$ are (1, 2) and 3.

$\therefore$ length of $\perp$ from (1, 2) on $L = 0$

$$\text{is } \frac{|1 + 4 + 2|}{\sqrt{1 + 4}} = \frac{7}{\sqrt{5}} \neq \text{radius}$$

$\therefore$ (b) option is wrong

L is also the common chord of C_1 and C_2

$\therefore$ (c) option is correct.

$\therefore$ centres of $C_1 = 0$ and $C_2 = 0$ are (1, 2) and (-1, -2)

$\therefore$ slope of line joining centres of circles $C_1 = 0$ & $C_2 = 0$ is

$$\frac{-2-2}{-1-1} = \frac{4}{2} = 2 = m_1 \text{ (say)}$$

And slope of $L = 0$ is $-\frac{1}{2} = m_2$ (say)

$$\therefore m_1 m_2 = -1$$

Hence, L is perpendicular to the line joining centres of C_1 and C_2

$\therefore$ (d) option is correct

56. A, B, C

In the given situation $a^2 - 4ab + b^2 = 0$ where P is (a, b).

57. A, C

The figure shows circle touching the x-axis at A(3, 0) and having intercept BD = $BD = 2\sqrt{7}$ on the y-axis from the figure, the abscissa of center is 3

$$\text{Also } BC = \text{Radius of the circle} = \sqrt{CM^2 + BM^2} = \sqrt{9+7} = 4$$

Therefore, AC = 4

So, the center of the circles are

C(3, 4) and C'(3, -4)

Hence, the equations of circles are

$$(x-3)^2 + (y \pm 4)^2 = 16$$

$$\text{Or } x^2 + y^2 - 6x \pm 8y + 9 = 0$$

58. Key. D

59. Key. C

58. Equation of common chord is $2x - 2y = 0$

Equation of family of circle is

$$x^2 + y^2 + 2x + \lambda(2x - 2y) = 0$$

Centre of circle is $(-\lambda-1, +\lambda)$

Centre lies on $y = x$

$$\lambda = -\lambda - 1$$

$$2\lambda = -1$$

$$\lambda = -\frac{1}{2}$$

$$\text{Equation is } x^2 + y^2 + x + y = 0$$

59. Let the second circle $x^2 + y^2 + 2gx + 2fy = 0$

Hence, $x^2 + y^2 + 2gx + 2fy = 0$ lies equal roots $f + g = 0$

Equation of common chord is

$$2(g-3)x + 2(-g-4)y + 7 = 0$$

$$(-6x - 8y + 7) + g(2x - 2y) = 0$$

Passes through the intersection point of

$$-6x - 8y + 7 = 0 \text{ and } 2x - 2y = 0$$

$$\Rightarrow \left(\frac{1}{2}, \frac{1}{2} \right)$$

$$\cos 60^\circ = \frac{\sqrt{(h+1)^2 + (k-1)^2}}{2}$$

60. Key.D

61. Key. B

62. Key.C

Sol. 60,61,62

The point of contact is at a distance of r units from $(-r, 0)$ on a line of inclination 60°

$$\therefore \text{Point of contact} = (-r + r \cos 60^\circ, 0 + r \sin 60^\circ)$$

$$= \left(-\frac{r}{2}, \frac{r\sqrt{3}}{2} \right) \quad C_1 \text{ centre} = (-r + 2r \cos 60^\circ, 0 + 2r \sin 60^\circ) = (0, r\sqrt{3})$$

DCT's are parallel to line of centres.

Equation of DTCs are of the form : $x\sqrt{3} - y + K = 0$ to find K use $CP=r$.

Transverse common tangent is $\perp r$ to line of centers and passes through point of contact.

63. Key: 1

$$\text{Hint: } \left(\frac{x^2}{4} + y^2 - 1 \right) + \lambda \left(\frac{x^2}{a^2} + y^2 - 1 \right) = 0$$

$$x^2 \left(\frac{a^2 + 4\lambda}{4a^2(1+\lambda)} \right) + y^2 = 1$$

Clearly radius is 1 unit

64. Key. 6

Sol. Centre of circle $S_1 = (2, 4)$

Centre of circle $S_2 = (4, 2)$

Radius of circle $S_1 =$ radius of circle $S_2 = 4$

$\therefore$ equation of circle S_2

$$(x - 4)^2 + (y - 2)^2 = 16$$

$$\Rightarrow x^2 + y^2 - 8x - 4y + 4 = 0 \quad \dots (i)$$

Equation of circle touching $y = x$ at $(1, 1)$ can be taken as

$$(x - 1)^2 + (y - 1)^2 + \lambda(x - y) = 0$$

$$\text{or, } x^2 + y^2 + x(\lambda - 2) + y(-\lambda - 2) + 2 = 0 \quad \dots (ii)$$

As this is orthogonal to S_2

$$\Rightarrow 2\left(\frac{\lambda - 2}{2}\right) \cdot (-4) + 2\left(\frac{-\lambda - 2}{2}\right) \cdot (-2) = 4 + 2$$

$$\Rightarrow -4\lambda + 8 + 2\lambda + 4 = 6$$

$\therefore$ required equation of circle is

$$x^2 + y^2 + x - 5y + 2 = 0.$$

$$\text{Radius} = \sqrt{\frac{1}{4} + \frac{25}{4} - 2} = \sqrt{\frac{26 - 8}{4}} = \sqrt{\frac{18}{4}} = \frac{3}{2}\sqrt{2}.$$

65. $ax + by + c = 0$

Also $a - 2b + c = 0$

Comparing we get $x=1, y=-2$

Thus variable lines are concurrent at $(1, -2)$

These lines are normal to the circle

Thus $(1, -2)$ is the centre of circle

So, $\alpha = 1, \beta = -2$

Now $(x - 1)^2 + (y + 2)^2 = \gamma$

$$\Rightarrow x^2 + y^2 - 2x + 4y + (5 - \gamma) = 0$$

And $x^2 + y^2 - 4x - 4y - 1 = 0$

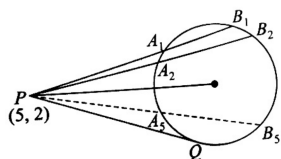
Above two circles intersect orthogonally

$$\Rightarrow 2(-1)(-2) + 2(2)(5 - \gamma) + (-1)$$

$$\Rightarrow \gamma = 8$$

So, $\alpha + \beta + \gamma = 1 + (-2) + 8 = 7$

66. 8



$$\begin{aligned} & \sum_{i=1}^5 PA_i^2 + \sum_{i=1}^5 PB_i^2 \\ &= \sum_{i=1}^5 (PA_i + PB_i)^2 - 2 \sum_{i=1}^5 PA_i + PB_i \\ &= 5^2 + 5^2 + 7^2 + 8^2 + 9^2 - \sum_{i=1}^5 PQ^2 \end{aligned}$$

(Where PQ = length of tangent from P to the circle)

$$\begin{aligned} &= 5^2 + 5^2 + 7^2 + 8^2 + 9^2 - 5(4) \\ &= 50 + 49 + 64 + 81 - 20 \\ &= 244 - 20 = 224 \end{aligned}$$

67. 6

Let O, O_1, O_2, O_3 be the centres and r be the radius of C_3 C and C_1 touches internally

$$\therefore OO_1 = 5 - 3 = 2$$

 C_1 and C_2 touches internally

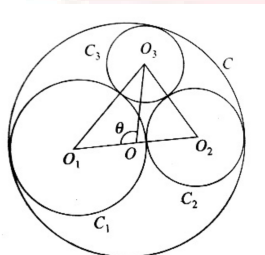
$$\therefore OO_2 = 5 - 2 = 3$$

 C_3 touches C_1 and C_2 externally

$$\therefore O_1O_3 = r + 3 \text{ and } O_2O_3 = r + 2$$

 C_3 touches C internally

$$\therefore OO_3 = 5 - r$$

In $\triangle OO_1O_3$ by cosine rule at vertex 'O' we get

$$O_1O_3^2 = OO_1^2 + OO_3^2 - 2OO_1OO_3 \cos \angle O_1OO_3$$

$$\therefore (r+3)^2 = 4 + (5-r)^2 - 2 \cdot 2(5-r) \cos \theta$$

In $\triangle OO_2O_3$ by cosine rule at vertex 'O' we get $O_2O_3^2 = OO_2^2 + OO_3^2 - 2OO_2OO_3 \cos \angle O_2OO_3$

$$(r+2)^2 = 9 + (5-r)^2 + 2 \cdot 3 \cdot (5-r) \cos \theta$$

By eliminating $\cos \theta$, we get $r = \frac{30}{19}$

68. 6

Parametric points on given circles are $A(20+2 \cos \alpha, 2 \sin \alpha)$ and $B(6 \cos \beta, 6 \sin \beta)$

Midpoint of AB is (h, k) $(h, k) = (10 + \cos \alpha + 3 \cos \beta, \sin \alpha + 3 \sin \beta)$

$$\therefore h - 10 = \cos \alpha + 3 \cos \beta$$

$$\text{And } k = \sin \alpha + 3 \sin \beta$$

Squaring and adding, we get $(h-10)^2 + k^2 = 10 + 6 \cos(\alpha - \beta)$

Or we have region bounded by $(x-10)^2 + y^2 = 16$ (where $\cos(\alpha - \beta) = 1$)

And $(x-10)^2 + y^2 = 4$ (when $\cos(\alpha - \beta) = -1$)

69. $\frac{1}{2} < a < \frac{1}{\sqrt{2}} \alpha + \beta = 2 + \sqrt{2}[\alpha + \beta] = 3$

Success



JEE ADVANCED WEEKEND TEST



Sri Chaitanya IIT Academy, India

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR-HYD

Sec: Jr.Super60-I

Time: 07:30 AM to 10:30 AM

Date: 20-01-18

Max.Marks:183

Name of the Student: _____

H.T. NO:

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2017_PAPER-II

IMPORTANT INSTRUCTIONS:

- 1) This booklet is your Question Paper.
- 2) Use the Optical Response Sheet (ORS) provided separately for answering the questions
- 3) Blank spaces are provided within this booklet for rough work.
- 4) Write your name, roll number and sign in the space provided on the back cover of this booklet.
- 5) You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET:

- 6) Darken the appropriate bubbles on the ORS by applying sufficient pressure. This will leave an impression at the corresponding place on the Candidate's sheet.
- 7) The ORS will be collected by the invigilator at the end of the examination.
- 8) Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
- 9) Write your name, roll number and code of the examination center, and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else** on the ORS. Darken the appropriate bubble under each digit of your roll number.

DARKENING THE BUBBLES ON THE ORS

- 10) Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
- 11) Darken the bubble **COMPLETELY**.
- 12) The correct way of darkening a bubble is as:
- 13) The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- 14) Darken the bubbles **ONLY** IF you are sure of the answer. There is **NO WAY** to erase or

“un-darken” a darkened bubble.

20-01-18_Jr.Super60-I_Jee Adv_WAT-32_Weekend Syllabus

MATHS: SYLLABUS COVERED FROM_16-01-18 TO 19-01-18

PHYSICS: SYLLABUS COVERED FROM_16-01-18 TO 19-01-18

CHEMISTRY: Preparation of alcohols, Properties and reactions of alcohols

JEE-ADVANCE-2017-P2-Model

Time: 07:30 to 10:30 AM

IMPORTANT INSTRUCTIONS

Max Marks: 183

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 7)	Questions with Single Correct Options	+3	-1	7	21
Sec – II (Q.N : 8 – 14)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	7	28
Sec – III (Q.N : 15 – 18)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	+3	0	4	12
Total				18	61

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 19 – 25)	Questions with Single Correct Options	+3	-1	7	21
Sec – II (Q.N : 26 – 32)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	7	28
Sec – III (Q.N : 33 – 36)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	+3	0	4	12
Total				18	61

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 37 – 43)	Questions with Single Correct Options	+3	-1	7	21
Sec – II (Q.N : 44 – 50)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	7	28
Sec – III (Q.N : 51 – 54)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	+3	0	4	12
Total				18	61

Sec: Jr.Super60-I

space for rough work

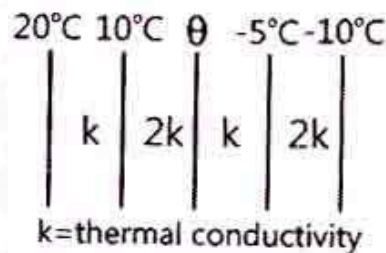
Page 2

SECTION – I
(SINGLE CORRECT CHOICE TYPE)

This section contains **7 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Marking scheme +3 for correct answer , 0 if not attempted and -1 in all other cases.

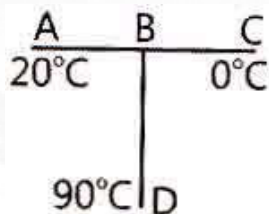
1. A thin walled metal tank of surface area $5m^2$ is filled with water and contains an immersion heater dissipating 1kW. The tank is covered with 4 cm thick layer of insulation whose thermal conductivity is 0.2 W/m/K. the outer face of the insulation is $25^{\circ}C$. Find the temperature of the tank in the steady state.
A) $35^{\circ}C$ B) $65^{\circ}C$ C) $45^{\circ}C$ D) $55^{\circ}C$
2. The figure shows the face and interface temperature of a composite slab containing of four layers of two materials having identical thickness. Under stady state condition, find the value of temperature θ



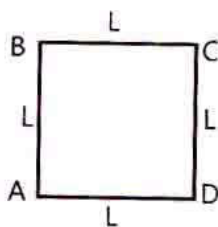
- A) $5^{\circ}C$ B) $2^{\circ}C$ C) $1^{\circ}C$ D) $4^{\circ}C$

3. Three conducting rods of same material and cross-section are shown in figure.

Temperature of A,D and C are maintained at 20°C , 90°C and 0°C . Find the ratio of length BD and BC if there is no heat flow in AB

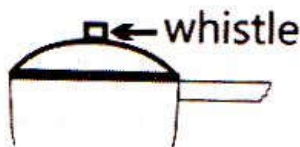


- A) 7.8 B) 1.2 C) 3.5 D) 3
4. In the square frame of side L of metallic rods, the corners A and C are maintained at T_1 and T_2 respectively. The rate of heat flow from A to C is W. if A and D are instead maintained at T_1 and T_2 respectively, find the total rate of heat flow.



- A) $\frac{1}{4}W$ B) $\frac{1}{3}W$ C) $\frac{3}{4}W$ D) $\frac{4}{3}W$

5. One end of copper rod of uniform cross-section and of length 1.5 meters is in contact with melting ice and the other end with boiling water. At what point from the boiling water side along the length should a temperature of 200°C be maintained, so that in steady state, the mass of ice melting is equal to that of steam produced in the same interval of time ? Assume that the whole system is insulated from the surroundings. Use $L_f = 80$ units and $L_v = 540$ units
- A) 10.34 cm B) 7.4 cm C) 1.34 cm D) 9.34 cm
6. An empty pressure cooker of volume 10 liters contains air at atmospheric pressure 10^5 Pa and temperature of 27°C . It contains a whistle which has area of 0.1 cm^2 and weight of 100 gm. What should be temperature of air inside so that the whistle is just lifted up?



- A) 127°C B) 227°C C) 327°C D) 427°C

7. Spheres P and Q are uniformly constructed from the same material which is a good conductor of heat and the radius of Q is thrice the radius of P, the rate of fall of temperature of P is x times that of Q when both are at the same surface temperature. The value of x is
- A) $1/4$ B) $1/3$ C) 3 D) 4

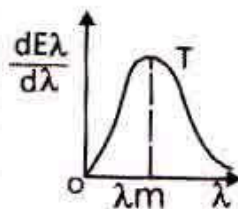
SECTION-II**(ONE OR MORE OPTIONS CORRECT TYPE)**

This section contains 7 multiple choice questions. Each question has four choices (A) (B), (C) and (D) out of which **ONE** or **MORE THAN ONE** are correct.

Marking scheme: +4 for all correct options, 0 if not attempted and -2 in all wrong cases.

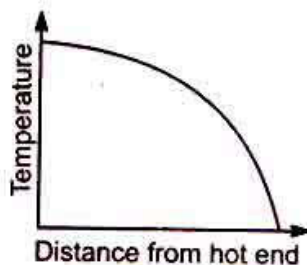
8. Two metallic spheres A and B are made of same material and have got identical surface finish. The mass of sphere A is four times that of B. Both the spheres are heated to the same temperature and placed in a room having lower temperature but thermally insulated from each other.
- A) The ratio of initial rate of heat loss to the surroundings, of A to that of B is $2^{4/3}$
- B) The ratio of initial rate of heat loss to the surroundings, of A to that of B is $2^{2/3}$
- C) The ratio of the initial rate of cooling of A to that of B is $2^{-2/3}$
- D) The ratio of the initial rate of cooling of A to that of B is $2^{-4/3}$

9. Two bodies A and B have thermal emissivity of 0.01 and 0.81 respectively. The outer surface areas of the two bodies are the same. The two bodies radiate energy at the same rate. The wavelength λ_B , corresponding to the maximum spectral radiance in the radiation from B, is shifted from the wavelength corresponding to the maximum spectral radiance in the radiation from A by $1.00 \mu\text{m}$. If the temperature of A is 5802K,
- A) temperature of B is 1934 K B) $\lambda_B = 1.5 \mu\text{m}$
 C) temperature of B is 11604 K D) temperature of B is 2901K
10. The figure shows a radiant energy spectrum graph for a black body at a temperature T. Choose the correct statement (s)



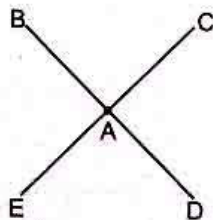
- A) The radiant energy is not equally distributed among all the possible wavelengths
 B) For a particular wavelength the spectral intensity is maximum
 C) The area under the curve is equal to the rate at which heat is radiated by the body at that temperature
 D) only B & C correct

11. The ends of a long homogeneous bar are maintained at different temperatures and there is no loss to heat from the sides of the bar due to conduction or radiation. The graph of temperature against distance of the bar when it has attained steady state is shown here. The graph shows:



- A) the temperature gradient is not constant
- B) the bar has uniform cross- sectional area
- C) the cross- sectional area of the bar increases as the distance from the hot end increases
- D) the cross- sectional area of the bar decreases as the distance from the hot end increases

12. Four identical rods which have thermally insulated lateral surfaces are joined at point A. points B, C, D and E are connected to large reservoirs. If heat flows into the junction from point B at rate of 1 W and from point C at 3 W inside, flows out from D at 5 W, which relation (s) is/are correct for temperature of these points ?



- A) $T_A < T_E$ B) $T_B = T_C$ C) $T_C > T_D$ D) $T_B = T_E$
13. Two spherical black-bodies A and B, having radii r_A and r_B , where $r_B = 2r_A$ emit radiations with peak intensities at wavelengths 400 nm and 800 nm respectively. If their temperature are T_A and T_B respectively in Kelvin scale, their emissive powers are E_A and E_B and energies emitted per second are P_A and P_B then:
- A) $T_A/T_B = 2$ B) $P_A/P_B = 4$ C) $E_A/E_B = 8$ D) $E_A/E_B = 4$
14. The net rate of heat loss by a hot body depends upon:
- A) temperature of body B) temperature of surroundings
C) material of body D) nature of the surface

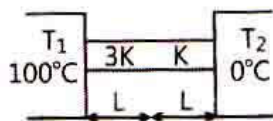
SECTION – III
(PARAGRAPH TYPE)

This section contains **2 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph For Questions 15 and 16

Two rods A and B of same cross-sectional area A and length L connected in series and having thermal conductivities as $3K$ and K respectively, between a source ($T_1 = 100^\circ\text{C}$) and a sink ($T_2 = 0^\circ\text{C}$) as shown in figure. The rod is laterally insulated



15. The ratio of the thermal resistance of the rod is
- A) $\frac{R_A}{R_B} = \frac{1}{3}$ B) $\frac{R_A}{R_B} = 3$ C) $\frac{R_A}{R_B} = \frac{3}{4}$ D) $\frac{4}{3}$
16. If T_A and T_B are the temperature drops across the rod A and B, If G_A and G_B are the temperature gradients across the rod A and B, then
- A) $\frac{T_A}{T_B} = \frac{3}{1}, \frac{G_A}{G_B} = \frac{3}{1}$ B) $\frac{T_A}{T_B} = \frac{1}{3}, \frac{G_A}{G_B} = \frac{1}{3}$
- C) $\frac{T_A}{T_B} = \frac{3}{4}, \frac{G_A}{G_B} = \frac{3}{4}$ D) $\frac{T_A}{T_B} = \frac{4}{3}, \frac{G_A}{G_B} = \frac{4}{3}$

Paragraph For Questions 17 and 18

In fluids heat transfer takes place and molecules of the medium take very active part. The molecules take energy from high temperature zone and move towards low temperature zone. This method is known as convection, when we require heat transfer with fast phase, we use some mechanism to make the flow of fluid on the body fast. The rate of loss of heat is proportional to velocity of fluid (v). and temperature difference (ΔT) between the body and fluid, of course more surface area of body, more rate of loss of heat. We can write the rate of loss of heat as $\frac{dQ}{dt} = KA v \Delta T$ Where K is Positive constant. Now answer the following questions.

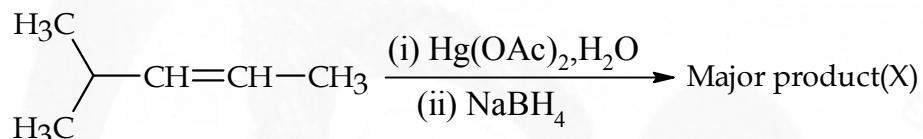
17. A body is being cooled with fluid when we increase the velocity of fluid 4 times and decrease the temperature difference $1/2$ time, the rate of loss of heat increases
A) four times B) two times C) six times D) no change
18. In the above question, if mass of the body increased two times, without change in any of the other parameters, the rate of cooling
A) Decreases B) increases
C) No effect of change of mass D) None of these

SECTION – I
(SINGLE CORRECT CHOICE TYPE)

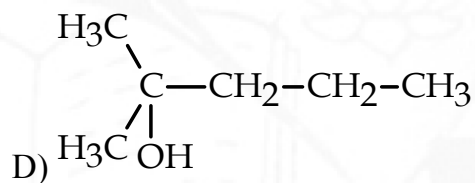
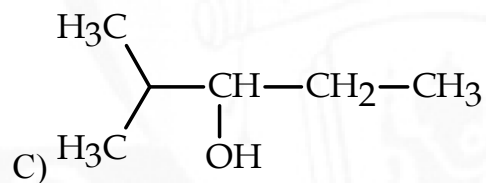
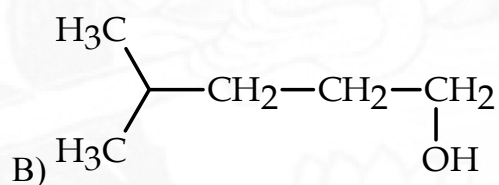
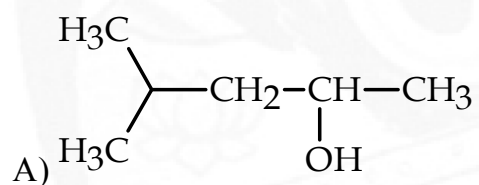
This section contains **7 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Marking scheme +3 for correct answer , 0 if not attempted and -1 in all other cases.

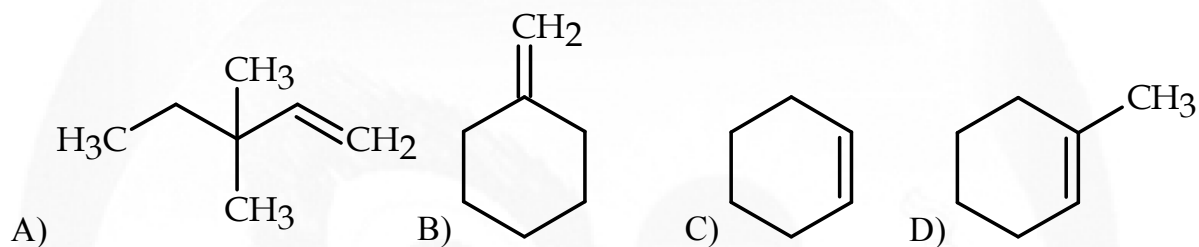
19. In the reaction

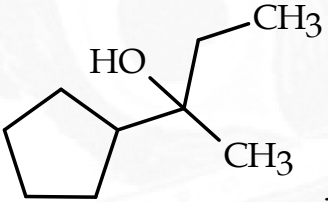


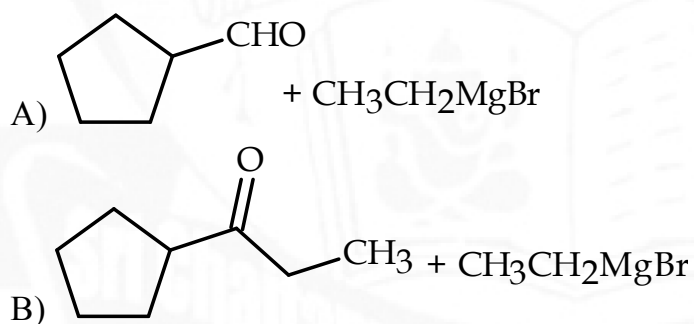
X will be

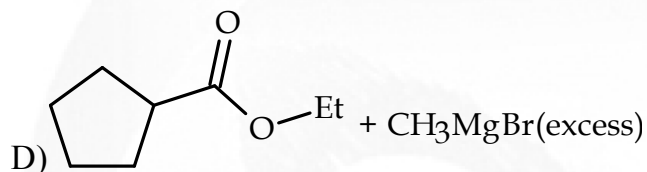
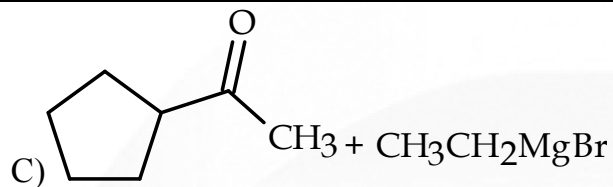


20. Which of the following alkenes give same product in HBO, oxymercuration – demercuration and acid catalysed hydration

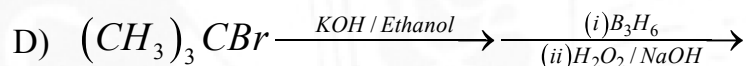
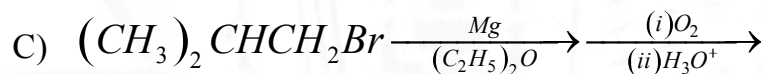


21. For preparation of , which of the following combination is most suitable

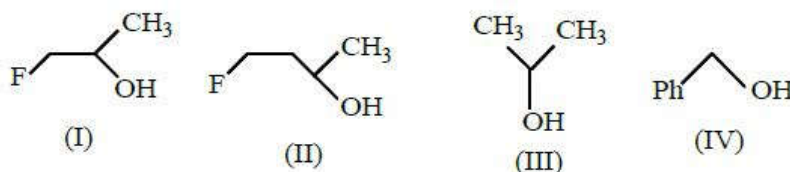




22. Primary alcohol can easily be prepared from primary alkyl halide via S_N2 reaction with aqueous NaOH . However, similar method does not work for the preparation of tertiary alcohol. Which reaction can be used for the efficient preparation of tertiary alcohol (tertiary butanol) from tertiary butyl bromide?



23. The order of reactivity of the following alcohols towards conc. HCl is



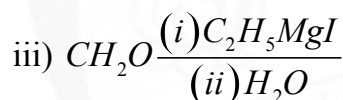
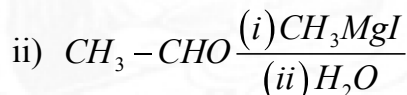
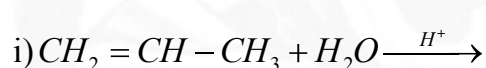
A) I > II > III > IV

B) I > III > II > IV

C) IV > III > II > I

D) IV > III > I > II

24. Which one/ones of the following reactions will yield 2-propanol? Choose the right answer from (A), (B), (C) and (D)



A) I and II

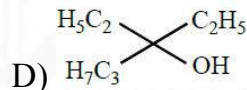
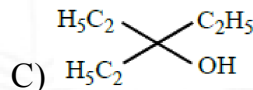
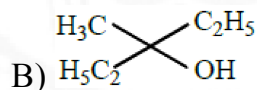
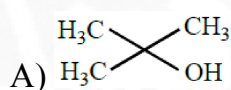
B) II and III

C) III and I

D) II and IV

25.

Ethyl ethanoate $\xrightarrow[(i) H_3O^+]{(i) CH_3MgBr (excess)}$ P. The product P is



SECTION-II

(ONE OR MORE OPTIONS CORRECT TYPE)

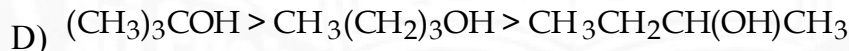
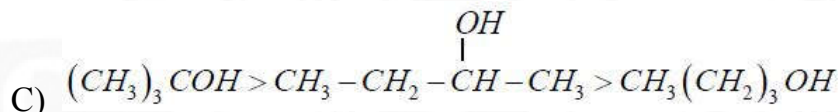
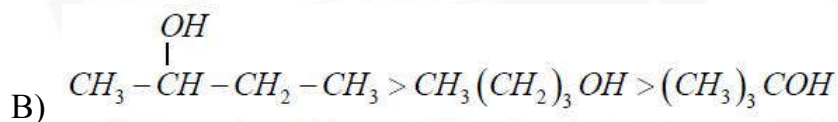
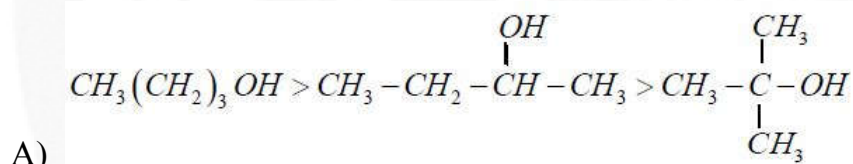
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Marking scheme: +4 for all correct options, 0 if not attempted and -2 in all wrong cases.

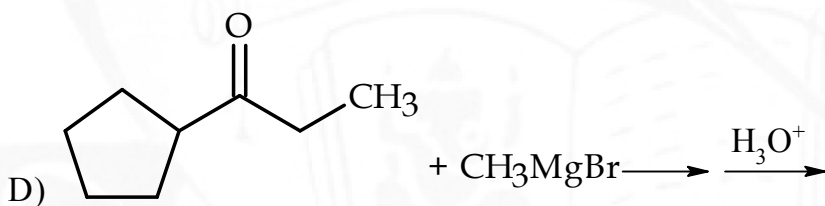
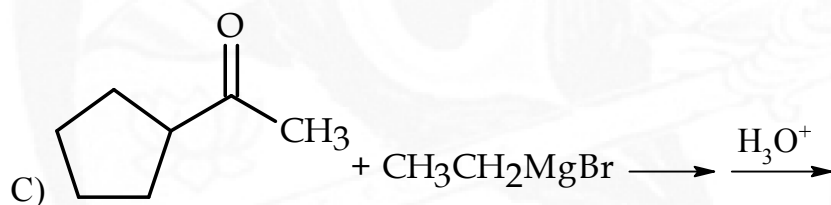
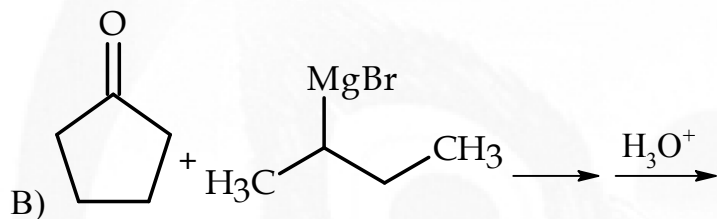
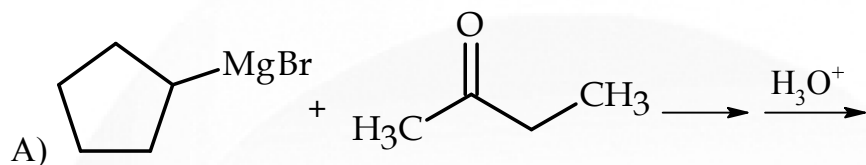
26. Which of the following can be reduced to alcohol on reaction with LiAlH_4 followed by acidification?



27. The correct order of decreasing ease of acid catalysed esterification.

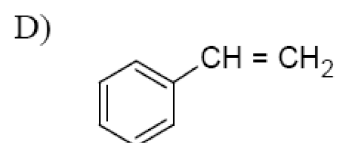
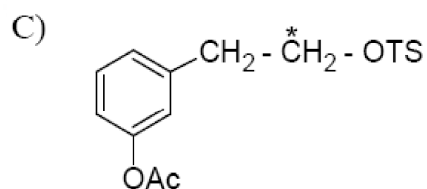
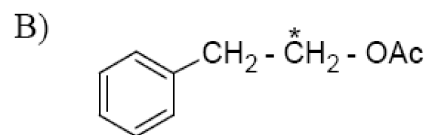
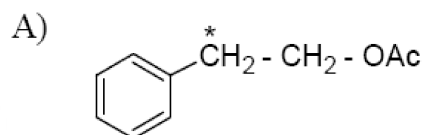
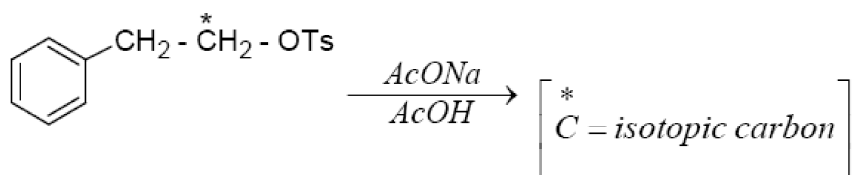


28. Which of the following Grignard's synthesis can result into 2 – cyclopentyl -2 butanol ?

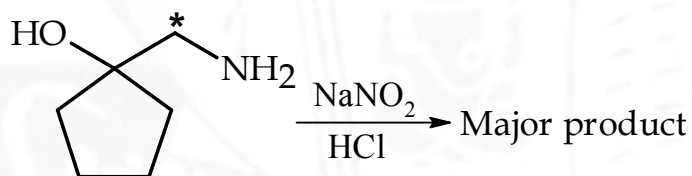


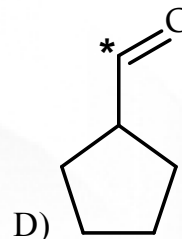
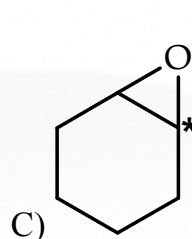
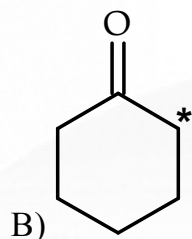
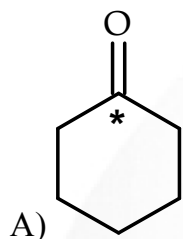
29.

Product (s) formed during this reaction are



30. The major product of the following reaction is





31. The reagent used to prepare alcohol from carbonyl compounds is/are
 A) $LiAlH_4$ B) Na/C_2H_5OH C) $NaBH_4$ D) CH_3MgBr
32. Consider the following substituted ethanol $G-CH_2CH_2OH$ which group(s) when present as G gives greater equilibrium of gauche conformer than its anti counterpart ?
 A) $-OH$ B) $-F$ C) $-NH_2$ D) $-NO_2$

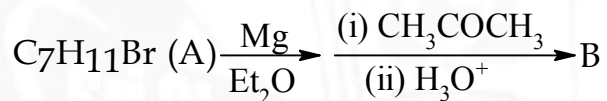
SECTION – III (PARAGRAPH TYPE)

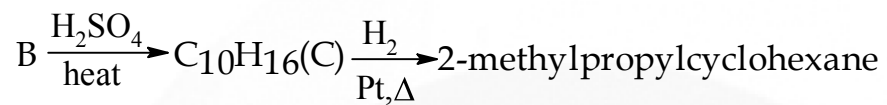
This section contains **2 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

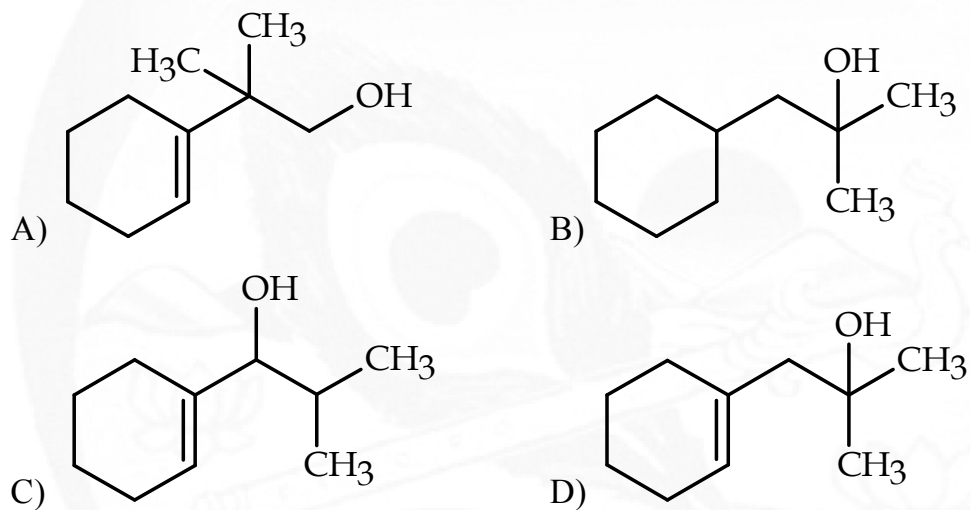
Paragraph For Questions 33 and 34

Consider the following sequence of reaction, compound

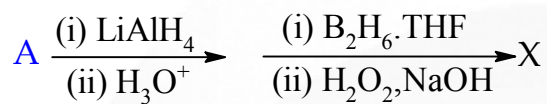




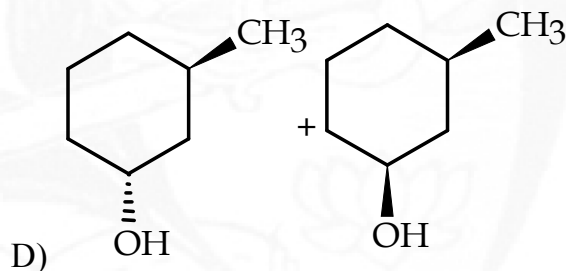
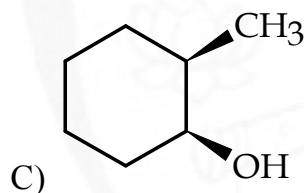
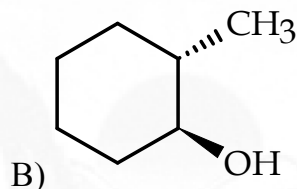
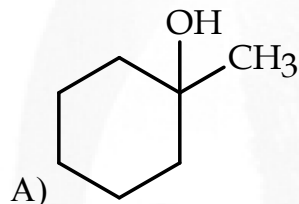
33. The structure of compound B is



34.



Hence, X is

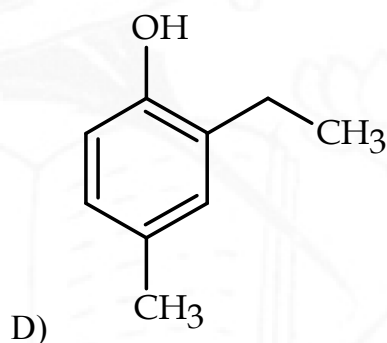
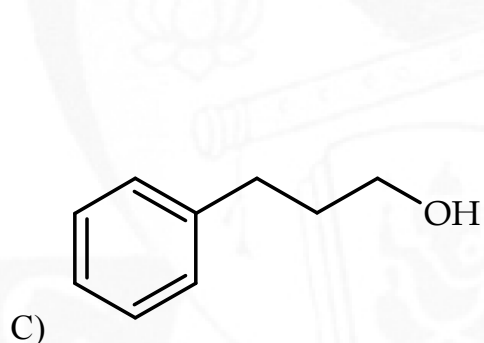
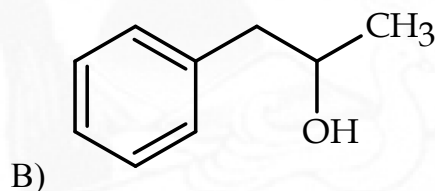
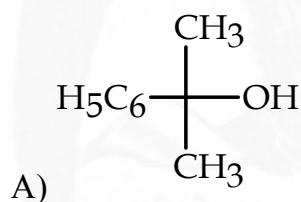
**Paragraph for Questions 35 and 36**

An organic compound X ($C_9H_{12}O$) gives the following reactions :

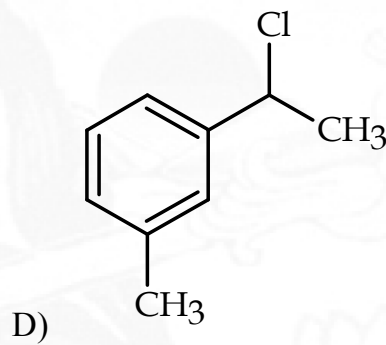
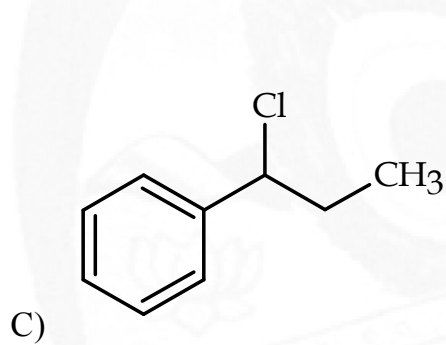
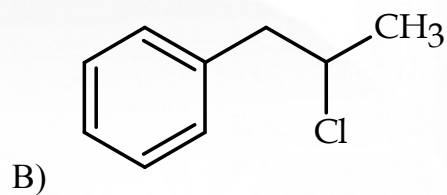
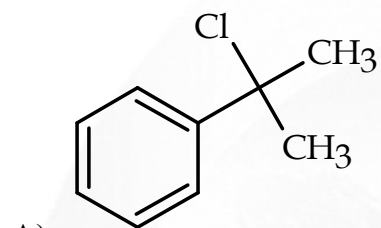
- Na- Slow gas bubble formation
- Acetic anhydride – Pleasant smelling liquid

- iii) $\text{CrO}_3 - \text{H}_2\text{SO}_4$ – Blue – green solution
- iv) Hot KMnO_4 – Benzoic acid
- v) $\text{Br}_2 - \text{CCl}_4$ – No decolouration
- vi) can give pseudonitrol acid in Victor Meyer's test.
- vii) X rotates the plane polarized light

35. The structure of X is



36. If X is treated with HCl in the presence of $ZnCl_2$, the major product would be



SECTION – I
(SINGLE CORRECT CHOICE TYPE)

This section contains **7 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Marking scheme +3 for correct answer , 0 if not attempted and -1 in all other cases.

37. Consider a parabola $y^2 = 4ax$, the length of focal chord is l and the length of the perpendicular from vertex to the chord is p then
- A) $l.p$ is constant B) $l.p^2$ is constant
C) $l^2.p$ is constant D) none of these
38. Two straight lines $y - b = m_1(x + a)p$ and $y - b = m_2(x + a)$ are the tangents of $y^2 = 4ax$. Then $m_1 m_2 =$
- A) $\frac{1}{2}$ B) -1 C) 1 D) 2
39. The triangle PQR of area 'A' is inscribed in the parabola $y^2 = 4ax$ such that the vertex P lies at the vertex of the parabola and the base QR is a focal chord. The modulus of the difference of the ordinates of the points Q and R is:
- A) $\frac{A}{2a}$ B) $\frac{A}{a}$ C) $\frac{2A}{a}$ D) $\frac{4A}{a}$

40. TP & TQ are tangents to the parabola, $y^2 = 4ax$ at P & Q. If the chord PQ passes through the fixed point $(-a, b)$ then the locus of T is:
- A) $ay = 2b(x - b)$ B) $bx = 2a(y - a)$ C) $by = 2a(x - a)$ D) $ax = 2b(y - b)$
41. The straight line joining any point P on the parabola $y^2 = 4ax$ to the vertex and perpendicular from the focus to the tangent at P, intersect at R, then the equation of the locus of R is
- A) $x^2 + 2y^2 - ax = 0$ B) $2x^2 + y^2 - 2ax = 0$
C) $2x^2 + 2y^2 - ay = 0$ D) $2x^2 + y^2 - 2ay = 0$
42. If A_1B_1 and A_2B_2 are two focal chords of the parabola $y^2 = 4ax$, then the chords A_1A_2 and B_1B_2 intersect on
- A) directrix B) axis
C) tangent at vertex D) none of these
43. The mirror image of the parabola $y^2 = 4x$ in the tangent to the parabola at the point $(1, 2)$ is
- A) $(x - 1)^2 = 4(y + 1)$ B) $(x + 1)^2 = 4(y + 1)$
C) $(x + 1)^2 = 4(y - 1)$ D) $(x - 1)^2 = 4(y - 1)$

SECTION-II

(ONE OR MORE OPTIONS CORRECT TYPE)

This section contains 7 multiple choice questions. Each question has four choices (A) (B), (C) and (D) out of which **ONE** or **MORE THAN ONE** are correct.

Marking scheme: +4 for all correct options, 0 if not attempted and -2 in all wrong cases.

44. Let PQ be a chord of the parabola $y^2 = 4x$. A circle drawn with PQ as a diameter passes through the vertex V of the parabola. If area of triangle PVQ = 20 sq units, then the coordinates of P are
A) (16,8) B) (16,-8) C) (-16,8) D) (-16,-8)
45. The focus and directrix of the parabola $9x^2 - 24xy + 16y^2 - 20x - 15y - 60 = 0$ is
A) $\left(-\frac{43}{25}, -\frac{129}{100}\right)$ B) $\left(\frac{43}{25}, \frac{129}{100}\right)$ C) $4x + 3y + \frac{53}{4} = 0$ D) $4x + 3y - \frac{53}{4} = 0$
46. Variable circle is described to pass through point (1,0) and tangent to the curve $y = \tan(\tan^{-1} x)$. The locus of the centre of the circle is a parabola whose:
A) length of the latus rectum is $2\sqrt{2}$
B) axis of symmetry has the equation $x + y = 1$
C) vertex has the co-ordinates $(3/4, 1/4)$
D) None of these

47. The locus of the midpoint of the focal distance of a variable point moving on the parabola, $y^2 = 4ax$ is a parabola whose
- A) latus rectum is half the latus rectum of the original parabola
B) vertex is $\left(\frac{a}{2}, 0\right)$
C) directrix is y-axis
D) focus has the co-ordinates (a,0)
48. P is a point on the parabola $y^2 = 4x$ and Q is a point on the line $2x + y + 4 = 0$. If the line $x - y + 1 = 0$ is the perpendicular bisector of PQ, then the co-ordinates of P can be:
- A) (1,-2) B) (4,4) C) (9,-6) D) (16,8)
49. If two distinct chords of a parabola $y^2 = 4ax$ passing through the point (a, 2a) are bisected by line $x + y = 1$, Then the length of the latusrectum cannot be:
- A) 2 B) 4 C) 5 D) 7
50. Let there be two parabolas with the same axis, focus of each being exterior to the other and the latusrecta being 4a and 4b. The locus of the middle points of the intercepts between the parabolas made on the lines parallel to the common axis is a
- A) Straight line if $a=b$ B) Parabola if $a \neq b$
C) Parabola $\forall a, b \in \mathbb{R}$ D) none of these

SECTION – III
(PARAGRAPH TYPE)

This section contains **2 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph For Questions 51 and 52

If l, m are variable real number such that $5l^2 + 6m^2 - 4lm + 3l = 0$, then variable line

$lx + my = 1$ always touches a fixed parabola, whose axes is parallel to x-axis.

51. Vertex of the parabola is

- A) $\left(-\frac{5}{3}, \frac{4}{3}\right)$ B) $\left(-\frac{7}{4}, \frac{3}{4}\right)$ C) $\left(\frac{5}{6}, -\frac{7}{6}\right)$ D) $\left(\frac{1}{2}, -\frac{3}{4}\right)$

52. Focus of the parabola is

- A) $\left(\frac{1}{6}, -\frac{7}{6}\right)$ B) $\left(\frac{1}{3}, \frac{4}{3}\right)$ C) $\left(\frac{3}{2}, -\frac{3}{2}\right)$ D) $\left(-\frac{3}{4}, \frac{3}{4}\right)$

Paragraph For Questions 53 and 54

Let the two parabola $y^2 = 4ax$ and $x^2 = 4ay, a > 0$ intersect at O and A (O being origin).

Parabola P whose directrix is the common tangent to the two parabolas and whose focus is the point which divides OA internally in the ratio $(1 + \sqrt{3}) : (7 - \sqrt{3})$

53. The equation of the parabola P is:

A) $(x-y)^2 = (2+\sqrt{3})a(x+y-(\sqrt{3})a)$

B) $(x-y)^2 = (2+\sqrt{3})a(2x+2y-(2+\sqrt{3})a)$

C) $(x-y)^2 = (2+\sqrt{3})a(2x+2y-(\sqrt{3})a)$

D) $(x-y)^2 = (2-\sqrt{3})a(x+y-(1+\sqrt{3})a)$

54. Extremities of latusrectum of P are:

A) $\left(\frac{a}{2}, \frac{(3+2\sqrt{3})a}{2}\right), \left(\frac{(3+2\sqrt{3})a}{2}, \frac{a}{2}\right)$

B) $\left(-\frac{a}{2}, \frac{(3-\sqrt{3})a}{2}\right), \left(\frac{(3-\sqrt{3})a}{2}, -\frac{a}{2}\right)$

C) $\left(\frac{a}{2}, \frac{(3-\sqrt{3})a}{2}\right), \left(\frac{(3-\sqrt{3})a}{2}, \frac{a}{2}\right)$

D) $\left(-\frac{a}{2}, \frac{(3+2\sqrt{3})a}{2}\right), \left(\frac{(3+2\sqrt{3})a}{2}, -\frac{a}{2}\right)$



Sri Chaitanya IIT Academy, India

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR-HYD

Sec: Jr.Super60-I

Time: 07:30 AM to 10:30 AM

Jee-Advanced

2017_P2 MODEL

Date: 20-01-18

Max.Marks:183

KEY SHEET

PHYSICS

1	B	2	A	3	C	4	D	5	A
6	C	7	C	8	AC	9	AB	10	AB
11	AD	12	ACD	13	AB	14	ABD	15	A
16	B	17	B	18	A				

CHEMISTRY

19	C	20	C	21	C	22	B	23	C
24	A	25	A	26	ABCD	27	A	28	ACD
29	AB	30	B	31	ABCD	32	ABCD	33	D
34	B	35	B	36	C				

MATHS

37	B	38	B	39	C	40	C	41	B
42	A	43	C	44	AB	45	AC	46	BC
47	ABCD	48	AC	49	BCD	50	AB	51	A
52	B	53	C	54	D				

SOLUTIONS**PHYSICS**

1. continuously 1 kW of heat is being dissipated from 25°C tank

$$\therefore \frac{dQ}{dt} = 10^3 = \frac{-KA[25 - T]}{\ell}$$

$$\Rightarrow \frac{dQ}{dt} = 10^3 = \frac{-0.2 \times 5 \times [25 - T]}{4 \times 10^{-2}}$$

$$\Rightarrow 25 - T = -40; \Rightarrow T = 65^{\circ}\text{C}$$

2. For 1st layer $\frac{dQ}{dt} = \frac{-KA\Delta T}{\ell} = \frac{-KA(10 - 20)}{\ell} = \frac{+KA \times 10}{\ell}$

$$\text{For 2nd layer } \frac{dQ}{dt} = \frac{-(2K)A(\theta - 10)}{\ell} = \frac{-2AK}{\ell}(\theta - 10)$$

Rate for both layers must be equal

$$\therefore \frac{kA \times 10}{\ell} = \frac{-2kA}{\ell}(\theta - 10); \Rightarrow \theta = 5^{\circ}\text{C}$$

3. if $\left(\frac{dQ}{dt}\right)_{AB} = 0$

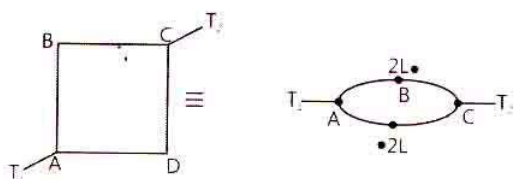
Then rate of heat flow from D to B must be equal to rate heat flow from B to C

$$\text{i.e. } \left(\frac{dQ}{dt}\right)_{DB} = \left(\frac{dQ}{dt}\right)_{BC}$$

$$\Rightarrow -\frac{kA(T_B - T_D)}{\ell_{DB}} = \frac{-kA(T_C - T_B)}{\ell_{BC}}$$

$$\Rightarrow \frac{(20 - 90)}{\ell_{DB}} = \frac{(0 - 20)}{\ell_{BC}}; \Rightarrow \frac{\ell_{BD}}{\ell_{BC}} = \frac{-70}{-20} = 3.5$$

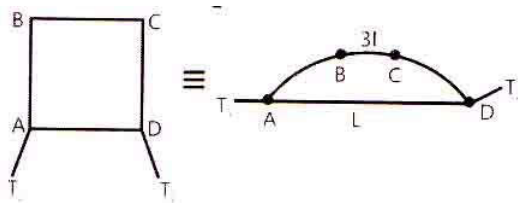
- 4.



$$\therefore \left(\frac{dQ}{dt}\right)_{AC} = \left(\frac{dQ}{dt}\right)_{ABC} + \left(\frac{dQ}{dt}\right)_{ADC}$$

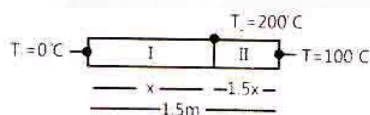
$$\Rightarrow W = \frac{-kA(T_2 - T_1)}{2L} + \frac{-kA(T_2 - T_1)}{2L}$$

$$\Rightarrow W = \frac{-kA(T_2 - T_1)}{L}$$



$$\begin{aligned} \therefore \left(\frac{dQ}{dt} \right)_{AD} &= \frac{-kA(T_2 - T_1)}{3L} + \frac{-kA(T_2 - T_1)}{L} \\ &= \frac{-4kA(T_2 - T_1)}{L} = \frac{4}{3}W \end{aligned}$$

5.



Mass of ice melting per second = mass of steam produced per sec

$$\begin{aligned} \Rightarrow \frac{-kA(0 - 200)}{x \times 80} &= \frac{-kA(100 - 200)}{(1.5 - x)(540)} \\ \Rightarrow \frac{1.5 - x}{x} &= \frac{1}{2} \times \frac{80}{540} \\ \Rightarrow x &= \frac{27 \times 1.5}{29} = 1.3966m \end{aligned}$$

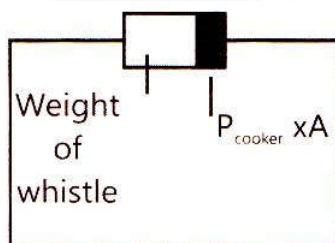
$$\therefore 1.5 - x = 0.1034m = 10.34cm$$

6. Mass of whistle = 100 gm = 0.1 kg

$$\therefore \text{Weight of whistle} = 1 \text{ N}$$

To just lift the whistle. Pressure in pressure cooker must be equal to = Atmospheric pressure + pressure due to weight of whistle

$$= 10^5 + \frac{1N}{0.1 \times 10^{-4} m^2} = 2 \times 10^5 Pa$$



Free - body diagram of whistle Patm A

By Force – balance

$$P_{\text{atm}} \times A + \text{weight of whistle} = P_{\text{cooker}} \times A$$

$$\Rightarrow P_{\text{cooker}} = P_{\text{atm}} + \frac{\text{weigh of whistle}}{A}$$

$$\text{Initially, it is given } P = 10^5 \text{ Pa}, V = 10 \text{ L}, T = 300 \text{ K}$$

$$\therefore P.V. = nRT; \Rightarrow nR = \frac{10 \times 10^5}{300} \text{ Pa.L / K} \dots\dots 300 \text{ K}$$

$$\text{Finally, we require } P = 2 \times 10^5 \text{ Pa}, V = 10 \text{ L}, T = ?$$

$$\therefore \text{ By gas equation : - } PV = nRT$$

$$\Rightarrow 2 \times 10^5 \times 10 = \frac{10 \times 10^5}{300} \times T [\text{using (1)}]$$

$$\Rightarrow T = 600 \text{ K} = 327^\circ \text{ C}$$

$$7. \quad r_\theta = 3r_p$$

$$P = eA\sigma T^4$$

$$\Rightarrow mc \frac{dT}{dt} = eA\sigma T^4$$

(C : specific heat, m : mass of sphere)

$$\Rightarrow \frac{dT}{dt} = \frac{-eA\sigma T^4}{mc} = \frac{-eA\sigma T^4}{(pV)c} = \frac{-e4\pi r^2 \sigma T^4}{\frac{4}{3}\pi r^3 c}$$

(V : volume of sphere)

$$\Rightarrow \frac{dT}{dt} = \frac{1}{r} \times \left[\frac{-3e\sigma T^4}{c} \right]$$

[Quantity in parenthesis is Constant for both spheres]

$$\therefore \frac{\left(\frac{dT}{dt}\right)_P}{\left(\frac{dT}{dt}\right)_Q} = \frac{\frac{1}{r_P}}{\frac{1}{r_Q}} = 3 = \times$$

$$8. \quad \rho = \frac{m}{V}$$

$$\Rightarrow \rho \times \frac{4}{3}\pi r^3 = m; \Rightarrow r \propto (m)^{1/3}$$

and Area of sphere (A) $\propto r^2$

$$\therefore \frac{A_A}{A_B} = (4)^{2/3} = (2)^{4/3}$$

$$\therefore \text{Ratio of heat loss} = \frac{eA_A\sigma(T-T_0)^4}{eA_B\sigma(T-T_0)^4}$$

$$= \frac{A_A}{A_B} = (2)^{4/3}$$

By Newton's law of cooling :

$$\frac{dQ}{dt} = ms \frac{dT}{dt} = -k(T-T_0)$$

Where $k = 4eA\sigma T_0^3$

$$\therefore \frac{dT}{dt} \propto \frac{A}{m}$$

$$\therefore \frac{\left(\frac{dT}{dt}\right)_A}{\left(\frac{dT}{dt}\right)_B} = \frac{\frac{A_A}{m_A}}{\frac{A_B}{m_B}} = \frac{(2)^{4/3}}{4} = 2^{-2/3}$$

9. $eA = 0.01$ and $eB = 0.81$

$$A_A = A_B$$

$$E_A = E_B$$

$$\Rightarrow eA\sigma A_A T_A^4 = e_B\sigma A_B T_B^4$$

$$\Rightarrow 0.01T_A^4 = 0.81T_B^4$$

$$\Rightarrow T_B = \frac{1}{3} \times T_A$$

$$\Rightarrow T_B = \frac{1}{3} \times 5802 = 1934 \text{ K}$$

By Wien's displacement law

$$\lambda_m T = \text{const.} = 2.93 \times 10^{-3} \text{ mK}$$

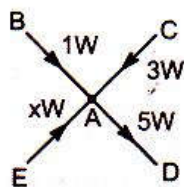
$$\therefore \lambda_{m_A} = 0.5 \mu\text{m}$$

Since, it is given in the question that

$$\lambda_{m_B} = 1 \mu\text{m} + \lambda_{m_A}$$

$$\therefore \lambda_{m_B} = 1.5 \mu m$$

10. Area under the curve gives the rate at which heat per unit surface is radiated by the body i.e. total rate of heat radiation = (Area under the curve) $\times$ (surface area of the body)
11. In graph $\frac{dT}{dx}$ is not constant it is changing. $\frac{dT}{dx}$ is increasing continuously then A is decreasing.



12.

Heat flow

At A

$$x + 1 + 3 = 5$$

 $x = 1$ Heat in flows from E

$$\therefore T_E > T_A$$

$$\therefore T_C > T_A > T_D$$

$$T_B - T_A = T_E - T_A$$

$$\therefore T_B = T_E$$

$$13. \frac{T_A}{T_B} = \frac{\lambda_B}{\lambda_A} = \frac{800}{400} = 2$$

$$\frac{E_A}{E_B} = \frac{\sigma T_A^4}{\sigma T_B^4} = 16$$

$$\frac{P_A}{P_B} = \frac{\sigma T_A^4 \times 4\pi r_A^2}{\sigma T_B^4 \times 4\pi r_B^2} = 4$$

$$14 \quad \text{Heat loss} = [\text{Heat loss}]_{\text{radiation}} + [\text{Heat loss}]_{\text{convection}}$$

$$= eA\sigma T^4 + k[T - T_0]^n$$

Where e = surface property ; T = body temperature T_0 = surrounding temperature

15. $A_A = A; \quad A_B = A$

$$\ell_A = \ell; \quad \ell_B = \ell$$

$$k_A = 3k; \quad k_B = k$$

$$\therefore Ra = \frac{\ell_A}{k_A A_A} = \frac{\ell}{3kA}; Ra = \frac{\ell_B}{k_B A_B} = \frac{\ell}{k_A}$$

$$\therefore \frac{R_A}{R_B} = \frac{1}{3}$$

16. Rate at which heat flows from A = Rate at which heat flows from B

$$\Rightarrow \left(\frac{dQ}{dt} \right)_A = \left(\frac{dQ}{dt} \right)_B$$

$$\Rightarrow \frac{T_A}{R_A} = \frac{T_B}{R_B} \Rightarrow \frac{T_A}{T_B} = \frac{R_A}{R_B} = \frac{1}{3}$$

$$G_A = \frac{T_A}{L_A} = \frac{T_A}{L} \text{ and } G_B = \frac{T_B}{L_B} = \frac{T_B}{L}$$

$$\therefore \frac{G}{G_B} = \frac{T_A}{T_B} = \frac{1}{3}$$

17. $\left(\frac{dQ}{dt} \right)_{\text{initially}} = KA v \Delta T$

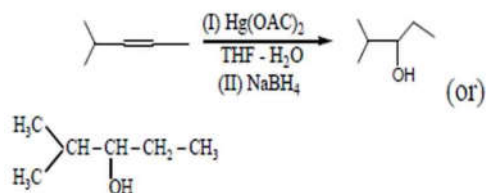
$$\left(\frac{dQ}{dt} \right)_{\text{finally}} = KA(4V) \left(\frac{\Delta T}{2} \right) = 2 \left(\frac{dQ}{dt} \right)_{\text{initially}}$$

18. If all the parameters are kept constant then

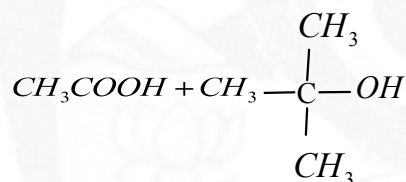
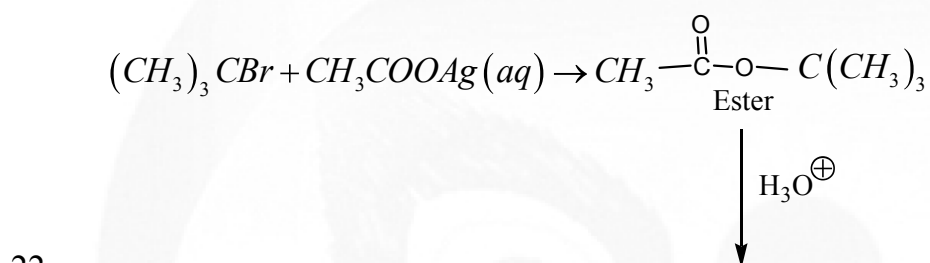
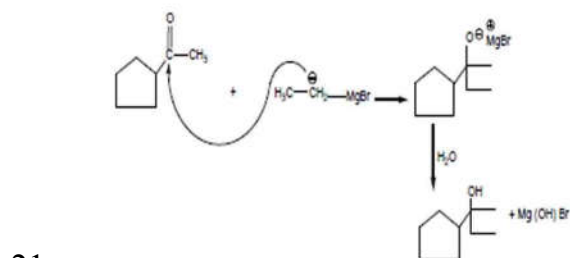
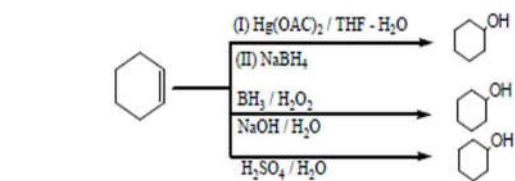
$$\frac{dQ}{dt} = ms \frac{dT}{dt} = kA v \Delta T$$

$$\therefore \frac{dT}{dt} = \frac{kA v \Delta T}{ms}$$

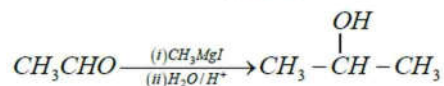
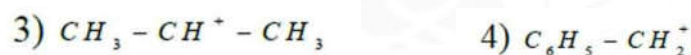
CHEMISTRY



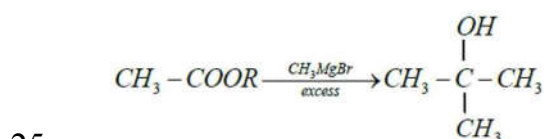
19.



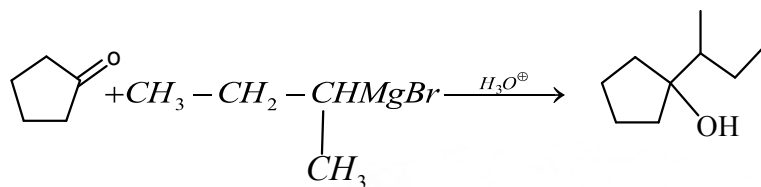
23. When Alcohol reacts with Hx, the carbocations formed in these cases are



24. α - propanol

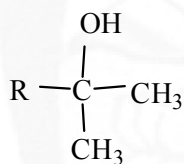


26. conceptual
27. case of acid catalysed esterification $1^0 > 2^0 > 3^0$
28. Option (b) does not give 2-cyclopentyl-2-butanol as



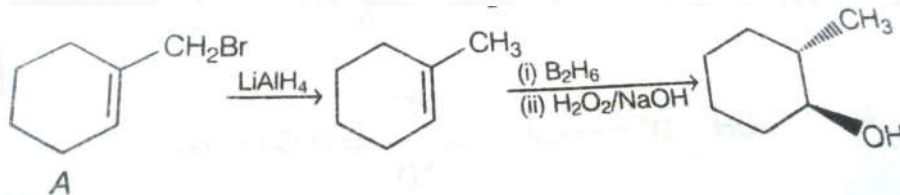
1-(1-methyl propyl) cyclopentanol

29. ngp MECHANISM
30. conceptual
31. All are converts carbonyl compounds to alcohols
32. All these groups increases the stability of gauche conformer due to stable intramolecular H-bonding.
33. B is an alcohol formed by the attack of Grignard's reagent on acetone. Hence, alcohol must have the skeleton

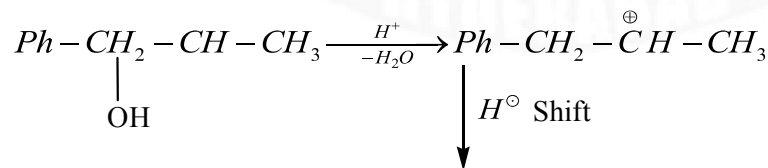


Also, R is C_7H_{11} with two degree of unsaturation it must be cyclohexenyl methyl not cyclohexyl methyl.

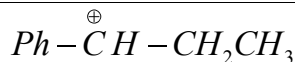
34. Hydroboration oxidation at double bond gives anti-Markownikoff's addition of H_2O in syn orientation.



35. CONCEPTUAL



- 36.



Benzylic

**MATHS**

37. ℓ = length of focal chord $= a\left(t + \frac{1}{t}\right)^2$, Let $P(at^2, 2at), Q\left(\frac{a}{t^2}, \frac{-2a}{t}\right)$ be the ends of

focal chord Equation of PQ is $y\left(t - \frac{1}{t}\right) = 2(x - a)$

$$\therefore p = \frac{2a}{\sqrt{\left(t - \frac{1}{t}\right)^2 + 2^2}} = \frac{2a}{\left(t + \frac{1}{t}\right)}$$

$$\therefore \ell p^2 = \frac{4a^2}{\left(t + \frac{1}{t}\right)^2} a\left(t + \frac{1}{t}\right)^2 = 4a^3$$

38. Let the equation of tangent passing through $p(x_1, y_1)$ is

$$y = mx + \frac{a}{m} \Rightarrow (y_1 - mx_1) = \frac{1}{m} \Rightarrow x_1 m^2 - (y_1)m + 1 = 0$$

$$\text{Let } m_1 \text{ and } m_2 \text{ are roots} \Rightarrow m_1 + m_2 = \frac{y_1}{x_1} \text{ and } m_1 m_2 = \frac{1}{x_1}$$

$$\text{but } m_1 = \tan \theta_1 \text{ and } m_2 = \tan \theta_2 \text{ and } \theta_1 + \theta_2 = \frac{\pi}{4} \Rightarrow \frac{m_1 + m_2}{1 - m_1 m_2} = 1$$

$$\Rightarrow \frac{(y_1/x_1)}{1 - \frac{1}{x_1}} = 1 \Rightarrow y_1 = x_1 - 1$$

$\therefore$ locus is $y = x - 1$

39. Q R is a focal chord $\Rightarrow Q = (at_1^2, 2at_1)$ and $R = (at_2^2, 2at_2) \Rightarrow t_1 t_2 = -1$

$$A = t_2 \begin{vmatrix} 0 & 0 & 1 \\ at_1^2 & 2at_1 & 1 \\ at_2^2 & 2at_2 & 1 \end{vmatrix} = |a^2 t_1 t_2 (t_1 - t_2)| = -|a^2 (t_1 - t_2)| \Rightarrow |2a(t_1 - t_2)| = \frac{2A}{a}$$

40. Chord of contact of $T(x_1y_1)$ is PQ

i.e $yy_1 = 2a(x + x_1)$ passes through $(-a, b)$

$$\Rightarrow by_1 = 2a(-a + x_1)$$

$\therefore$ locus is $by = 2a(x - a)$

41. Tangent at $p(t)$ is $yt = x + at^2$ _____ (1)

Line $\perp$ er to (1) and passing through $(a, 0)$ is

$$y - 0 = -t(x - a) \Rightarrow y = t(a - x)$$
 _____ (2)

$$\text{Equation of OP is } y = \frac{2}{t}x$$
 _____ (3)

Eliminating t from (2) and (3) we get $2x^2 + y^2 - 2ax = 0$

42. A_1B_1 is a focal chord $\Rightarrow A_1 = (at_1^2, 2at_1) \Rightarrow B_1 = \left(\frac{a}{t_1^2}, \frac{-2a}{t_1}\right)$

A_2B_2 is a focal chord $\Rightarrow A_2 = (at_2^2, 2at_2) \Rightarrow B_2 = \left(\frac{a}{t_2^2}, \frac{-2a}{t_2}\right)$

$$\text{Equation of } A_1A_2 \text{ is } y(t_1 + t_2) - 2x - 2at_1t_2 = 0$$
 _____ (1)

$$\text{Equation of } B_1B_2 \text{ is } y\left(\frac{-1}{t_1} - \frac{1}{t_2}\right) - 2x - 2a\left(\frac{-1}{t_1}\right)\left(\frac{-1}{t_2}\right) = 0$$

$$\Rightarrow y(t_1 + t_2) + 2xt_1t_2 + 2a(t_1t_2 + 1) = 0$$
 _____ (2)

$$(1), (2) \Rightarrow (x + a)(t_1t_2 + 1) = 0 \Rightarrow x + a = 0$$

43. Equation of tangent at $(1, 2)$ is $x - y + 1 = 0$ _____ (1)

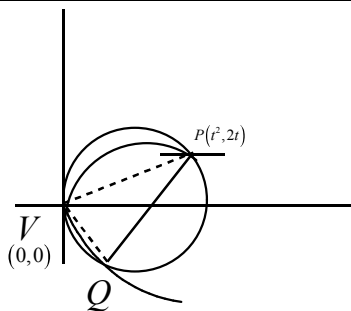
Let $P = (t^2, 2t)$ is a point on $y^2 = 4x$

Image of P w.r.t (1) is $((2t - 1), t^2 + 1)$

$$x = 2t - 1 \text{ and } y = t^2 + 1$$

$$\Rightarrow (x + 1)^2 = 4(y - 1)$$

44.



$$y^2 = 4x \quad (1)$$

$$\text{Slope of } PV = \frac{2}{t}$$

$$\text{Equation Q V is } y = \left(-\frac{t}{2}\right)x \quad (2)$$

$$(1), (2) \Rightarrow Q = \left(\frac{16}{t^2}, -\frac{8}{t}\right)$$

$$\text{Area of } \Delta PVQ = \frac{1}{2} \cdot PV \cdot VQ = 20$$

$$\Rightarrow PV^2 \cdot QV^2 = 40^2$$

$$\Rightarrow (t^2 - 16)(t^2 - 1) = 0 \Rightarrow t = \pm 4, \pm 1$$

$$\Rightarrow Q(16, 8) \text{ (or) } (16, -8)$$

$$45. \quad (3x - 4y)^2 - 5(4x + 3y + 12) = 0$$

$$\Rightarrow \left(\frac{3x - 4y}{5}\right)^2 \cdot 25 = 5\left(\frac{4x + 3y + 12}{5}\right) \cdot 5$$

$$\Rightarrow \left(\frac{3x - 4y}{5}\right)^2 = \left(\frac{4x + 3y + 12}{5}\right) \quad (1)$$

$$\text{Is of the form } Y^2 = 4AX \text{ where } Y = \frac{3x - 4y}{5}, X = \frac{4x + 3y + 12}{5}, 4A = 1$$

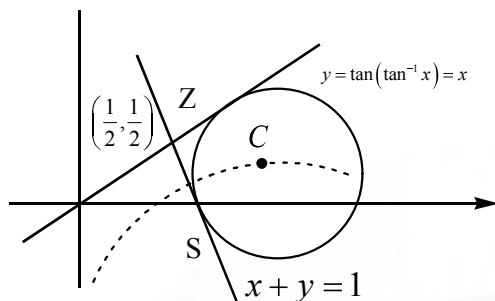
$$\Rightarrow \text{Directrix is } X = -A \Rightarrow \frac{4x + 3y + 12}{5} = -\frac{1}{4} \Rightarrow 4x + 3y + \frac{53}{4} = 0 \quad (2)$$

$$\text{Axis is } Y = 0 \Rightarrow 3x - 4y = 0 \quad (3)$$

$$\text{Focus is on } X = A \Rightarrow \frac{4x + 3y + 12}{5} = \frac{1}{4} \Rightarrow 4x + 3y + \frac{43}{4} = 0 \quad (4)$$

$$(3), (4) \Rightarrow S = \left(\frac{-43}{25}, \frac{-129}{100} \right)$$

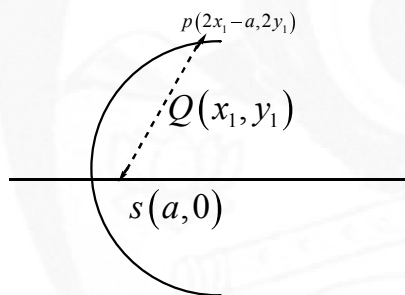
46.



Center of the circle C is equidistant from $y=x$ and $(1, 0) \Rightarrow y=x$ directrix and $(1,0)$ is the focus $\therefore 4a = 2(SZ) = \sqrt{2}$

Axis is $x + y = 1$, vertex $= \left(\frac{3}{4}, \frac{1}{4} \right)$

47.



Let $Q(x_1, y_1)$ be the mid point of SP $\Rightarrow P = (2x_1 - a, 2y_1)$ lies on $y^2 = 4ax$

$$\Rightarrow 4y_1^2 = a(2x_1 - a) \Rightarrow y_1^2 = 2ax_1 - a^2$$

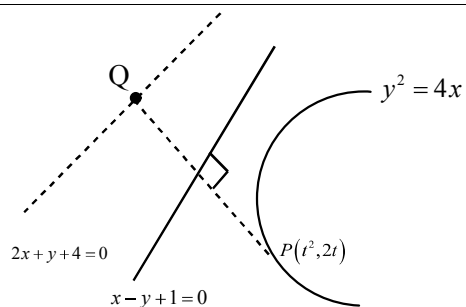
$$\text{Locus is } y^2 = 2a \left(x - \frac{a}{2} \right)$$

$$\text{Vertex} = \left(\frac{a}{2}, 0 \right)$$

$$\text{Directrix is } x - \frac{a}{2} = -\frac{a}{2} \Rightarrow x = 0$$

$$\text{Focus} = (a, 0)$$

48.



$$y^2 = 4x \quad (1)$$

$$2x + y + 4 = 0 \quad (2)$$

$$\text{Let } P = (t^2, 2t)$$

But Q is image of P .w.r.t $x - y + 1 = 0$

$$\frac{h - t^2}{1} = \frac{k - 2t}{-1} = -2 \frac{(t^2 - 2t + 1)}{2}$$

$$\Rightarrow Q = (2t - 1, t^2 + 1)$$

$$Q \text{ lies on } (2) \Rightarrow t = -1, -3$$

$$\Rightarrow P = (1, -2) \text{ (or) } (9, -6)$$

49. Let $P(a, 2a)$ and Q be the ends of a chord of $y^2 = 4ax$ _____ (1)

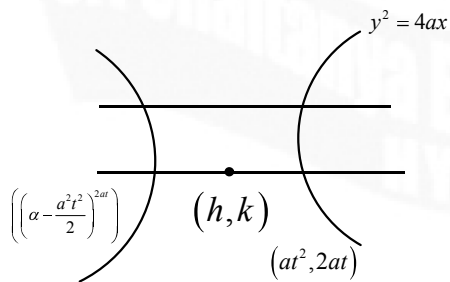
Let $R(x_1, y_1)$ be the mid point of PQ lies on $x + y = 1$ _____ (2)

$$\Rightarrow Q = (2x_1 - a, 2 - 2x_1 - 2a) \text{ lies on } (1)$$

$$\Rightarrow x_1^2 - 2x_1 + (2a^2 - 2a + 1) = 0 \text{ has real roots } \Rightarrow \Delta > 0 \Rightarrow a(a - 1) > 0$$

$$\Rightarrow a \in (0, 1) \Rightarrow 4a \in (0, 4)$$

50.



$$K = 2at \text{ and } 2h = at^2 + \alpha - \frac{a^2 t^2}{b}$$

$$\Rightarrow y^2 \left(\frac{b-a}{4ab} \right) \cdot \frac{k^2}{4a^2} + \alpha = 2x$$

If $a = b$ then $2x = \alpha$ a straight line

If $a \neq b$ then $y^2(b-a) = 8abx - 4ab\alpha$ a parabola

PASSAGE I

The equation of parabola is $(y-a)^2 = 4b(x-c)$ _____ (1)

$\ell x + my = 1$ can be written as

$$y-a = \frac{-\ell}{m}(x-c) + \frac{\ell-am-\ell c}{m} \text{ _____ (2)}$$

$$2 \text{ touch (1)} \Rightarrow \frac{\ell-am-\ell c}{m} = \frac{b}{-\ell/m} \Rightarrow c\ell^2 - bm^2 - a\ell m - \ell = 0 \text{ _____ (3)}$$

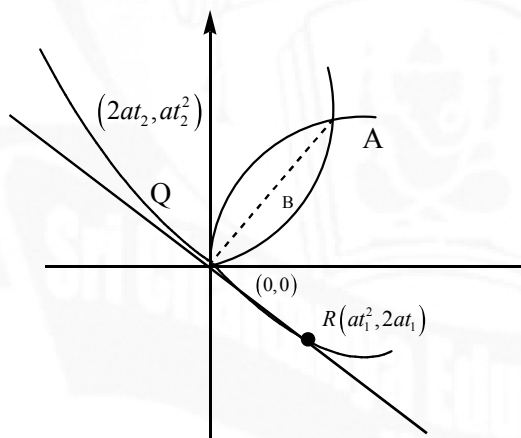
$$\text{but } 5\ell^2 + 6m^2 - 4\ell m + 3\ell = 0 \text{ _____ (4)}$$

$$(3), (4) \Rightarrow \frac{c}{5} = \frac{-b}{6} = \frac{a}{-4} = \frac{-1}{3}$$

$$\therefore \text{ The parabola is } \left(y - \frac{4}{3} \right)^2 = 8 \left(x + \frac{5}{3} \right)$$

Answers 51) A 52) B

PASSAGE II



$$\text{Equation of tangent at Q is } xt_2 - y = at_2^2 \quad (1)$$

$$\text{P is } xt_1 - y = at_1^2 \quad (2)$$

$$(1) \text{ and } (2) \text{ are identical} \Rightarrow -t_1 = \frac{-1}{t_2} = \frac{at_1^2}{at_2}$$

$$\Rightarrow t^4 = -t \Rightarrow t = -1$$

$\therefore$ common tangent is $x + y + a = 0$

$$A = (4a, 4a)$$

$$B = \left(\left(1 + \frac{\sqrt{3}}{2} \right) a, \left(1 + \frac{\sqrt{3}}{2} \right) a \right)$$

$$\text{Latusrectum of parabola} = 2 \left[\frac{\left(1 + \frac{\sqrt{3}}{2} \right) a + \left(1 + \frac{\sqrt{3}}{2} \right) a}{\sqrt{2}} \right] = \sqrt{2} (2 + \sqrt{3}) a$$

$$\text{Equation of parabola P is } \frac{(y-x)^2}{2} = \sqrt{2} (2 + \sqrt{3}) a \left[\frac{x + y - \left(\frac{1 + \sqrt{3}}{2} \right) a}{\sqrt{2}} \right]$$

$$(y - x^2) = (2 + \sqrt{3}) a (2x + 2y - (1 + \sqrt{3}) a)$$

Final the extremities of latusrectum

Answer

53) C

54) D

Success



JEE ADVANCED WEEKEND TEST



Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr. Super60-I

Time: 07:30 AM to 10:30 AM

Date: 27-01-18

Max. Marks: 180

Name of the Student: _____

I.D. NO:

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2014_PAPER-I

IMPORTANT INSTRUCTIONS:

- 1) This booklet is your Question Paper.
- 2) Use the Optical Response Sheet (ORS) provided separately for answering the questions
- 3) Blank spaces are provided within this booklet for rough work.
- 4) Write your name, roll number and sign in the space provided on the back cover of this booklet.
- 5) You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET:

- 6) Darken the appropriate bubbles on the ORS by applying sufficient pressure. This will leave an impression at the corresponding place on the Candidate's sheet.
- 7) The ORS will be collected by the invigilator at the end of the examination.
- 8) Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
- 9) Write your name, roll number and code of the examination center, and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else** on the ORS. Darken the appropriate bubble under each digit of your roll number.

DARKENING THE BUBBLES ON THE ORS

- 10) Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
- 11) Darken the bubble **COMPLETELY**.
- 12) The correct way of darkening a bubble is as:
- 13) The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- 14) Darken the bubbles **ONLY** IF you are sure of the answer. There is **NO WAY** to erase or

"un-darken" a darkened bubble.

27-01-18_Jr.Super60-I_Jee Adv_WAT-33_Weekend Syllabus

MATHS: SYLLABUS COVERD FROM 22-01-18 TO 26-01-2018

PHYSICS: SYLLABUS COVERD FROM 22-01-18 TO 26-01-2018

CHEMISTRY: Preparation, properties of phenols and ethers

JEE-ADVANCE-2014-P1-Model

Time: 07:30 AM to 10:30 AM

IMPORTANT INSTRUCTIONS

Max Marks: 180

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 10)	Questions with Multiple Correct Choice	3	0	10	30
Sec – II(Q.N : 11 – 20)	Questions with Integer Answer Type	3	0	10	30
Total				20	60

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 21 – 30)	Questions with Multiple Correct Choice	3	0	10	30
Sec – II(Q.N : 31 – 40)	Questions with Integer Answer Type	3	0	10	30
Total				20	60

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 41 – 50)	Questions with Multiple Correct Choice	3	0	10	30
Sec – II(Q.N : 51 – 60)	Questions with Integer Answer Type	3	0	10	30
Total				20	60

Sec: Jr. Super60-I

Space for rough work

Page 2

SECTION-1

(ONE OR MORE OPTIONS CORRECT TYPE)

This section contains 10 multiple choice questions. Each question has four choices (A) (B), (C) and (D) out of which **ONE** or **MORE THAN ONE** are correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

1. The equation of a transverse wave travelling in a rope is given by

$$y = A \cos \frac{\pi}{2} [kx - \omega t - \alpha] \text{ where } A = 0.6m, k = 0.005cm^{-1}, \omega = 8.0s^{-1} \text{ and } \alpha \text{ is a non-}$$

vanishing constant. Then for this wave,

- A) the wavelength of the wave is $\lambda = 8m$
 B) the maximum velocity v_m of a particle of the rope will be ; $v_m = 7.54m/s$
 C) magnitudes of velocity and slope of y-x graph for an element of rope are inversely proportional to each other .
 D) magnitudes of velocity and slope of y-x graph for an element of rope are directly proportional to each other.
2. A wire of $9.8 \times 10^{-3} kg/m$ passes over a frictionless light pulley fixed on the top of a frictionless inclined plane which makes an angle of 30° with the horizontal. Masses m and M are tied at the two ends of wire such that ' m ' rests on the plane and M hangs

freely vertically downwards. The entire system is in equilibrium and a transverse wave propagates along the wire with a velocity of 100 m/s.

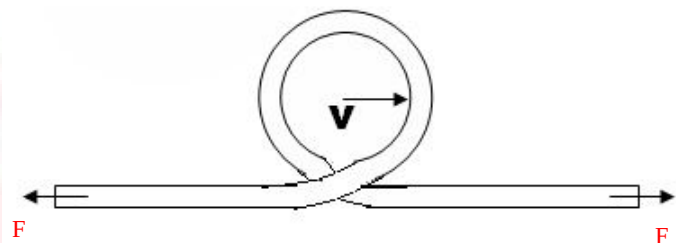
- A) $m = 20\text{kg}$ B) $M = 5\text{kg}$ C) $\frac{m}{M} = \frac{1}{2}$ D) $\frac{m}{M} = 2$

3. $y(x, t) = \frac{0.8}{[(4x+5t)^2 + 5]}$ represents a moving pulse, where x and y are in the metres

and t is in seconds, then

- A) pulse is moving in $+x$ direction
 B) in 2s it will travel a distance of 2.5 m
 C) its maximum displacement is 0.16 m
 D) it is a symmetric pulse
4. A transverse wave travelling on a taut string is represented by $Y = 0.01\sin 2\pi(10t - x)$ Y and x are in meters and t in seconds. Then,
- A) the speed of the wave is 10m/s.
 B) closest points on the string which differ in phase by 60° are $(1/6)\text{m}$ apart.
 C) maximum particle velocity is $\pi / 4 \text{ m/s}$.
 D) the phase of a certain point on the string changes by 120° in $(1/20)$ seconds.

5. Two identical point particles A and B are placed in front of a concave mirror of focal length 20 cm, at distances 10 cm and 30 cm respectively. The particles oscillate perpendicular to the principal axis, such that the displacement equation for both the particles is given by $Y_A = Y_B = 0.1 \sin(\pi t)$. The correct statement (s) is/are
- A) displacement equation of the image of particle A is $Y'_A = 0.2 \sin(\pi t + \pi/2)$
- B) displacement equation of the image of particle B is $Y'_B = 0.2 \sin(\pi t + \pi)$
- C) time period of oscillation of both the images is 2s.
- D) relative vertical displacement of image of A with respect to image of B is $Y'_{AB} = 0.4 \sin(\pi t)$
6. A long, heavy, flexible rope with mass per unit length ' ρ ' is stretched by a constant force F . A sudden movement causes a circular loop of radius R to form at one end of the rope. The loop rolls along the rope with speed ' v ' as shown in the figure. Then



A) $v = \sqrt{\frac{F}{\rho}}$

B) $v = \sqrt{\frac{F}{2\rho}}$

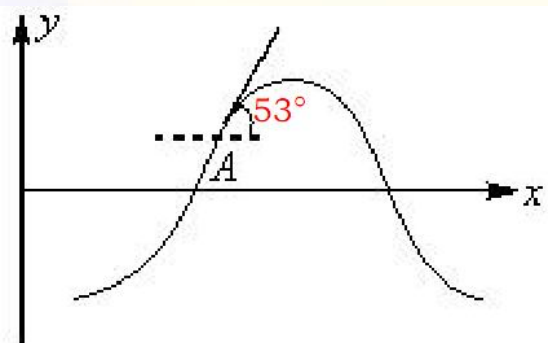
C) the kinetic energy carried by the loop is $2\pi RF$ D) the kinetic energy carried by the loop is $\pi R \rho v^2$ 7. Which of the following functions of x and t represent a progressive wave?

A) $y = \sin(4t - 3x)$ B) $y = \frac{1}{4 + (4t - 3x)^2}$ C) $y = \frac{1}{4t + 3x}$ D) $y = 3x^2 + 4t$

8. One end of a string of length 'L' is tied to ceiling of a lift accelerating upwards with an acceleration '2g'. The other end of the string is free. The linear mass density of the string varies linearly from 0 to ' λ ' from bottom to top. A transverse pulse is generated near the bottom of string.

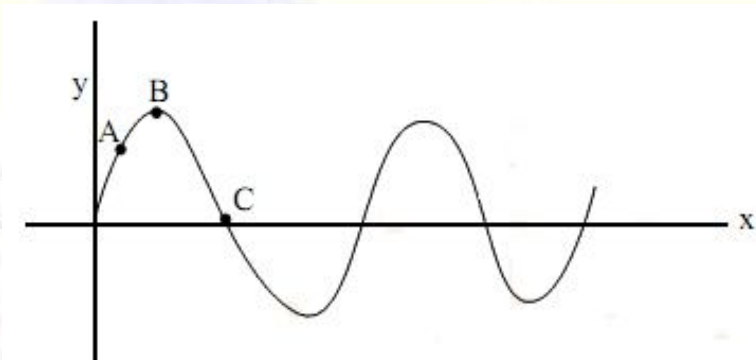
A) Acceleration of the pulse w.r.t. ground is $\frac{3g}{4}$.B) Relative to string, acceleration of the pulse on the string will be $3g/4$ every where.C) The time taken by a pulse to reach from bottom to top will be $\sqrt{8L/3g}$.D) The time taken by a pulse to reach from bottom to top will be $\sqrt{4L/3g}$.

9. y-x curve at an instant for a transverse wave travelling along x axis on a string is shown. Angle made by tangent with horizontal at the point A on the curve is 53° , as shown.



- A) Transverse velocity of the particle at point A is parallel to positive y direction if the wave is travelling along positive x-axis.
- B) Transverse velocity of the particle at point A is parallel to positive y direction if the wave is travelling along negative x-direction.
- C) Magnitude of transverse velocity of the particle at point A is greater than the wave speed
- D) Magnitude of transverse velocity of the particle at point A is less than the wave speed

10. A plane progressive transverse wave is passing through a medium along the positive x-axis. At the given instant of time, the nature of the displacement(y), position(x) graph is shown in figure. Which of the following statements are correct? (Consider A, B, C are three particles of the vibrating string).



- A) Elastic potential energy per unit length of the string at the point B is more than that at points A and C
- B) Elastic potential energy per unit length of the string at the point C is more than that at points A and B
- C) Velocity of the particle at A is along negative y direction.
- D) Velocity of the particle at C is along positive y direction.

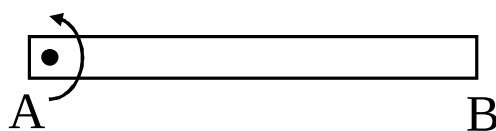
SECTION-II
INTEGER TYPE

This section contains 10 questions. The answer to each question is a **single digit integer, ranging** from 0 to 9 (both inclusive).

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

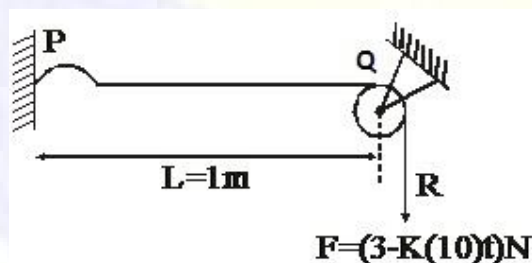
11. The speed of a transverse wave, going on a wire having a length 50cm and mass 5g is 80 m/s. the area of cross section of the wire is 1.0 mm^2 and its Young's modulus is $8 \times 10^{11} \text{ N/m}^2$. Find the extension (in $\times 10^{-2} \text{ mm}$) of the wire over its natural length.
12. A uniform rope of mass 0.1kg and length 2.45m hangs from a ceiling. The speed of a transverse wave in the rope at a point 0.5m distant from the lower end is $\sqrt{2.45 \times n}$ where $n =$
13. A 4.0 kg block is suspended from the ceiling of an elevator through a string having a linear density of $19.2 \times 10^{-3} \text{ kg/m}$. The speed (with respect to the string) with which a wave pulse can proceed on the string if the elevator accelerates up at the rate of 2.0 m/s^2 is $12.5n$. What is the value of n ? Take $g = 10 \text{ m/s}^2$ and $m_{\text{string}} \ll 4 \text{ kg}$.

14. An ant with mass ' m ' is standing peacefully on top of a horizontal, stretched rope. The rope has mass per unit length ' μ ' and is under tension F . Without warning, a student starts a sinusoidal transverse wave of wavelength ' λ ' propagating along the rope. The wave is in vertical plane. What minimum wave amplitude (in mm) can make that ant feel weightless momentarily? Assume m is so small that the presence of the ant has no effect on the propagation of the wave. [Given: $\lambda = 0.5m$, $\mu = 0.1 \text{ kg/m}$, $F = 3.125N$, take $g = \pi^2$]
15. Two small pulses are created on the rope at points A & B respectively as shown in the diagram.



The rope is rotating on horizontal plane with constant angular velocity about end A where as end B is free. S_A & S_B are the distances covered by pulse A & pulse B respectively in rope frame before they cross each other. If $k = \frac{S_A}{S_B}$, find the value of $(k-1)^2$.

16. As shown in figure, string PQR is stretched by force $F = (3 - 10 Kt)$ N, where K is a constant and t is time in seconds. At time $t = 0$, a small pulse is generated at the fixed end P of the string. Find the value of K (in N/s) if the value of force becomes zero as the pulse reaches point Q. (Given mass per unit length of string is 0.03 kg/m)



17. Two small blocks each having a mass of 3.2 kg are connected by wire CD and the system is suspended from the ceiling by another vertical wire AB with AB wire connected (fixed) to the mid point of wire CD. The linear mass density of the wire AB is 10 g/m and that of CD is 80 g/m . If the ratio of speeds of transverse pulse in AB to CD is $4/n$ find n .
18. The speed of a transverse wave, going on a uniform wire having a length 50 cm and mass 5.0 g , is 80 m/s . The area of cross-section of the wire is 1.0 mm^2 and its Young's modulus is $16 \times 10^9 \text{ N/m}^2$. Find the extension (in mm) of the wire over its natural length.

19. An ant sitting on a string can pull the string with a maximum force up to three times of its own weight. A sine wave of amplitude $A = \frac{10}{\pi^2}$ in vertical plane is generated in the string. Find the least time period of oscillations of string elements so that ant will not fall from the string. (Neglect the variation of tension in the string due to weight of the ant $g = 10 \text{ m/s}^2$).
20. A simple harmonic wave of amplitude 2mm is travelling along positive x-axis. At a given instant, displacements for two different particles on the wave are 1 mm and $\sqrt{2} \text{ mm}$ along positive Y direction. If wavelength of the wave is 48 mm then find minimum separation between mean positions of the particles (in mm).

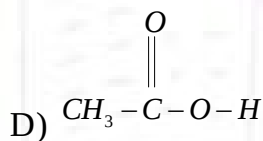
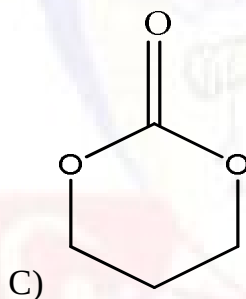
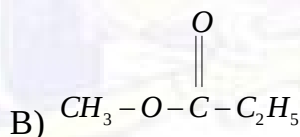
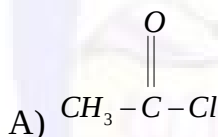
SECTION-1

(ONE OR MORE OPTIONS CORRECT TYPE)

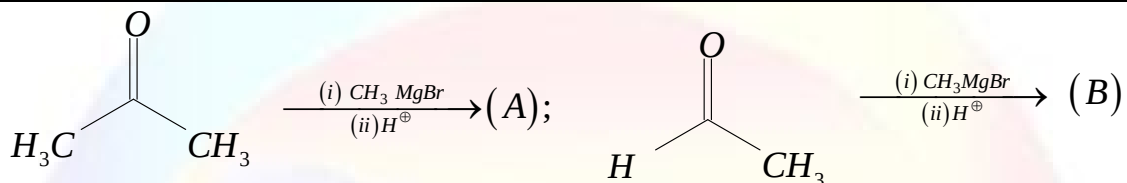
This section contains 10 multiple choice questions. Each question has four choices (A) (B), (C) and (D) out of which **ONE** or **MORE THAN ONE** are correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

21. Which of the following compounds when reacts with PhMgBr (excess) followed by acidification will yield 3° alcohol?



22.



Which of the following reagent can be used to differentiate between A and B?

A) Victor-Mayer test

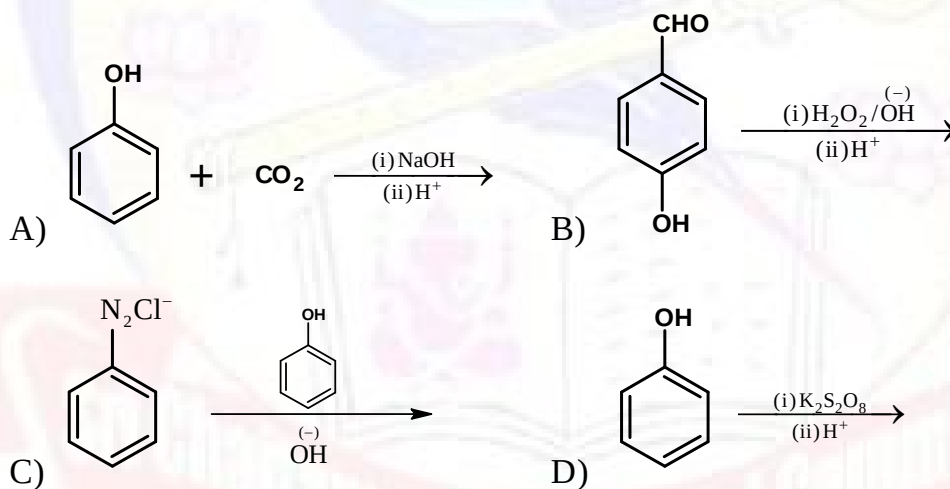
B) Sodium metal

C) Lucas reagent

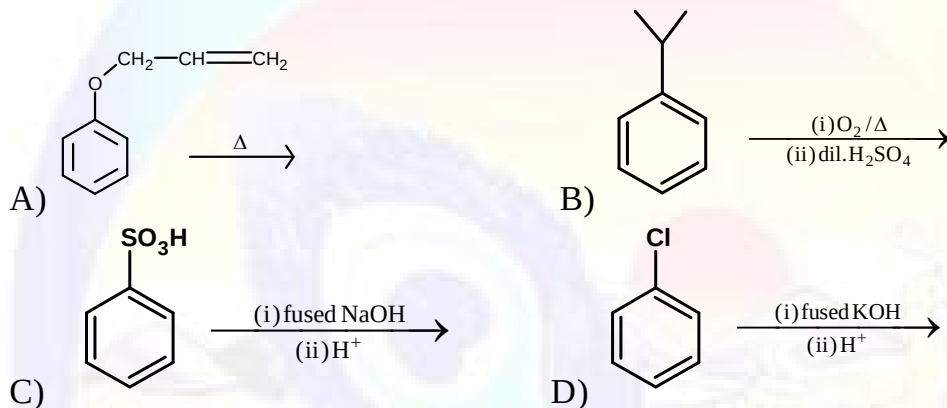
D) $\text{K}_2\text{Cr}_2\text{O}_7 / \text{H}^{\oplus}$

23.

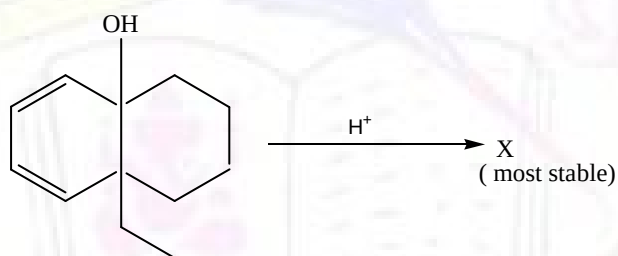
Which reaction can convert phenol to hydroquinone



24. In which of the following reactions, a phenolic compound is formed as product ?

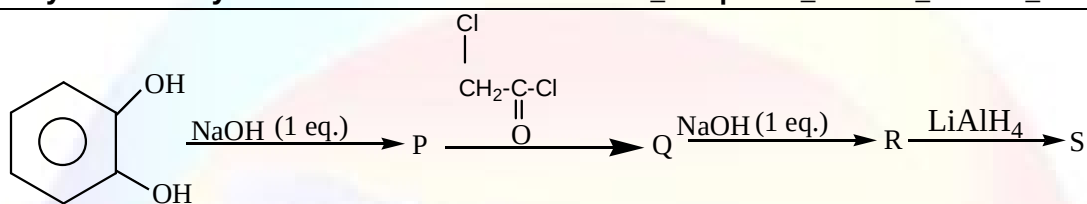


25. Which of the following changes are involved in the formation of most stable product X under given condition?

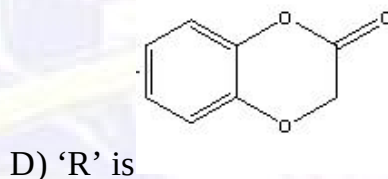
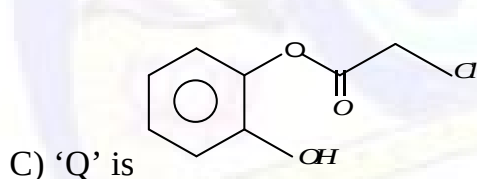
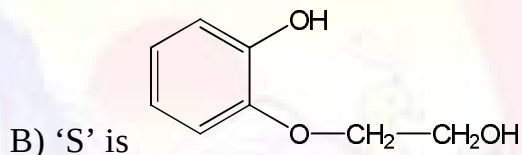
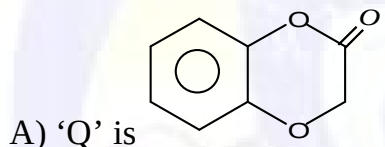


- A) Delocalisation of charge
- B) 1, 2 shift of Ethyl group
- C) Aromatisation of ring in which ethyl substituent is present
- D) Aromatisation of ring in which ethyl substituent is absent

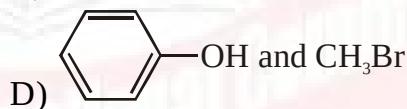
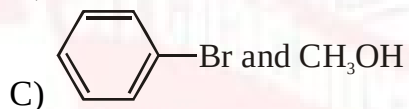
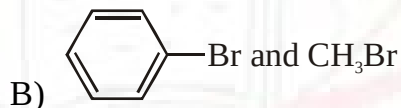
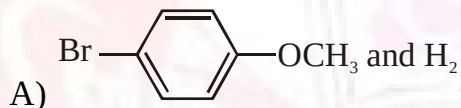
26.



select the correct statement(s)



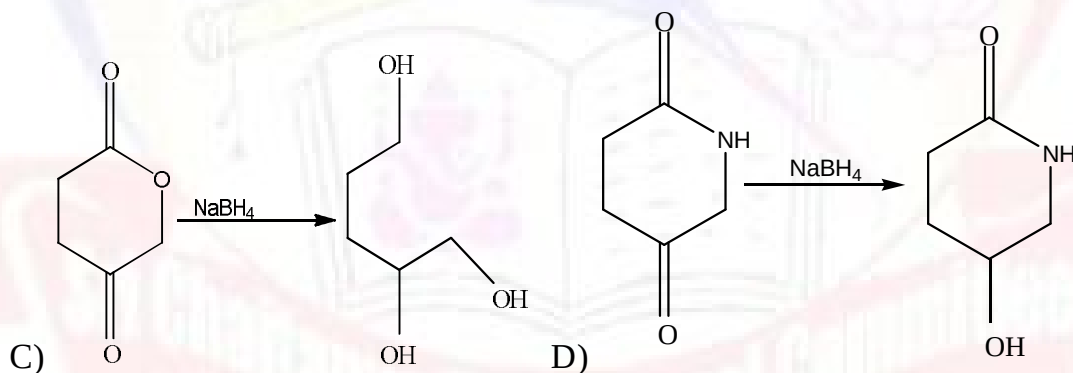
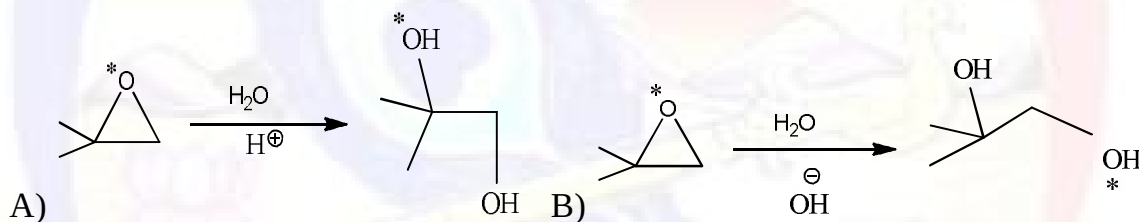
27. In the reaction the products are



28. Which of the following will result in the formation of an ether as major product?



29. Select incorrect reaction(s):



30. Diethyl ether is prepared through

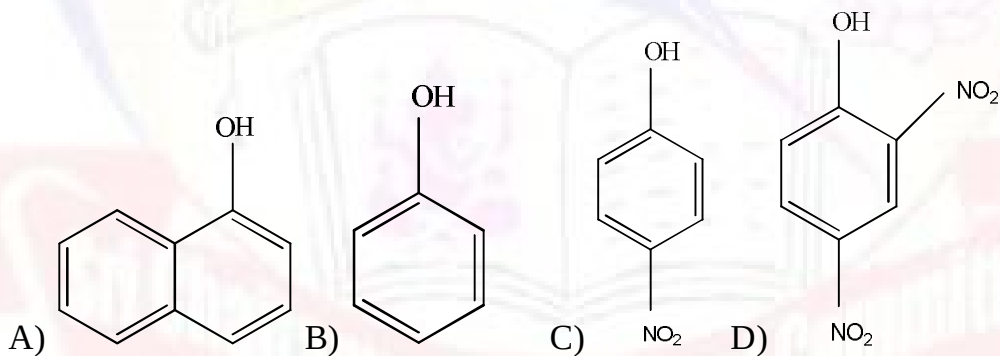
- A) The reaction between ethanol and conc. H_2SO_4 at $140^\circ C$
- B) The reaction between ethyl bromide and sodium ethoxide
- C) heating of ethyl bromide with dry silver oxide
- D) Heating ethanol with conc. H_2SO_4 at $170^\circ C$

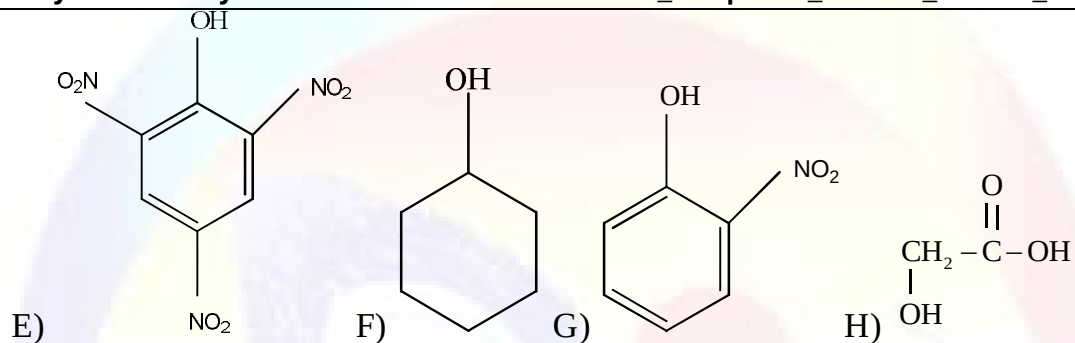
SECTION-II
INTEGER TYPE

This section contains 10 questions. The answer to each question is a **single digit integer, ranging** from 0 to 9 (both inclusive).

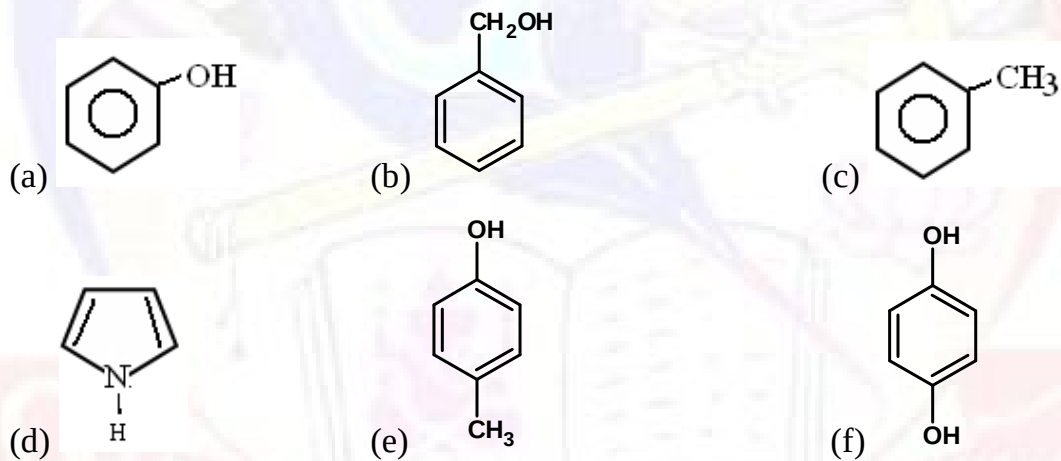
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

31. At room temperature how many of the following compounds can decompose $NaHCO_3$ to liberate CO_2 ?

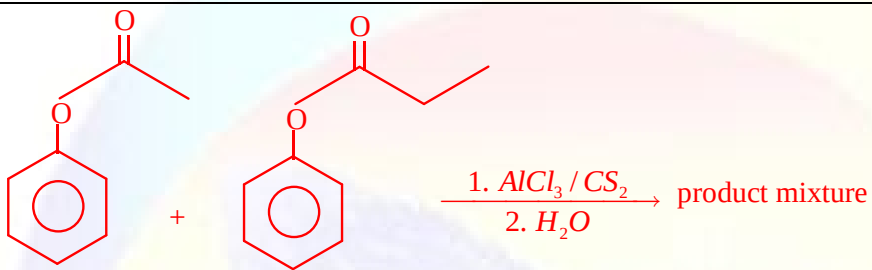




32. How many of the following compounds give Reimer-Tiemann reaction?



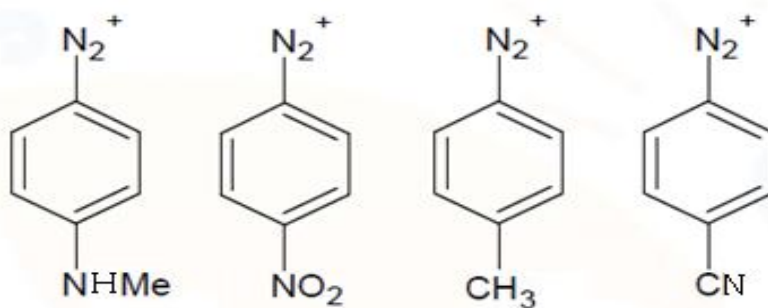
33.



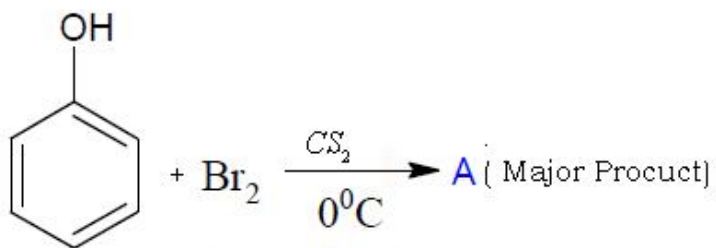
How many compounds are exist in the product mixture?

34. The diazonium ions give diazo-coupling with phenol in the presence of dilute NaOH.

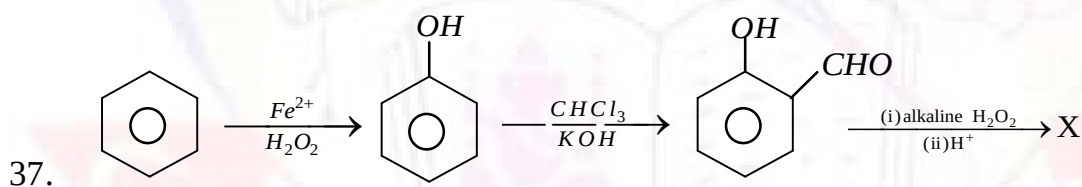
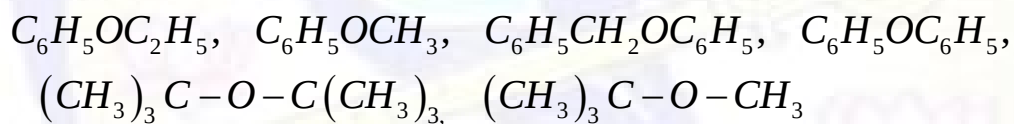
How many of the following are more reactive than Benzene diazonium chloride?



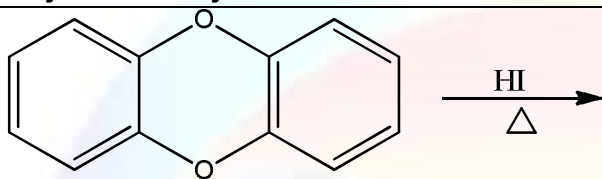
35. Number of bromine atoms in **A** are **X**, therefore X is



36. How many of the following ethers cannot be prepared by Williamson's synthesis?



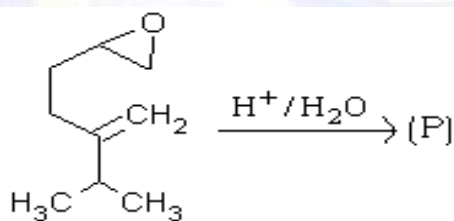
The number of hydroxyl groups present in the product **X** is/are



38. (A)

Number of moles of HI consumed per mole of (A) is X. Identify the value of X

39. Phenol $\xrightarrow[\text{Dust}]{\text{Zn}}$ X $\xrightarrow[\text{Anhydrous AlCl}_3]{\text{C}_2\text{H}_5\text{Cl}}$ Y (Monoalkylated product); the number of structural benzenoid isomers of Y (including Y) are



40.

Five membered ring is major product then the sum of degrees of alcoholic groups present in P is

SECTION-1

(ONE OR MORE OPTIONS CORRECT TYPE)

This section contains 10 multiple choice questions. Each question has four choices (A) (B), (C) and (D) out of which **ONE** or **MORE THAN ONE** are correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

41. 'P' is a point which moves in XY-plane such that the point 'P' is nearer to the centre of a square than any of the sides. The four vertices of the square are $(\pm a, \pm a)$. The region in which 'P' will move is bounded by parts of parabola/parabolas having possible equations is/are
- A) $x^2 = a^2 + 2ay$ B) $y^2 + 2ax = a^2$ C) $y^2 = a^2 + 2ax$ D) $x^2 = a^2 + ay$
42. Let $y^2 = 4ax$ be a parabola and $x^2 + y^2 + 2bx = 0$ be a circle. If parabola and circle touch each other externally then
- A) $a > 0, b > 0$ B) $a > 0, b < 0$ C) $a < 0, b < 0$ D) $a < 0, b > 0$
43. TP and TQ are tangents to parabola $y^2 = 4x$ and normal's at P and Q intersect at a point R on the curve. The locus of the centre of the circle circumscribing triangle TPQ is a parabola whose:
- A) Vertex is $(1, 0)$ B) foot of directrix $\left(\frac{7}{8}, 0\right)$
- C) Length of latus rectum $\frac{1}{4}$ D) focus is $\left(\frac{9}{8}, 0\right)$

44. Variable chords of the parabola $y^2 = 4ax$ subtend a right angle at the vertex. Then:
- A) Locus of feet of the perpendicular from the vertex on these chords is a circle
 - B) Locus of the middle points of the chords is a parabola
 - C) Variable chords passes through a fixed point on the axis of the parabola
 - D) None
45. A parabola which touches x-axis at $(1,0)$ and y-axis at $(0,2)$. Then
- A) Vertex is $\left(\frac{16}{25}, \frac{2}{25}\right)$
 - B) Vertex is $\left(\frac{16}{5}, \frac{2}{5}\right)$
 - C) Focus is $\left(\frac{4}{5}, \frac{2}{5}\right)$
 - D) Focus is $\left(\frac{4}{25}, \frac{2}{25}\right)$
46. Equation of chord joining at $A(t_1)$, $B(t_3)$ on $y^2 = 4ax$ subtends a right angle at $C(t_2)$ and $D(t_4)$ on the same parabola. Then
- A) $\left|\frac{t_2+t_4}{t_1+t_3}\right| = 1$
 - B) $t_2t_4 - t_1t_3 = 0$
 - C) $\left|\frac{t_2+t_4}{t_1+t_3}\right| = \frac{1}{2}$
 - D) $t_2t_4 - t_1t_3 = 4$
47. If two distinct chords of a parabola $y^2 = 4ax$ passing through the point $(a, 2a)$ are bisected by the line $x + y = 1$, then the length of the latus rectum cannot be
- A) 5
 - B) 6
 - C) 9
 - D) 11

48. The tangent PT and normal PN to the parabola $y^2 = 4ax$ at a point P on it meet its axis at points T and N respectively. The locus of the centroid of the triangle PTN is a parabola whose ($a > 0$)
- A) Vertex is $\left(\frac{2a}{3}, 0\right)$ B) focus is $(2a, 0)$
- C) Length of latus rectum is $\frac{4a}{3}$ D) directrix is $x = \frac{a}{3}$
49. Normal's drawn at points P, Q and R lying on the parabola $y^2 = 4x$ intersect at $(3, 0)$. Then
- A) Centroid of $\Delta^{le} PQR = \left(\frac{3}{2}, 0\right)$ B) Circumcentre of $\Delta^{le} PQR = \left(\frac{5}{3}, 0\right)$
- C) Area of $\Delta^{le} PQR = 2$ D) Radius of circumcircle of $\Delta^{le} PQR$ is $\frac{5}{2}$
50. Through a point $P(-2, 0)$, tangents PQ and PR are drawn to the parabola $y^2 = 8x$. Two circles each passing through the focus of the parabola and one touching parabola at Q and other at R are drawn. Which of the following points(s) with respect to the triangle PQR lie (s) on the common chord of the two circles?
- A) Centroid B) orthocenter C) incentre D) Circumcentre

SECTION-II
INTEGER TYPE

This section contains 10 questions. The answer to each question is a **single digit integer, ranging** from 0 to 9 (both inclusive).

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

51. The radius of circle passing through the focus of parabola $x^2 = 4y$ and touches it at $(6, 9)$ is equal to $\lambda\sqrt{10}$ then $\lambda = \dots\dots\dots$
52. Abscissa of two points P and Q on the parabola $y^2 = 8x$ are roots of equation $x^2 - 17x + 11 = 0$. Tangents at P and Q meet at point T. Distance of T from the focus of the parabola.
53. A line 'L' with slope 2 cuts parabola $y^2 = 8x$ to form a chord. If the midpoint of chord lies on $x = 4$ then the equation of L is $y = 2x + k$. Then $\frac{|k|}{3} = \dots\dots\dots$
54. Let $L_1 : x + y = 0$ and $L_2 : x - y = 0$ are tangent to a parabola whose focus is $S(1, 2)$. If the length of latus-rectums of the parabola can be expressed as $\frac{m}{\sqrt{n}}$ (m and n are co prime). Then the value of $m + n - 4$ is $\dots\dots\dots$
55. A pair of variable straight lines $5x^2 + 3y^2 + \alpha xy = 0$ ($\alpha \in R$), cut $y^2 = 4x$ at two points P and Q. If the locus of the point of intersection of tangents to the given parabola at P and Q is $Kx = 12$ ($K \in N$). Then $K = \dots\dots\dots$

56. m_1, m_2, m_3 are the slope of normal's ($m_1 < m_2 < m_3$) drawn through the point $(9, -6)$ to the parabola $y^2 = 4x$; and $A = (a_{ij})$ is square matrix of order 3 such that
- $$a_{ij} = \begin{cases} 1 & \text{if } i \neq j \\ m_i & \text{if } i = j \end{cases} \quad \text{Then } |A| = \dots\dots\dots$$
57. A tangent is drawn to the parabola $y^2 = 4x$ at the point 'P' whose abscissa lies in the interval $[1, 4]$. The maximum possible area of the triangle formed by the tangent at 'P', ordinate of the point 'P'; and the x-axis is equal to K . Then the value of $\sqrt{K}$ is.....
58. If the co-ordinate of the point on the parabola $y = x^2 + 7x + 2$, which is nearest to the straight line $y = 3x - 3$ is represented as (K_1, K_2) . Then the value of $\frac{|K_1 + K_2|}{2} = \dots\dots\dots$
59. If the lengths of two perpendicular tangents from a point to a parabola are of 3 and 4 units and the length of latus rectum of the parabola is K units. Then $[K] = \dots\dots$
 ($[.]$ denotes G.I.F)
60. Let S be the focus of the parabola $y^2 = 8x$ and PQ be the common chord of the circle $x^2 + y^2 - 2x - 4y = 0$ and the given parabola. The area of triangle PQS is.....



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KEY SHEET

PHYSICS

1	ABD	2	AD	3	BCD	4	AB	5	BCD
6	AC	7	AB	8	BC	9	BC	10	BCD
11	4	12	2	13	4	14	2	15	2
16	2	17	1	18	2	19	1	20	2

CHEMISTRY

21	ABC	22	ACD	23	BD	24	ABCD	25	ABC
26	BCD	27	D	28	C	29	ABC	30	ABC
31	3	32	4	33	4	34	2	35	1
36	2	37	2	38	0	39	4	40	4

MATHS

41	ABC	42	AC	43	ABD	44	ABC	45	AC
46	AD	47	ABCD	48	ACD	49	CD	50	ABCD
51	5	52	7	53	2	54	7	55	5
56	8	57	4	58	5	59	4	60	4

SOLUTIONS**PHYSICS**

1. Conceptual

$$2. \quad v = \sqrt{\frac{T}{\mu}} \Rightarrow 10^4 = \frac{T}{9.8 \times 10^{-3}} \Rightarrow T = 98$$

$$T = Mg \Rightarrow M = 10, T = mg \sin 30^\circ$$

$$\Rightarrow m = 20$$

$$\therefore \frac{m}{M} = 2$$

3. Conceptual

4. Conceptual

5. Displacement of image of particle A is in phase with the displacement of particle A hence the displacement equation of particle A's image is

$$Y'_A = 0.2 \sin(\pi t) \quad \dots(1)$$

Displacement of image of particle B is out of phase with the displacement of particle B therefore the displacement equation of particle B's image is

$$Y'_B = 0.2 \sin(\pi t + \pi) \quad \dots(2)$$

Relative vertical displacement of A.w.r.t. to B is

$$\begin{aligned} Y_{AB}' &= Y'_A - Y'_B \\ &= 0.2 \sin(\pi t) - 0.2 \sin(\pi t + \pi) \\ &= 0.4 \sin(\pi t) \quad \dots(3) \end{aligned}$$

$$6. \quad V = \sqrt{\frac{F}{\mu}}$$

$$E = \frac{1}{2} MV^2 + \frac{1}{2} I_C \omega^2$$

7. Conceptual

8. Conceptual

9. Conceptual

10. Conceptual

$$11. \quad V = \sqrt{\frac{T \times L}{m}}$$

$$\therefore \text{Extension } e = \frac{FL}{AY}$$

$$12. \quad \therefore T = \frac{Mx}{L}g$$

$$\text{So, } v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{Mgx}{L\left(\frac{M}{L}\right)}} = \sqrt{gx}$$

$$\text{hence } x = 0.5 \quad \therefore n = 2$$

$$13. \quad 12.5 \times n = \sqrt{\frac{4(10+2)}{19.2 \times 10^{-3}}} \quad \therefore n = 4$$

14. Conceptual

15. At any distance x

$$T = \frac{m}{L}(L-x)\omega^2 \frac{L+x}{2} = \frac{m\omega^2}{2L}(L^2 - x^2)$$

But $T = \mu v^2$ in rope frame

$$\therefore v^2 = \frac{\omega^2}{2}(L^2 - x^2)$$

$\therefore$ pulse will be meeting at a distance $\frac{AB}{\sqrt{2}}$ from A.

$$\therefore k = \frac{\frac{AB}{\sqrt{2}}}{AB - \frac{AB}{\sqrt{2}}} = \frac{1}{\sqrt{2} - 1} = \sqrt{2} + 1$$

$$\Rightarrow (k-1)^2 = 2$$

$$16. \quad v = \sqrt{\frac{3-(10K)t}{3 \times 10^{-2}}} = \frac{dx}{dt}$$

$$\int_0^L dx = \frac{10}{\sqrt{3}} \int_0^{t_0} \sqrt{3-10kt} \cdot dt \text{ where } t_0 = \frac{3}{10K}$$

$$17. \quad v_{AB} = \sqrt{\frac{T_{AB}}{\mu_{AB}}} = \sqrt{\frac{64}{10 \times 10^{-3}}} = \sqrt{6400} = 80 \text{ m/s}$$

$$v_{CD} = \sqrt{\frac{T_{CD}}{\mu_{DC}}} = \sqrt{\frac{32}{80 \times 10^{-3}}}$$

$$\frac{v_{AB}}{v_{CD}} = \frac{80}{20} = 4:1$$

18. The linear mass density is

$$\mu = \frac{5 \times 10^{-3}}{50 \times 10^{-2}} = 1.0 \times 10^{-2} \text{ kg/m}$$

The wave speed is $v = \sqrt{\frac{T}{\mu}}$

Thus, the tension is $T = \mu v^2 = (1.0 \times 10^{-2}) \times 6400 = 64 \text{ N}$

The Young's modulus is given by $Y = \frac{T/A}{\Delta L/L}$

The extension is, therefore, $\Delta L = \frac{TL}{AY} = \frac{64 \times 0.50}{1.0 \times 10^{-6} \times 16 \times 10^9} = 2 \text{ mm.}$

19. At the highest point

$$mg + 3mg = m(a_{\max})$$

$$\Rightarrow 4mg = m\omega^2 A$$

$$\Rightarrow \omega = \sqrt{\frac{4g}{A}} = 2\pi$$

$$\Rightarrow T = \frac{2\pi}{\omega}$$

$$\frac{y}{A} = \sin 2\pi \left(\frac{t}{T} - \frac{X}{\lambda} \right)$$

20. $y_1 = 1, A = 2,$

$$2\pi \left(\frac{t}{T} - \frac{x_1}{\lambda} \right) = \sin^{-1} \left(\frac{1}{2} \right) = \frac{\pi}{6} \text{ rad}$$

$$y_1 = \sqrt{2}, A = 2,$$

$$2\pi \left(\frac{t}{T} - \frac{x_2}{\lambda} \right) = \sin^{-1} \left(\frac{1}{\sqrt{2}} \right) = \frac{\pi}{4} \text{ rad}$$

$$\frac{t}{T} - \frac{x_1}{\lambda} = \frac{1}{12}$$

$$\frac{t}{T} - \frac{x_2}{\lambda} = \frac{1}{8}$$

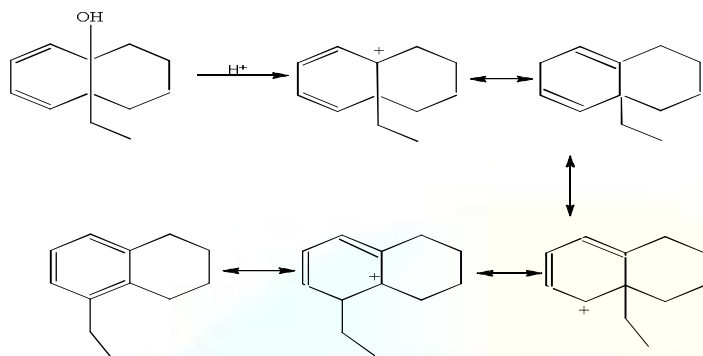
$$\text{subtract } \frac{x_1}{\lambda} - \frac{x_2}{\lambda} = \frac{1}{24}$$

$$(x_1 - x_2) = \frac{\lambda}{24}$$

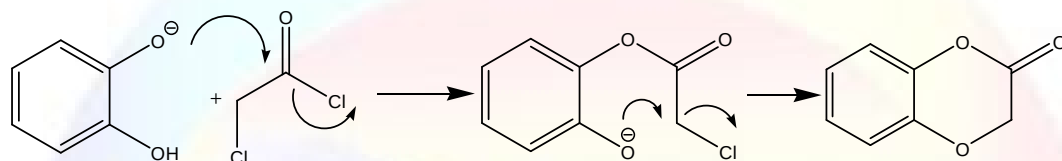
CHEMISTRY

21. Conceptual
22. Conceptual
23. Conceptual

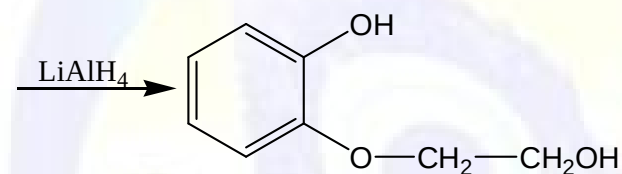
24. Conceptual



25.

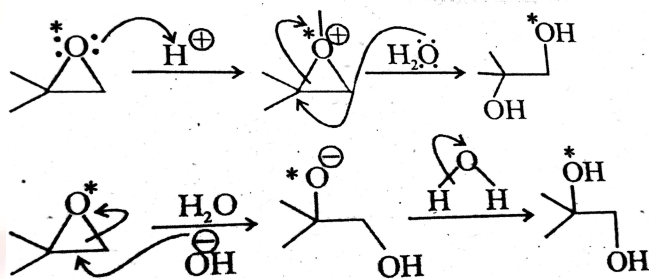


26.



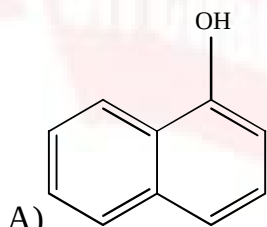
27. Conceptual

28. Conceptual



29.

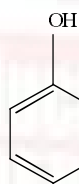
30. Conceptual



31.

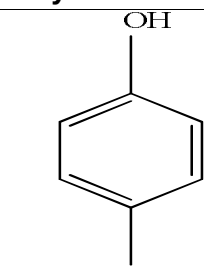
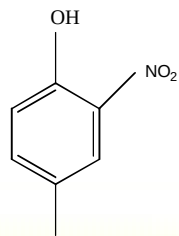
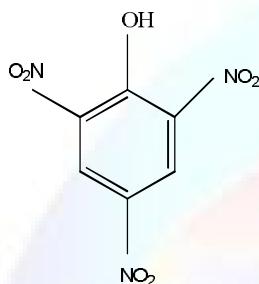
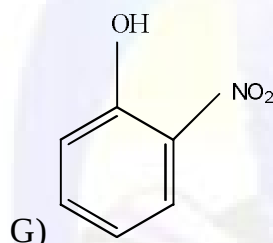
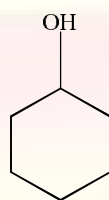
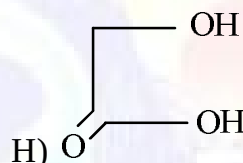
A)

PKa 9.31



B)

NO₂ PKa 8.28

C) $\text{PKa } 7.15$ D) $\text{PKa } 3.96$ E) $\text{PKa } 0.38$ F) $\text{PKa } 18$ G) $\text{PKa } 7.12$ H) $\text{PKa } 3.83$

32. Conceptual

33. Conceptual

34. Electron withdrawing groups will increase the acidity

35. Conceptual

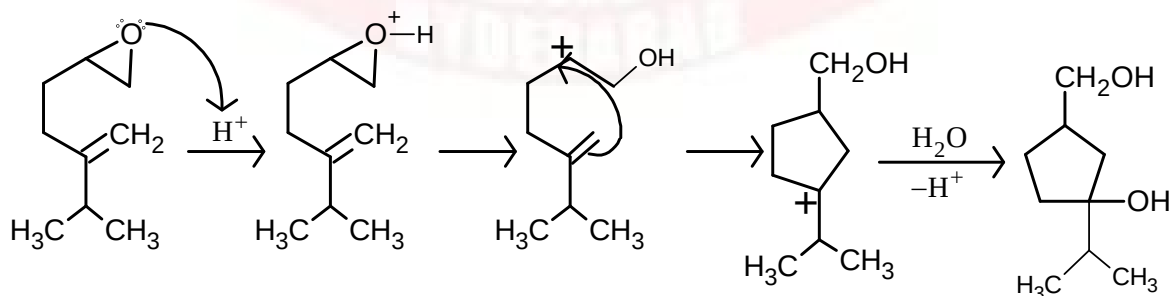
36. Conceptual

37. Conceptual

38. Conceptual



39.



40.

MATHS

41. Let 'O' be the centre of square ABCD and

$$\text{Let } P = (x, y)$$

$$\text{By the given data } \sqrt{x^2 + y^2} < |a - x|, \sqrt{x^2 + y^2} < |a + x|, \sqrt{x^2 + y^2} < |a - b| \text{ and } \sqrt{x^2 + y^2} < |a + y|$$

42. For externally touching a and b must have same sign

43. Locus of centre of triangle circumscribing triangle TPQ is

$$y^2 = \frac{1}{2}(x-1)$$

$$\therefore \text{Vertex} = V = (1, 0)$$

$$L.L.R = 4a = \frac{1}{2}$$

44. Conceptual

45. Conceptual

46. $A = (at_1^2, 2at_1)$ $B = (at_2^2, 2at_2)$, $P = (at^2, 2at)$

$$\angle APB = 90^\circ \Rightarrow \frac{4}{(t_1 + t)(t + t_3)} = 1 \rightarrow (1)$$

Let t_2, t_3 be the roots of (1)

$$\therefore t_2 + t_4 = -(t_1 + t_3)$$

$$\left| \frac{t_2 + t_4}{t_1 + t_3} \right| = 1 \quad \text{and} \quad t_2 t_4 = t_1 t_3 + 4$$

$$\therefore t_2 t_4 - t_1 t_3 = 4$$

47. Given point $P = (a, 2a)$ let other entity be $Q = (at^2, 2at)$

$$\text{Midpoint of } PQ = \left(\frac{a + at^2}{2}, a(1+t) \right) \text{ lies as } x + y = 1$$

$$\therefore \frac{a + at^2}{2} + a(1+t) = 1$$

$$\Rightarrow at^2 + 2at + (3a - 2) = 0$$

$$\text{Here } \Delta > 0 \Rightarrow a \in (0, 1) \\ \Rightarrow 4a \in (0, 4)$$

48. Let $G = (h, k)$

$$h = \frac{at^2 + (-at^2) + 2a + at^2}{3}, \quad k = \frac{2at}{3}$$

$$\therefore 3h = 2a + a\left(\frac{3k}{2a}\right)^2$$

$$\Rightarrow 3h = 2a + \frac{9k^2}{4a} \Rightarrow 9k^2 = 4a(3h - 2a)$$

$$\text{Returns } y^2 = \frac{4a}{3}\left(x - \frac{2a}{3}\right)$$

$$\therefore V = \left(\frac{x}{3}, 0\right) \text{ directs } x - \frac{2a}{3} = -\frac{a}{3} \Rightarrow x = \frac{a}{3}$$

49. Normal at 't' is $y + xt = 2at + at^3$

(3,0) lies on it $t^3 - t = 0 \therefore t = 0, 1, -1$

$$\therefore P = (1, 2), Q = (0, 0), R = (1, -2)$$

$$\therefore \text{Area} = \frac{1}{2} \times 1 \times 4 = 0$$

$$\text{Centroid} = \left(\frac{2}{3}, 0\right) \Rightarrow S = \left(\frac{5}{2}, 0\right) \Rightarrow R = \frac{5}{2}$$

50. Conceptual

51. Tangent at (6,9) is $3x - y - 9 = 0$

$$\text{Equation of circle is } (x-6)^2 + (y-9)^2 + \lambda(3x - y - 9) = 0$$

(0,1) lies on it $\Rightarrow \lambda = 10$

$$\therefore \text{circle is } r = 5\sqrt{10} \therefore \lambda = 5$$

52. Use result $ST = \sqrt{SP \cdot SQ}$

53. (4, y) be the mid point

$$\therefore \text{equation of chord is } S_1 = S_{11}$$

$$\therefore yy_1 - 4x = y_1^2 - 16$$

$$\text{Given slope} = 2 \therefore y_1 = 2$$

54. Feet of the perpendicular from focus upon any tangent to parabola lies on the tangent at the vertex.

$$\text{Equation of } SN_1 \text{ is } x + y = k \Rightarrow 1$$

$$(1, 2) \text{ lies on it } \therefore 1 \Rightarrow x + y = 3$$

Solve $x + y = 3$ and $x = y \therefore N_1 = \left(\frac{3}{2}, \frac{3}{2}\right)$

Equation of SN_2 is $x - y = t \Rightarrow 2$

$(1, 2)$ lies on it $\therefore 2 \Rightarrow y - x = 1$

Solve $y - x = 1$ and $x + y = 0 \therefore N_2 = \left(\frac{-1}{2}, \frac{1}{2}\right)$

$\therefore$ Equation of tangent at vertex = equation of N_1N_2

$= 2x - 4y + 3 = 0$

L.L.R = 4(distance from $S(1, 2)$ to tangent at vertex)

L.L.R = $4 \times \frac{|2 - 8 + 3|}{\sqrt{4 + 16}} = \frac{4 \times 3}{2\sqrt{5}} = \frac{6}{\sqrt{5}}$

55. Equation of a chord of contact is $S_1 = 0$

i.e. $yy_1 = 2(x + x_1) \Rightarrow \frac{yy_1 - 2x}{2x_1} = 1 \Rightarrow 1$

Horizontal curve $y^2 = 4x$ with 1

$y^2 - 4x \left(\frac{yy_1 - 2x}{2x_1} \right) = 0$

$\Rightarrow y^2 x_1 - 2y_1 xy + 4x^2 = 0$

$\Rightarrow 4x^2 - 2y_1 xy + y^2 x_1 = 0 \Rightarrow 1$

Equation of $5x^2 + 2xy + 3y^2 = 0 \Rightarrow 2$

1&2 are identical $\frac{4}{5} = \frac{-2y_1}{\alpha} = \frac{x_1}{3}$

$x_1 = \frac{12}{5} \Rightarrow 5x_1 = 12 \Rightarrow 5x = 12$

56. Equation is $y = mx - 2m - m^3$

$(9, -6)$ lies on it $m^3 - 7m - 6 = 0$

Roots are -1, -2, 3

$\therefore A = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & 3 \end{pmatrix} \therefore |A| = 8$

57. Area of triangle $PTQ = \left| \frac{1}{2} \times (2t^2) \times (2t) \right| = 2|t^3|$

$$\text{Given } t^2 \in [1, 4] \Rightarrow t \in [-2, -1] \cup [1, 2]$$

$$\Delta_{\max} = 16 \text{ for } t = \pm 2$$

$$\therefore K = 16 \Rightarrow \sqrt{k} = \sqrt{16} = 4$$

$$58. \text{ Parabola is } \left(x + \frac{7}{2}\right)^3 = 4\left(\frac{1}{4}\right)\left(y + \frac{41}{4}\right)$$

Slope of tangent parallel to given line = m = 3

$$P.C = \left[2am - \frac{7}{2}, am^2 - \frac{41}{4}\right] = [-2, -8] = (k_1, k_2)$$

$$\therefore \frac{|k_1 + k_2|}{2} = 5$$

$$59. \left(a + at^2\right)^2 \left(\frac{t^2 + 1}{t^2}\right) = 9 \text{ and } \left(a + at^2\right)^2 \left(\frac{t^2 + 1}{t^2}\right) = 16$$

$$\therefore t^2 = \frac{9}{16} \therefore 4a = \frac{4 \times 144}{125} \therefore [4a] = 4$$

$$60. x = 2t^2; y = 4t, \text{ substitute in circle}$$

$$\therefore 4t^4 + 16t^2 - 4t^2 - 16t = 0 \Rightarrow t = 0 \text{ or } 1$$

$$\therefore P = (0, 0); Q(2, 4); S = (2, 0)$$

$$\text{Area} = \frac{1}{2}|0 - 8| = 4 \text{ units}$$